Microsoft Certified Machine Learning Operations (MLOps) Engineer Associate

160 Real Exam Question and Answers



AI-300



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1) Case Study -
This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.
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To start the case study -
To display the first question in this case study, select the "Next" button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the "Question" button to return to the question.

Background -
Fabrikam Inc. is a mid-sized healthcare analytics company that provides population health dashboards and predictive insights to regional hospital systems across the United States. Fabrikam Inc. customers rely on near real time analytics to monitor patient flow, staffing needs, and readmission risks. They use multiple traditional forecasting machine learning models for predictions.

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Fabrikam Inc. has an established Microsoft Azure footprint. The company uses Jupyter Notebooks that run on a local server as the primary development environment. The data science team is experiencing scalability, asset management and code management issues with the current development platform. Fabrikam Inc. plans to migrate to a cloud-based development environment to mitigate the issues.
Additionally, the company plans to implement a Retrieval-Augmented Generation (RAG)-based chat application for client support. Leadership requires the application to be developed and deployed with a low operational risk.

Current Environment -
Fabrikam Inc. operates a single Azure subscription that has the following components:
Azure Data Lake Storage Gen2 that contains de-identified clinical and operational datasets
Azure AI Search indexing curated analytical documents and reference materials
A small set of Python-based training scripts maintained by data scientists
Azure OpenAI Service with deployed foundational models
A Microsoft Foundry resource for building a RAG-based solution
Evaluation data has manually defined expected responses.
The current challenges faced by the data science team include the following:
Model training jobs are run manually from notebooks.
Experiment tracking is inconsistent
Model versions are registered without standardized metadata.
Deployment is performed manually by data scientists, with limited rollback capability.
The team has no standardized evaluation process for generative AI outputs.
The environment currently allows public network access. Authentication relies on user accounts rather than managed identities.
Compute targets are manually created and shared across experiments. This has led to resource contention during peak usage.

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Business Requirements -
Fabrikam Inc. has the following business requirements for the modernization initiative:
Provide a conversational interface that answers analytics questions by using internal documents and datasets.
Ensure that sensitive healthcare-related data is not exposed outside the Fabrikam Inc. Azure tenant.
Enable repeatable and auditable model training and deployment processes.
Support experimentation to compare prompt strategies and fine-tuned models.
Align the model with the ranked preferences and optimize behavior for the long term.
Minimize disruption to existing analytics workloads during rollout.

Technical Requirements -
To support the business goals, Fabrikam Inc. identifies these technical requirements:
Use Azure Machine Learning workspaces to centrally manage data assets, models, and environments.
Implement experiment tracking and model versioning for all training jobs.
Orchestrate training and evaluation by using pipelines rather than manually running notebooks.
Deploy traditional machine learning models with support for staged rollout and rollback.
Improve RAG-based solution output quality.
Use the existing evaluation datasets that are based on real data with input-output pairs.
Apply advanced fine-tuning techniques only when prompt engineering is insufficient

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Issues and Constraints -
Fabrikam Inc. must comply with internal security policies that require the company to restrict network access and avoid long-lived secrets. The data science team has limited Azure DevOps experience, so solutions must favor managed services and automation over custom infrastructure.
Cost predictability is important. Leadership prefers serverless or managed compute options where possible but is willing to approve dedicated compute for stable production workloads.

Problem Statement -
Fabrikam Inc. must design and implement an Azure-based AI operations solution that enables reliable training, evaluation, deployment, and iteration of generative AI models. The solution must support experimentation and gradual rollout while ensuring governance, security, and operational stability. The data science and platform teams must collaborate to deliver this solution by using Azure Machine Learning and Microsoft Foundry capabilities.
You need to standardize how Fabrikam Inc. manages machine learning assets.
Which action should you perform first?

A. Register assets in the Azure Machine Learning registry.

B. Create a shared Azure Machine Learning workspace.

C. Deploy a managed online endpoint.

D. Create a new Microsoft Foundry project.

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You need to isolate training workloads while remaining cost-aware to address Fabrikam Inc.'s issues, constraints, and technical requirements.
What should you implement?

A. Training jobs that run on a single shared compute cluster

B. Fixed-size compute cluster

C. Dedicated compute clusters per experiment

D. Managed compute targets with autoscaling

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Correct Answer: *D*

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3) HOTSPOT -
A team trains an MLflow model that scores customer churn risk. The model will be consumed by different downstream systems.
One system requests predictions synchronously during customer interactions.
Another system submits files containing millions of records for scheduled scoring.
You need to deploy the model by using managed inference options that match each usage pattern.
Which option should you use for each usage pattern? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

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Managed inference options

Requirement	Deployment option
Low-latency synchronous predictions	Batch endpoint Job-based training pipeline Real-time endpoint Registered model artifact
High-volume scheduled scoring	Batch endpoint Online endpoint with autoscaling Managed compute cluster Model registry version

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Managed inference options

Requirement	Deployment option
Low-latency synchronous predictions	Real-time endpoint
High-volume scheduled scoring	Batch endpoint

Options for Low-latency synchronous predictions:
- Batch endpoint
- Job-based training pipeline
- Real-time endpoint
- Registered model artifact

Options for High-volume scheduled scoring:
- Batch endpoint
- Online endpoint with autoscaling
- Managed compute cluster
- Model registry version

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4) You manage an Azure Machine learning workspace. You develop a machine learning model.
You must deploy the model to use a low-priority VM with a pricing discount.
You need to deploy the model.
Which compute target should you use?

A. Azure Container Instances (ACI)

B. Azure Machine Learning compute clusters

C. Local deployment

D. Azure Kubernetes Service (AKS)

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Correct Answer: *B*

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5) A team manages an Azure Machine Learning workspace where they deploy models to online endpoints.
The team needs to introduce a new version of a model to production without disrupting existing users.
The team must validate the new version before full rollout.
You need to reduce risk during deployment.
What should you do?

A. Deploy the model to a batch endpoint.

B. Split traffic between deployments.

C. Replace the existing endpoint.

D. Route all traffic to the new deployment.

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Correct Answer: *B*

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6) You have a deployment of an Azure OpenAI Service base model.
You plan to fine-tune the model.
You need to prepare a file that contains training data.
Which file format should you use?

A. CSV

B. TSV

C. JSONL

D. JSON

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You need to prepare a file that contains training data for multi-turn chat.
Which file encoding method should you use?

A. ISO-8859-1

B. UTF-16

C. UTF-8

D. ASCII

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D. ASCII

Correct Answer: *C*

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8) You are fine-tuning a base language model to analyze customer feedback.
You label examples of support tickets. You must improve classification accuracy by configuring and fine-tuning the base model in Microsoft Foundry.
You need to configure and run fine-tuning.
What should you do first?

A. Use prompt flow to generate multiple prompt templates for evaluation.

B. Deploy the base model to an online endpoint before starting fine-tuning.

C. Enable tracing for all inference calls in the evaluation pipeline.

D. Format the dataset as a JSONL file with prompt-completion pairs and upload the file.

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Correct Answer: *C*

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9) A data science team trains a classification model that predicts loan approval outcomes.
Before registering the model, the team must ensure the following:
- Predictions must not disproportionately impact protected groups.
- Prediction errors can be evaluated across different data segments.
You need to assess whether the model meets Responsible AI expectations.
Which two approaches should you use? Each correct answer presents part of the solution.
Choose two.
NOTE: Each correct selection is worth one point.

A. Analyze error rates across defined demographic cohorts.
B. Evaluate feature importance for prediction transparency.
C. Measure endpoint latency under load.
D. Analyze error rates across the global cohort.
E. Validate inference schema compatibility.

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Correct Answer: A,B

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10) A team schedules weekly retraining of a model using Azure ML pipelines. They also want retraining triggered automatically when production data significantly deviates from training data distribution, without duplicating pipeline logic. What should they implement?

A. One pipeline triggered by schedule and data drift alerts
B. Azure Function to retrain model manually
C. Notebook-based retraining process
D. Two independent pipelines with shared scripts

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Correct Answer: A

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11) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear on the review screen.
You work in Microsoft Foundry with a prompt flow.
You must manually evaluate prompts and compare results across prompt variants.
You need to capture the inputs, outputs, token usage, and latencies for each flow run for the evaluation.
Solution: In Microsoft Foundry, turn on Tracing for the prompt flow of the project and execute test runs to produce trace data.
Does the solution meet the goal?

A. No
B. Yes

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Does the solution meet the goal?

A. No
B. Yes

Correct Answer: B

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12) A team develops multiple AI applications in Microsoft Foundry that rely on shared prompt templates.
The team requires a centralized way to track, version, and reuse prompt content across projects.
You need to recommend a solution to track and reuse prompt content.
Which approach should you recommend?

A. Embed prompts directly in application configuration files.
B. Store prompts as versioned files in a Git repository.
C. Register prompts as datasets in the Azure Machine Learning workspace.
D. Persist prompts in Azure Blob Storage with folder-level organization.

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Correct Answer: B

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13) Hotspot Question
A team is building a generative AI agent by using Retrieval-Augmented Generation (RAG) in Microsoft Foundry.
The team frequently updates prompt content. The team must be able to track changes across contributors while avoiding full application redeployments.
You need to enable rapid prompt iteration with traceability. Applications consuming the agent must be able to use updated prompts without requiring redeployment.
What should you configure for each requirement? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

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Prompt management configuration

Requirement	Configuration
Track prompt changes across teams.	Git repository Azure Blob Storage Azure Machine Learning Datastores
Apply prompt updates without redeployment.	CI/CD pipeline Environment variables Agent's latest published version reference

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Prompt management configuration

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Track prompt changes across teams.	Git repository
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Git repository
Azure Blob Storage
Azure Machine Learning Datastores

CI/CD pipeline
Environment variables
Agent's latest published version reference

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14) A company requires that only models meeting predefined performance thresholds are registered and deployed. The solution must be fully automated within the ML workflow. What should you implement?

A. GitHub pull request checks
B. Conditional logic in pipeline
C. Azure Monitor alerts
D. Manual approval gate

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C. Azure Monitor alerts
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Solution: Use the prompt flow SDK to enable tracing for the flow before executing runs. Then run the flow to generate traceable results.
Does the solution meet the goal?

A. No
B. Yes

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Does the solution meet the goal?

A. No
B. Yes

Correct Answer: A

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16) Drag and Drop Question
A team iterates prompts used by a generative AI agent. The team must support internal review before releasing changes.
The team must:
- Track prompt changes with a clear history for audit and rollback.
- Compare prompt variants in parallel without affecting the prompt used in the production environment.
You need to select the appropriate source control approach for each requirement.
What should you use for each requirement? To answer, move the appropriate source controls to the correct requirements. You may use each source control once, more than once, or not at all.
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Source controls

- Git repository
- Azure Machine Learning notebooks
- Azure Storage
- Branch-based workflow

Source control options for prompts

Requirement	Source control
Track prompt changes.	
Evaluate prompt variants.	

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Source controls

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Source control options for prompts

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Evaluate prompt variants.	Branch-based workflow

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17) An Azure Machine Learning workspace processes sensitive training data.
The workspace must NOT be accessible from the public internet.
You need to restrict network access.
Which configuration should you implement?

A. Service endpoints
B. Network security groups
C. Azure Firewall rules
D. Private endpoints

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Correct Answer: D

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18) A team runs training jobs by using multiple Azure Machine Learning pipelines.
The team must ensure that all runs use the same Python packages and system libraries. The solution must allow dependency updates to be versioned without modifying training code.
You need to configure the workspace so that runtime dependencies are consistent and reusable.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Configure Azure Machine Learning pipelines

- Reference the environment in training jobs and pipelines.
- Register the environment in the Azure Machine Learning workspace.
- Store dependency versions in a configuration file.
- Modify training scripts to install required packages at runtime.
- Define the required dependencies in an environment specification.
- Create a new version of the environment when dependencies change.

Answer Area

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Answer Area

1. Define the required dependencies in an environment specification.
2. Register the environment in the Azure Machine Learning workspace.
3. Create a new version of the environment when dependencies change.
4. Reference the environment in training jobs and pipelines.

19) During training, pipelines occasionally fail due to schema mismatch caused by upstream data changes. You need a robust and automated solution that prevents invalid data from reaching training steps. What is the BEST approach?

A. Ignore schema differences
B. Retrain manually when failure occurs
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Correct Answer: C

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20) A company is standardizing generative AI development across multiple teams.
Each team requires an isolated workspace. Governance and shared connections must be centrally managed.
You need to implement a Microsoft Foundry environment structure that supports centralized governance and team isolation.
Which type of configuration should you use for each requirement? To answer, move the appropriate configurations to the correct requirements. You may use each configuration once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
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21) Hotspot Question
You manage a Retrieval-Augmented Generation (RAG) system that retrieves internal policy documents from a vector index.
Recent analysis shows that:
- Retrieved results frequently include duplicated content from the same document.
- Retrieved chunks sometimes span unrelated policy sections.
You review the following retrieval and ingestion configurations:

Configuration 1

Parameter	Setting
Chunk size	1,200
Chunk overlap	400
Similarity threshold	0.80
Retrieval strategy	Vector
Top k	8

Configuration 2

Parameter	Setting
Chunk size	300
Chunk overlap	40
Similarity threshold	0.90
Retrieval strategy	Vector
Top k	8

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You need to reduce duplicated retrieval results and improve chunk relevance across policy sections. For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

RAG retrieval and chunking configurations

Statement	Yes	No
Configuration 1 produces duplicated content across retrieved chunks.	○	○
Configuration 2 enforces stricter similarity filtering that can exclude relevant chunks.	○	○
Lowering the similarity threshold is required to reduce semantic dilution.	○	○
Switching to keyword-only search will eliminate semantic dilution while preserving semantic relevance.	○	○

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22) A Retrieval-Augmented Generation (RAG) solution returns incomplete answers because relevant content is inconsistently retrieved from the knowledge source.
You need to improve RAG accuracy without changing the embedding model currently in use. You need to achieve this goal while minimizing operational costs.
Which two actions should you perform? Each correct answer presents part of the solution.
Choose two.
NOTE: Each correct selection is worth one point.

A. Increase token limits for all requests.
B. Implement an optimized re-ranker.
C. Optimize the length of embedding vectors.
D. Tune chunk size and overlap to match content structure.

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22) A Retrieval-Augmented Generation (RAG) solution returns incomplete answers because relevant content is inconsistently retrieved from the knowledge source.
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C. Optimize the length of embedding vectors.
D. Tune chunk size and overlap to match content structure.

Correct Answer: B,D

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23) You manage a Microsoft Foundry project. You build a multi-turn chatbot application.
You plan to filter your traces to identify issues while observing how the application is responding. The solution must not use an external knowledge base.
You need to select an evaluation metric.
Which built-in evaluator should you use?

A. RelevanceEvaluator

B. SimilarityEvaluator

C. QAEvaluator

D. CoherenceEvaluator

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Which built-in evaluator should you use?

A. RelevanceEvaluator

B. SimilarityEvaluator

C. QAEvaluator

D. CoherenceEvaluator

Correct Answer: D

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24) You have a Microsoft Foundry project.
You plan to use the Microsoft Foundry portal to fine-tune a base Azure OpenAI Service model that can accept both text and images as input.
You need to choose the suitable model.
Which model should you choose?

A. davinci-002

B. gpt-4o

C. gpt-35-turbo

D. gpt-4

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24) You have a Microsoft Foundry project.
You plan to use the Microsoft Foundry portal to fine-tune a base Azure OpenAI Service model that can accept both text and images as input.
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Which model should you choose?

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C. gpt-35-turbo

D. gpt-4

Answer: *B*

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25) A team maintains Infrastructure as Code (IaC) templates to provision Azure Machine Learning resources. Provisioning must be triggered by changes in the templates and executed without manual intervention.
You need to automate resource provisioning.
Which action should you take for each requirement? To answer, move the appropriate actions to the correct requirements. You may use each action once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

Actions

- Use an Azure service principal.
- Create a manual deployment script.
- Configure a GitHub Actions workflow.
- Run Azure CLI commands from a local machine.

Automate provisioning

Requirement	Action
Trigger provisioning on repository changes.	
Authenticate securely during workflow execution.	

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Actions

- Use an Azure service principal.
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Automate provisioning

Requirement	Action
Trigger provisioning on repository changes.	Configure a GitHub Actions workflow.
Authenticate securely during workflow execution.	Use an Azure service principal.

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26) A company is creating an internal tool that summarizes long meeting transcripts and extracts action items.
The model must:
Process text inputs up to 200k tokens long.
Generate concise summaries in seconds.
Support interactive testing before integration into the app.
You need to select, deploy, and test a model that supports summarization with low latency.
How should you complete the configuration plan? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

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Deploying and testing models for summarization

Requirement	Tool
Select a model.	Whisper Phi-3-Mini GPT-4.1-Mini Text-Embedding-Ada-002
Select a deployment type.	Batch endpoint Custom container Serverless endpoint Managed real-time (online) endpoint
Test the model.	Prompt flow Microsoft Foundry Chat playground Microsoft Foundry model comparison

Deploying and testing models for summarization

Requirement	Tool
Select a model.	Whisper Phi-3-Mini **GPT-4.1-Mini** Text-Embedding-Ada-002
Select a deployment type.	Batch endpoint Custom container Serverless endpoint **Managed real-time (online) endpoint**
Test the model.	Prompt flow **Microsoft Foundry Chat playground** Microsoft Foundry model comparison

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27) A team deploys a generative AI application that uses a model deployed in Microsoft Foundry. The application must support latency monitoring under production load.
You need to enable performance observability.
Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Enable performance observability

- Set up the model endpoint in Foundry.
- Enable logging.
- Apply a term blocklist.
- Generate token embeddings.
- Configure monitoring metrics.

Answer Area

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Answer Area

- Set up the model endpoint in Foundry.
- Configure monitoring metrics.
- Enable logging.

28) An organization operates a generative AI application in production by using Microsoft Foundry. The application serves live user traffic and is updated by a data scientist team regularly as prompts and models evolve.
The application intermittently times out during production use, which requires ongoing visibility into runtime behavior.
The team must also validate model quality and safety before releasing new updates to avoid introducing regressions.
You need to apply the correct mechanisms for continuous runtime monitoring and for release time validation.
Which mechanisms should you use for each requirement? To answer, move the appropriate mechanisms to the correct requirements. You may use each mechanism once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

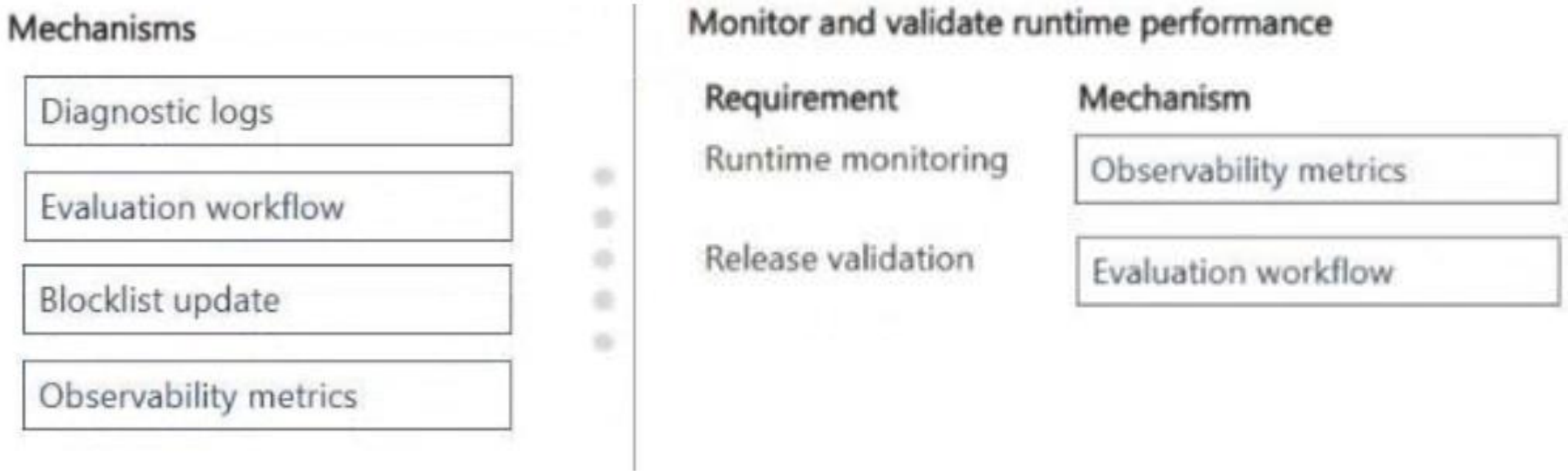
Mechanisms

- Diagnostic logs
- Evaluation workflow
- Blocklist update
- Observability metrics

Monitor and validate runtime performance

Requirement	Mechanism
Runtime monitoring	
Release validation	

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29) A machine learning model is deployed to production in Azure Machine Learning and is actively serving predictions for a business application. The model was trained by using a historical dataset that represented expected input patterns at the time of deployment.

The team working on the model must ensure the following:

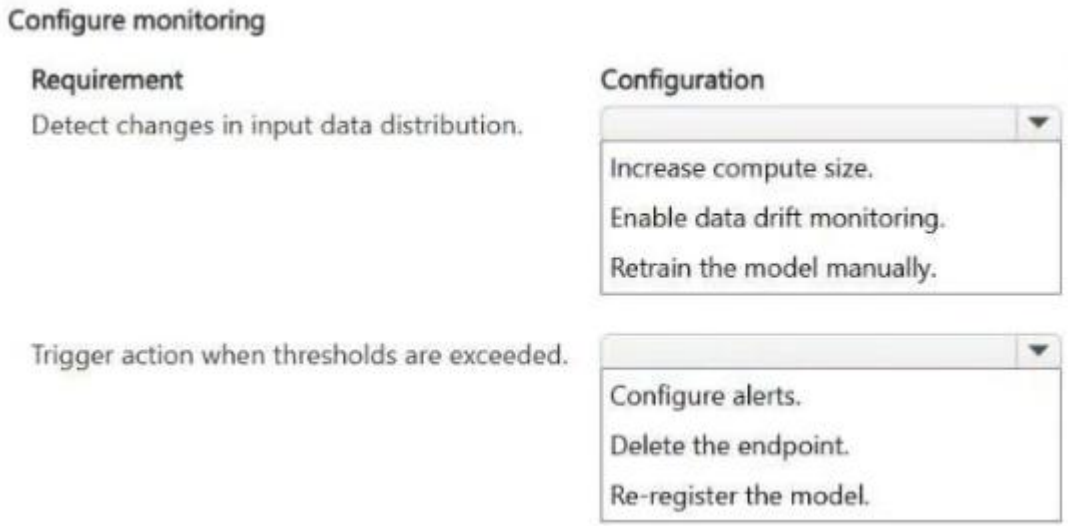
Changes in input data distribution are detected.

Appropriate actions are triggered when predefined thresholds are exceeded.

You need to configure monitoring to meet the requirements.

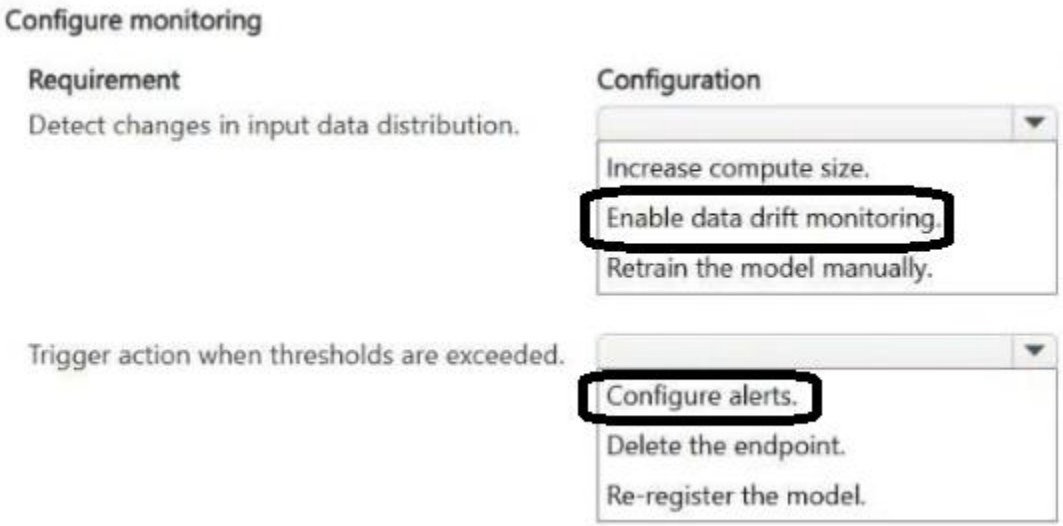
Which configuration should you use for each requirement? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



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Which configuration should you use for each requirement? To answer, select the appropriate options in the answer area.
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30) A team is developing a generative AI assistant. The team is experimenting with multiple prompt variants to improve the user experience.
When comparing prompt variants, the team plans to assess whether the generated responses are grammatically correct.
You need to evaluate the quality of the language from the generated responses.
Which evaluator should you use?

A. Coherence

B. Textual similarity

C. Grounded ness

D. Fluency

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D. Fluency

Answer: *D*

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31) Case Study -
This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.
To answer the case study questions, you will need to reference information that is provided in the case. Case studies and associated questions might contain exhibits or other resources that provide more information about the scenario described in the case. Information provided in an individual question does not apply to the other questions in the case study.
A Review Screen will appear at the end of this case study. From the Review Screen, you can review and change your answers before you move to the next exam section. After you leave this case study, you will NOT be able to return to it.

To start the case study -
To display the first question in this case study, select the "Next" button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the "Question" button to return to the question.

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Background -
Fabrikam Inc. is a mid-sized healthcare analytics company that provides population health dashboards and predictive insights to regional hospital systems across the United States. Fabrikam Inc. customers rely on near real time analytics to monitor patient flow, staffing needs, and readmission risks. They use multiple traditional forecasting machine learning models for predictions.
Fabrikam Inc. has an established Microsoft Azure footprint. The company uses Jupyter Notebooks that run on a local server as the primary development environment. The data science team is experiencing scalability, asset management and code management issues with the current development platform. Fabrikam Inc. plans to migrate to a cloud-based development environment to mitigate the issues.
Additionally, the company plans to implement a Retrieval-Augmented Generation (RAG)-based chat application for client support. Leadership requires the application to be developed and deployed with a low operational risk.

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Current Environment -
Fabrikam Inc. operates a single Azure subscription that has the following components:
Azure Data Lake Storage Gen2 that contains de-identified clinical and operational datasets
Azure AI Search indexing curated analytical documents and reference materials
A small set of Python-based training scripts maintained by data scientists
Azure OpenAI Service with deployed foundational models
A Microsoft Foundry resource for building a RAG-based solution
Evaluation data has manually defined expected responses.
The current challenges faced by the data science team include the following:
Model training jobs are run manually from notebooks.
Experiment tracking is inconsistent
Model versions are registered without standardized metadata.
Deployment is performed manually by data scientists, with limited rollback capability.
The team has no standardized evaluation process for generative AI outputs.
The environment currently allows public network access. Authentication relies on user accounts rather than managed identities. Compute targets are manually created and shared across experiments. This has led to resource contention during peak usage.

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Business Requirements -
Fabrikam Inc. has the following business requirements for the modernization initiative:
Provide a conversational interface that answers analytics questions by using internal documents and datasets.
Ensure that sensitive healthcare-related data is not exposed outside the Fabrikam Inc. Azure tenant.
Enable repeatable and auditable model training and deployment processes.
Support experimentation to compare prompt strategies and fine-tuned models.
Align the model with the ranked preferences and optimize behavior for the long term.
Minimize disruption to existing analytics workloads during rollout.

Technical Requirements -
To support the business goals, Fabrikam Inc. identifies these technical requirements:
Use Azure Machine Learning workspaces to centrally manage data assets, models, and environments.
Implement experiment tracking and model versioning for all training jobs.
Orchestrate training and evaluation by using pipelines rather than manually running notebooks.
Deploy traditional machine learning models with support for staged rollout and rollback.
Improve RAG-based solution output quality.
Use the existing evaluation datasets that are based on real data with input-output pairs.
Apply advanced fine-tuning techniques only when prompt engineering is insufficient

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Issues and Constraints -
Fabrikam Inc. must comply with internal security policies that require the company to restrict network access and avoid long-lived secrets. The data science team has limited Azure DevOps experience, so solutions must favor managed services and automation over custom infrastructure.
Cost predictability is important. Leadership prefers serverless or managed compute options where possible but is willing to approve dedicated compute for stable production workloads.

Problem Statement -
Fabrikam Inc. must design and implement an Azure-based AI operations solution that enables reliable training, evaluation, deployment, and iteration of generative AI models. The solution must support experimentation and gradual rollout while ensuring governance, security, and operational stability. The data science and platform teams must collaborate to deliver this solution by using Azure Machine Learning and Microsoft Foundry capabilities.
You need to recommend an experiment-tracking strategy that ensures consistent experiment results.
What should you recommend?

A. Azure Machine Learning job output logs

B. MLflow experiment tracking

C. Application Insights logs

D. Azure Monitor alerts

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What should you recommend?

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Answer: *B*

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32) You train a model in Azure Machine Learning.
You plan to capture experiment details for later comparison. The training code must log parameters and metrics for each run.
You review the following training script.

```
import mlflow
import mlflow.sklearn
from sklearn.linear_model import LogisticRegression
from sklearn.datasets import load_iris

X, y = load_iris(return_X_y=True)

mlflow.set_experiment("classification-experiment")

with mlflow.start_run():
    model = LogisticRegression(max_iter=200)
    model.fit(X, y)

    mlflow.log_param("max_iter", 200)
    y_pred = model.predict(X)
    accuracy = accuracy_score(y, y_pred)
    mlflow.log_metric("accuracy", accuracy)

    mlflow.sklearn.log_model(model, "model")
```

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You need to verify whether the training script meets the experiment tracking requirement. For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

MLflow experiment tracking

Statement	Yes	No
The code ensures that training parameters and metrics are tracked for each run.	○	○
The code uses the experiment name to define the artifact path of the logged model.	○	○
The code associates the run with a named experiment.	○	○

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The code associates the run with a named experiment.	○	○

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33) You deploy a model to production but do not have labeled data available for evaluating prediction accuracy. However, you must monitor model health continuously. What is the BEST strategy?

A. Use data drift detection
B. Disable monitoring
C. Monitor accuracy metrics
D. Perform manual evaluation only

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A. Use data drift detection
B. Disable monitoring
C. Monitor accuracy metrics
D. Perform manual evaluation only

Answer: A

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34) You are reviewing a dataset that will be used for an advanced fine-tuning job in Microsoft Foundry.
The fine-tuning job uses preference comparison data.
You review the following dataset excerpt.

Training record 1:

```
01 {"input":[{"role":"system","content":"You are a policy assistant."},
02           {"role":"user","content":"Summarize Policy X for managers."}],
03  "preferred_output":[{"role":"assistant","content":"Policy X applies to all managers and outlines approval
requirements."}],
04  "non_preferred_output":[{"role":"assistant","content":"Policy X applies to all managers and outlines approval
requirements."}]}
```

Training record 2:

```
01
02 {"input":[{"role":"system","content":"You are a policy assistant."},
03           {"role":"user","content":"What is the retention period?"}],
04  "preferred_output":[{"role":"assistant","content":"The retention period is seven years."}],
05  "non_preferred_output":[{"role":"assistant","content":"The retention period is five years."}]}
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Preference-based fine-tuning dataset validation

Statement	Yes	No
The first record provides a usable preference signal for preference-based fine-tuning.	○	○
The second record provides a usable preference signal for preference-based fine-tuning.	○	○
Each preferred and non-preferred output in the dataset contains a valid assistant message as required for direct preference optimization.	○	○
Both records must contain different preferred and non-preferred outputs to produce usable training signals.	○	○

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For each of the following statements, select Yes if the statement is true. Otherwise, select No.
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Preference-based fine-tuning dataset validation

Statement	Yes	No
The first record provides a usable preference signal for preference-based fine-tuning.	○	●
The second record provides a usable preference signal for preference-based fine-tuning.	●	○
Each preferred and non-preferred output in the dataset contains a valid assistant message as required for direct preference optimization.	○	●
Both records must contain different preferred and non-preferred outputs to produce usable training signals.	●	○

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35) You need to run large-scale inference jobs on millions of records periodically. Jobs are not latency-sensitive but must be cost-efficient and scalable. Which deployment option is MOST appropriate?

A. Notebook execution
B. Managed online endpoint
C. Local endpoint
D. Batch endpoint

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A. Notebook execution
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D. Batch endpoint

Answer: D

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36) A financial services company is deploying Microsoft Foundry to host generative AI workloads that process regulated customer data. The Microsoft Foundry environment must prevent any public network exposure while still allowing services managed by Microsoft Foundry to communicate with dependent Azure resources.
Security auditors require that all traffic to and from the Microsoft Foundry resource remain on private networks, with no public endpoints available.
You need to configure the Microsoft Foundry environment so that network access is restricted while maintaining full platform functionality.
Which two actions should you perform? Each correct answer presents part of the solution.
Choose two.
NOTE: Each correct selection is worth one point.

A. Disable all inbound network access.
B. Deploy the Microsoft Foundry resource in a separate Azure subscription.
C. Use API key authentication for all model endpoints.
D. Disable public network access to the Microsoft Foundry resource.
E. Configure a managed virtual network for the Microsoft Foundry resource.

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E. Configure a managed virtual network for the Microsoft Foundry resource.

Answer: A,E

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37) A company's platform engineers manage the resource settings and governance of Microsoft Foundry.
Developers must be able to create and update project assets but must not be able to change resource-level configurations.
You need to enforce least privilege access for the engineers and developers.
Which two actions should you perform? Each correct answer presents part of the solution.
Choose two.
NOTE: Each correct selection is worth one point.

A. Share a single API key across all teams.
B. Assign a resource-level Azure AI Administrator role to the platform engineers.
C. Disable Microsoft Entra ID authentication for the Microsoft Foundry resource.
D. Assign the Azure AI Developer role to the developers.

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Answer: B,D

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38) You create a binary classification model. You use the Fairlearn package to assess model fairness.
You must eliminate the need to retrain the model.
You need to implement the Fairlearan package.
Which algorithm should you use?

A. fairiearn.reductions.ExponentiatedGradient
B. fairlearn.preprocessing.CorrelationRemover
C. fairlearn.reductions.GridSearch
D. fairlearn.postprocessing.ThresholdOptimizer

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C. fairlearn.reductions.GridSearch
D. fairlearn.postprocessing.ThresholdOptimizer

Answer: D

The best algorithm to implement within the Fairlearn package for a binary classification model without retraining is the ThresholdOptimizer.
Why ThresholdOptimizer?
This algorithm is a post-processing technique specifically designed to mitigate unfairness after a model has already been trained. By setting the prefit parameter to True, it treats your existing model as a "black box," adjusting its decision thresholds for different groups to satisfy fairness constraints (like demographic parity or equalized odds) without changing the underlying model weights.
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/concept-fairness-ml

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39) A team accidentally deploys an outdated model version due to incorrect tagging. You need to enforce strict deployment governance and version control. What should you implement?

A. Rename models
B. Use version pinning and approval workflows
C. Increase logging
D. Store models locally

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40) A real-time endpoint experiences sporadic latency spikes. Investigation reveals instances scale down to zero during inactivity. You need to reduce latency without significantly increasing cost.
What should you configure?

A. Increase timeout settings
B. Set minimum instance count
C. Switch to batch endpoint
D. Reduce logging

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Answer: B

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41) You want to ensure ML pipelines produce identical results when executed in different regions or workspaces. Which factor is MOST critical to control for reproducibility?

A. Same compute SKU
B. Same dataset name
C. Versioned datasets and environments
D. Same user identity

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B. Same dataset name
C. Versioned datasets and environments
D. Same user identity

Answer: C

Reproducibility across environments depends on consistent versioning of datasets and environments. Differences in data or dependencies can lead to inconsistent results even if compute resources are identical. Proper asset versioning ensures experiments can be reliably reproduced in any workspace or region.

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42) A team develops and manages a conversational assistant by using Microsoft Foundry.
The team must be able to validate that the assistant does not produce hateful responses before the application is exposed to any users.
You need to evaluate the model output for hateful responses as part of a repeatable validation process.
Which evaluator should you configure first?

A. Protected material
B. Groundedness
C. Indirect attacks
D. Content safety

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Which evaluator should you configure first?

A. Protected material
B. Groundedness
C. Indirect attacks
D. Content safety

Answer: D

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43) An organization is deploying several generative AI workloads by using Microsoft Foundry. Each workload must meet different requirements related to data governance, task specialization, and operational cost control.
The organization requires models that meet the following requirements:
Model behavior aligns with the task being performed.
Data handling aligns with internal governance policies.
Operational complexity and cost are justified by workload needs.
You need to select the foundation model options that meet the requirements.
Which three models can you select? Each correct answer presents a complete solution. Choose three.
NOTE: Each correct selection is worth one point.

A. A model that is optimized for conversational reasoning when deploying an interactive assistant

B. The largest available model to simplify operational management

C. The smallest available model to minimize the usage cost

D. A model that supports multiple input types when workloads require combined text and image analysis

E. A model that offers enterprise governance controls when workloads process regulated business data

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E. A model that offers enterprise governance controls when workloads process regulated business data

Answer: *BCE*

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44) A team manages prompts that are used by a generative AI application built on Microsoft Foundry. Multiple developers contribute prompt updates, and changes must be reviewed and tracked over time.
The team requires that:
Prompt changes are reviewed before being applied to the version in production.
Previous prompt versions can be restored if issues occur.
Prompt updates follow the same governance practices as the application code.
You need to implement a controlled process for managing and updating prompts in production.
How should you manage prompt updates to meet the requirements? To answer, move the appropriate actions to the correct requirements. You may use each action once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

Actions

- Store prompts in a Git repository.
- Use gated pull requests for prompt changes.
- Use a prior agent version.
- Update prompts directly in production.

Manage prompt updates

Requirement	Action
Track changes and maintain history.	
Review updates before production use.	
Roll back to a specific prompt version.	

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Requirement	Action
Track changes and maintain history.	Store prompts in a Git repository.
Review updates before production use.	Use gated pull requests for prompt changes.
Roll back to a specific prompt version.	Use a prior agent version.

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45) Your ML pipeline contains independent feature engineering steps that currently execute sequentially, increasing overall runtime. You want to optimize execution without modifying logic.
What is the BEST solution?

A. Enable parallel step execution
B. Reduce dataset size
C. Increase compute size
D. Combine all steps

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C. Increase compute size
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Correct Answer: A

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46) You must ensure full reproducibility of experiments including dataset, code, and environment across multiple runs and workspaces. Which combination of practices is MOST appropriate?

A. Git only
B. Logging metrics only
C. Dataset versioning only
D. Environment + dataset + code versioning

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A. Git only
B. Logging metrics only
C. Dataset versioning only
D. Environment + dataset + code versioning

Correct Answer: D

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47) An Azure Machine Learning workspace contains multiple registered versions of a model that is used in production. An older model version must no longer be deployable, but it must remain available for compliance review and potential rollback.
You need to change the state of the model version to meet the requirements.
What should you do?

A. Archive the training dataset for the model version.
B. Delete the model version.
C. Unregister the model version.
D. Archive the model version.

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B. Delete the model version.
C. Unregister the model version.
D. Archive the model version.

Correct Answer: D

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48) You manage a Retrieval-Augmented Generation (RAG) system that uses Azure AI Search to retrieve documents from an indexed knowledge base.
The system must support the following retrieval requirements:
- Queries that include exact policy identifiers must return matching documents even when semantic similarity is low.
- Natural-language questions must prioritize semantically relevant documents even when keywords are not an exact match.
You need to configure the retrieval approach to meet the requirements.
How should you configure the retrieval behavior for each requirement? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

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Hybrid retrieval configuration

Retrieval requirement	Configuration
Exact identifier matching	Semantic ranking only Keyword-based retrieval disabled Vector similarity without keyword scoring Keyword-based retrieval with lexical scoring
Natural-language relevance	Exact-match filtering only Keyword-only search without embeddings Keyword-based retrieval with lexical scoring Semantic ranking over vector search results

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Hybrid retrieval configuration

Retrieval requirement	Configuration
Exact identifier matching	Keyword-based retrieval with lexical scoring
Natural-language relevance	Semantic ranking over vector search results

Configuration options:

- Semantic ranking only
- Keyword-based retrieval disabled
- Vector similarity without keyword scoring
- Keyword-based retrieval with lexical scoring
- Exact-match filtering only
- Keyword-only search without embeddings
- Keyword-based retrieval with lexical scoring
- Semantic ranking over vector search results

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49) An organization maintains separate Azure Machine Learning workspaces for development and production.
Both environments must use the same validated assets without duplicating them.
Assets must be shared across workspaces while maintaining centralized governance and version control.
You need to enable reuse of assets across workspaces without copying them.
What should you do?

A. Enable workspace-level Git integration and sync assets between repositories.
B. Publish the asset as a pipeline component.
C. Create a shared Azure Machine Learning environment that includes the asset.
D. Publish the asset to an Azure Machine Learning registry.

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Correct Answer: D

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50) A team is experimenting with traditional models for a classification workflow in Azure Machine Learning.
The team requires a consistent way to manage assets that are created during experimentation.
You need to ensure that artifacts can be reused and governed across projects.
Which asset should you register?

A. Model
B. Component
C. Environment
D. Pipeline

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You need to ensure that artifacts can be reused and governed across projects.
Which asset should you register?

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B. Component
C. Environment
D. Pipeline

Correct Answer: A

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51) A team deploys a model to a real-time endpoint in Azure Machine Learning. You deploy some updates to the endpoint.
The endpoint returns errors after the new deployment is released.
You need to restore the service as quickly as possible.
What should you do first?

A. Roll back traffic to the previous deployment.
B. Delete the endpoint and immediately redeploy it.
C. Change the authentication type to Azure Machine Learning token-based authentication.
D. Increase the compute size.

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D. Increase the compute size.

Correct Answer: A

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52) You train models on GPU-enabled clusters but deploy them on CPU-based endpoints. Recently, inference failures occur due to incompatible dependencies. What should you do to ensure consistency?

A. Use batch endpoints
B. Use same compute for training and inference
C. Define and reuse environment configurations
D. Increase endpoint compute size

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A. Use batch endpoints
B. Use same compute for training and inference
C. Define and reuse environment configurations
D. Increase endpoint compute size

Correct Answer: C

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53) Your model requires access to external APIs using sensitive credentials during inference. You must ensure credentials are not exposed in code, logs, or environment variables. What should you implement?

A. Hardcode credentials
B. Store in config files
C. Use Azure Key Vault with managed identity
D. Encrypt credentials locally

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A. Hardcode credentials
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D. Encrypt credentials locally

Correct Answer: C

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54) A pipeline step fails intermittently due to transient compute issues. You need to improve reliability without modifying core logic or increasing cost significantly. What is the BEST approach?

A. Increase compute size
B. Enable retry policy on pipeline step
C. Run pipeline manually
D. Remove failing step

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A. Increase compute size
B. Enable retry policy on pipeline step
C. Run pipeline manually
D. Remove failing step

Correct Answer: B

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55) A team is deploying machine learning models to a production inference endpoint in Azure Machine Learning.
The team requires a safe way to validate a new model version without disrupting existing users.
You need to recommend a deployment strategy for controlled testing of a new model version.
What should you configure?

A. traffic splitting between deployments
B. the model asset version in the registry
C. deployment to a separate staging endpoint
D. an evaluation script in Azure Machine Learning

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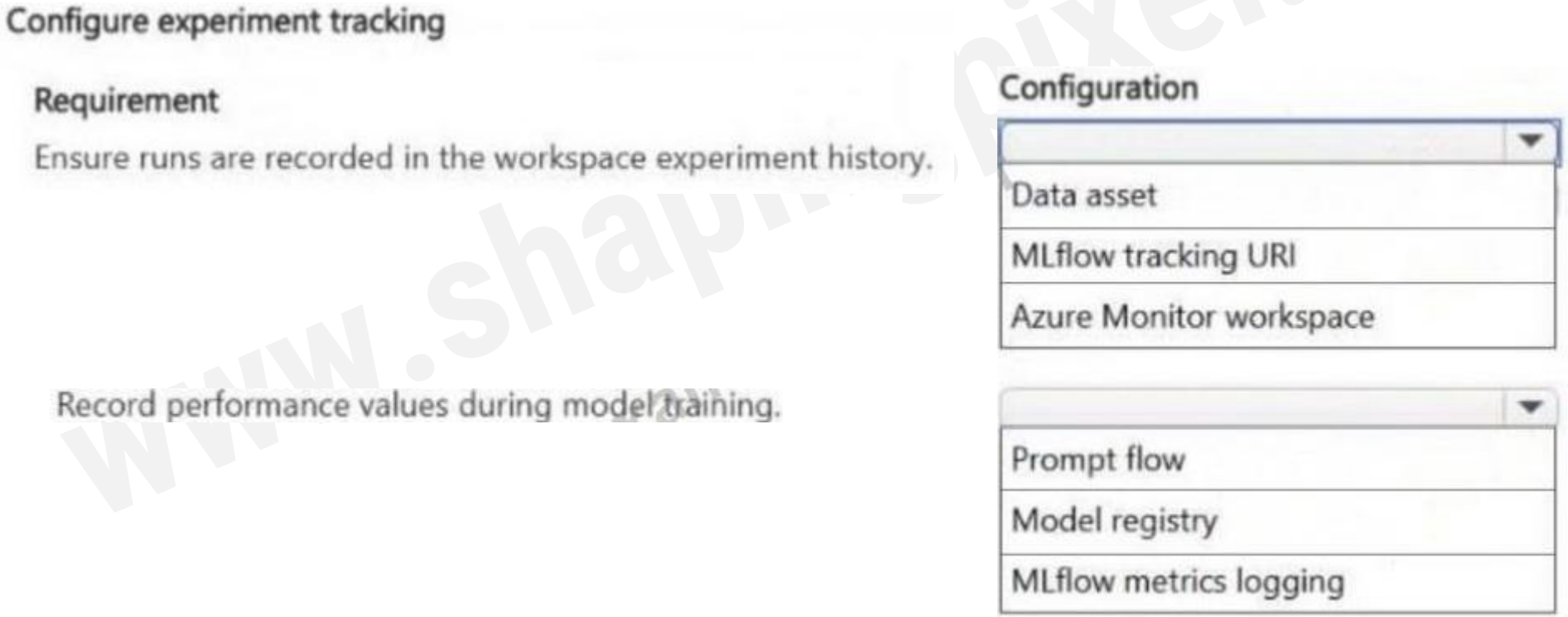
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A. traffic splitting between deployments
B. the model asset version in the registry
C. deployment to a separate staging endpoint
D. an evaluation script in Azure Machine Learning

Correct Answer: A

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56) You use Azure Machine Learning to train models across multiple experiments by using the same workspace.
You must record training runs in a centralized location to compare results from different jobs.
During training, performance values must be captured so they appear in the experiment run history.
You need to configure experiment tracking.
What should you configure for each requirement? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



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57) A company plans to deploy a foundation model in Microsoft Foundry.
The mode must support the following workloads:
- A customer support workload used across multiple regions
- A marketing workload that must remain within a specific region due to data residency requirements You need to select the deployment type.
Which deployment type should you use for each workload? To answer, move the appropriate deployment types to the correct requirements. You may use each deployment type once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
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NOTE: Each correct selection is worth one point.



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58) An organization validates generative AI applications during CI/CD Microsoft Foundry.
Evaluation must run automatically and block releases when quality thresholds are NOT met.
Manual evaluation is no longer acceptable.
Evaluation must use both predefined quality metrics and custom safety checks.
You need to implement an automated evaluation workflow that supports both built-in and custom metrics.
What should you do?

A. Enable application tracing to collect runtime telemetry.
B. Review evaluation results manually after deployment.
C. Monitor latency metrics during model inference.
D. Implement an evaluation step by using GitHub Actions.

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What should you do?

A. Enable application tracing to collect runtime telemetry.
B. Review evaluation results manually after deployment.
C. Monitor latency metrics during model inference.
D. Implement an evaluation step by using GitHub Actions.

Correct Answer: D

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59) You work in Microsoft Foundry with a prompt flow.
You must manually evaluate prompts and compare results across prompt variants.
You need to capture the inputs, outputs, token usage, and latencies for each flow run for the evaluation.
Solution: Create prompt variants and compare their outputs in the Evaluation experience.
Does the solution meet the goal?

A. Yes
B. No

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59) You work in Microsoft Foundry with a prompt flow.
You must manually evaluate prompts and compare results across prompt variants.
You need to capture the inputs, outputs, token usage, and latencies for each flow run for the evaluation.
Solution: Create prompt variants and compare their outputs in the Evaluation experience.
Does the solution meet the goal?

A. Yes
B. No

Correct Answer: B

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60) Multiple teams need access to approved models with version tracking, lineage, and governance controls. Models must be discoverable and reusable across projects. What Azure ML feature should you use?

A. Blob storage containers
B. Model registry
C. Git repositories
D. Data lake

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A. Blob storage containers
B. Model registry
C. Git repositories
D. Data lake

Correct Answer: B

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61). A team is validating a generative AI assistant for a company. The assistant generates responses by using internal knowledge sources.
The company requires assurance that responses are accurate, supported by sources, and related to the user prompts before enabling production access.
You need to implement quality metrics that confirm the assistant produces reliable and meaningful responses.
Which two evaluation metrics should you use? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Groundedness
B. Relevance
C. Harmfulness
D. Fairness
E. Tone

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61). A team is validating a generative AI assistant for a company. The assistant generates responses by using internal knowledge sources.
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Which two evaluation metrics should you use? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Groundedness
B. Relevance
C. Harmfulness
D. Fairness
E. Tone

Correct Answer: A,B

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62). Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear on the review screen.
You manage an Azure Machine Learning workspace. The Python script named script.py reads an argument named training_data. The training_data argument specifies the path to the training data in a file named dataset1.csv.
You plan to run the script.py Python script as a command job that trains a machine learning model.
You need to provide the command to pass the path for the dataset as a parameter value when you submit the script as a training job.
Solution: python train.py --training_data training_data
Does the solution meet the goal?

A. No
B. Yes

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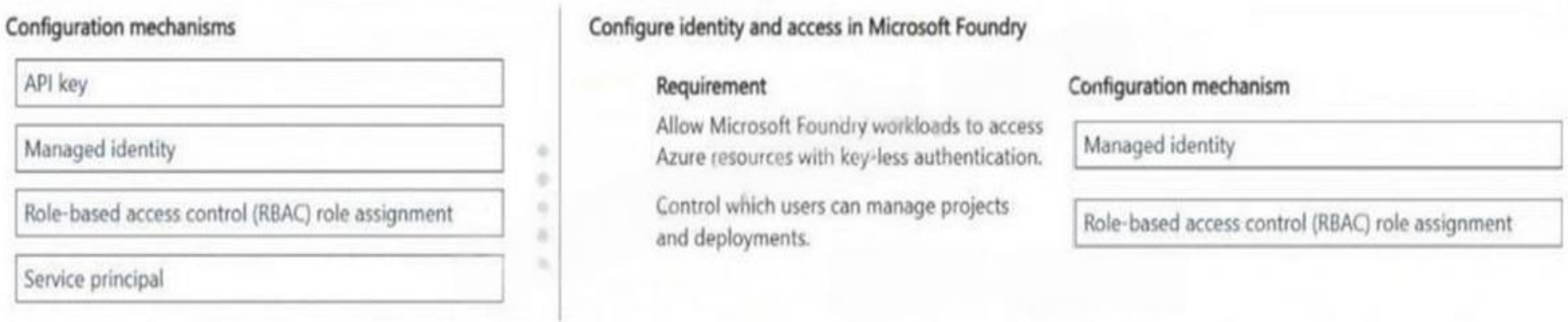
Correct Answer: A

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63). Drag and Drop Question
An organization uses Microsoft Foundry to develop generative AI projects that access shared Azure resources such as storage accounts and vector databases.
The organization s security policy requires eliminating secret key-based authentication and enforcing least-privilege access.
You must configure identity and access so that:
Services authenticate without stored credentials.
Permissions are scoped appropriately across projects and shared resources.
You need to configure the appropriate identity or access mechanism for each requirement.
What should you configure in Microsoft Foundry to meet each requirement? To answer, move the appropriate configuration mechanisms to the correct requirements. You may use each configuration mechanism once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Correct Answer:



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You need to provide the command to pass the path for the dataset as a parameter value when you submit the script as a training job.
Solution: python script.py --trainingdata ${{inputs.training_data}}
Does the solution meet the goal?

A. No
B. Yes

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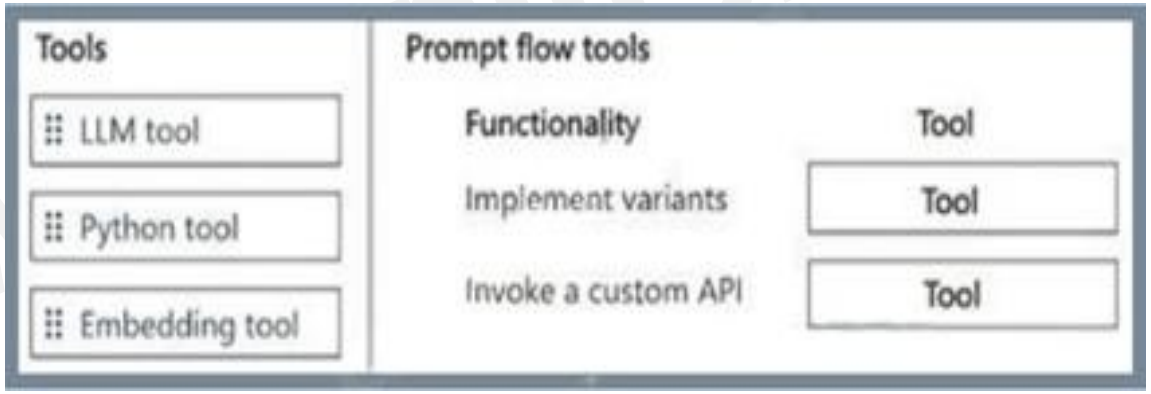
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Solution: python script.py --trainingdata ${{inputs.training_data}}
Does the solution meet the goal?

A. No
B. Yes

Correct Answer: B

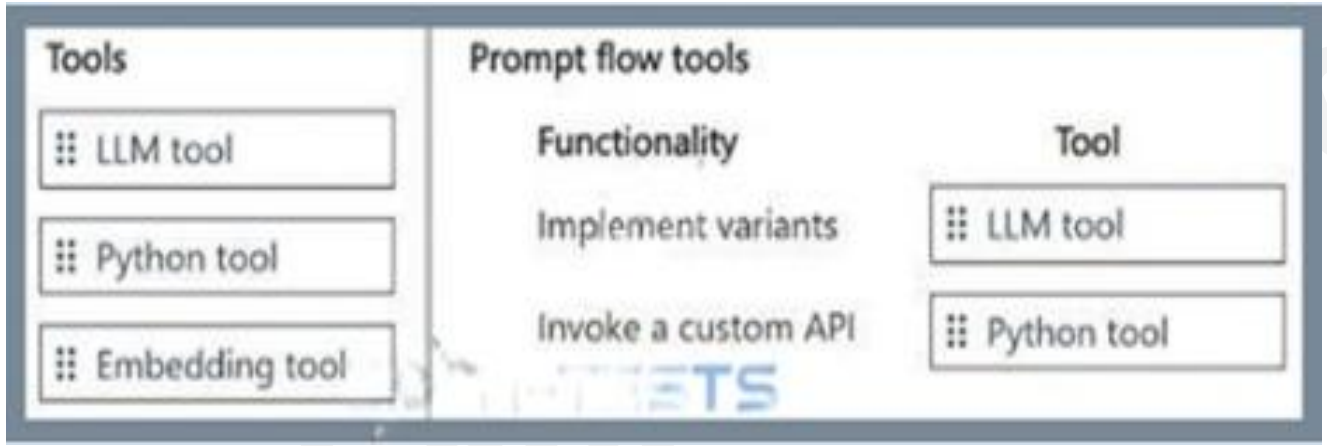
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65). Drag and Drop Question
You develop a Prompt flow in Microsoft Foundry project.
You plan to use variants and invoke a custom API in the flow.
You need to add tools to the flow that will implement the planned functionality. Your solution must minimize development efforts.
Which tools should you use? To answer, move the appropriate tools to the correct functionalities.
You may use each tool once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



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Correct Answer:



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Does the solution meet the goal?

A. No
B. Yes

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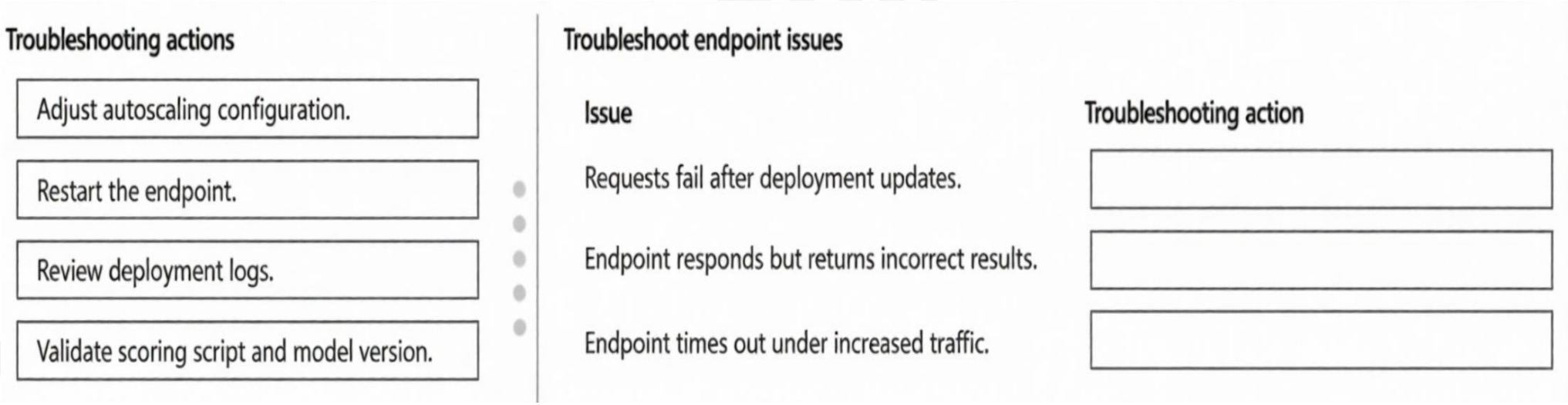
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Does the solution meet the goal?

A. No
B. Yes

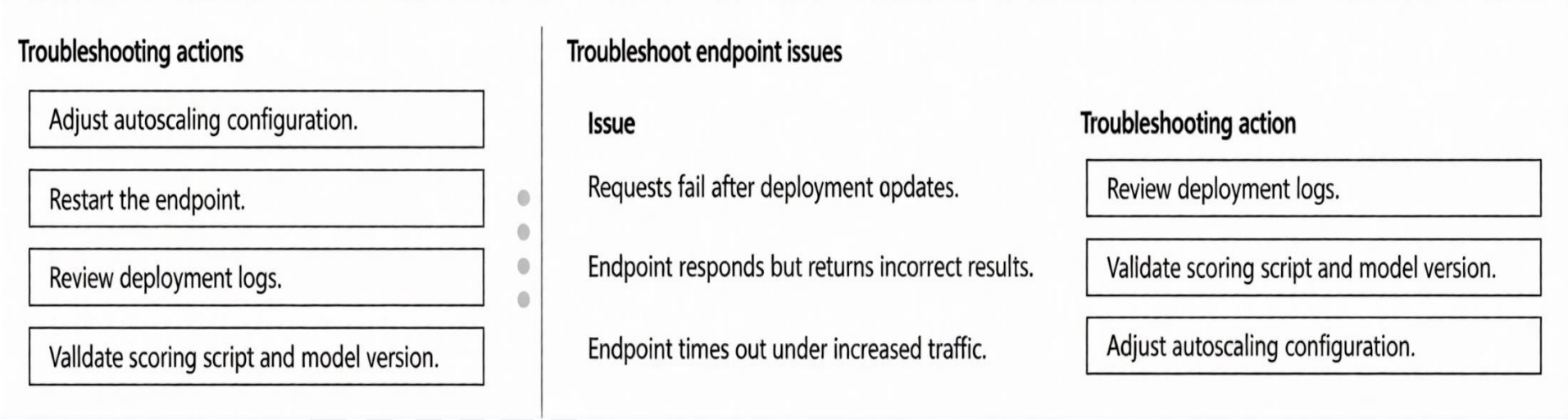
Correct Answer: A

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67). Drag and Drop Question
A real-time endpoint is deployed in Azure Machine Learning to serve predictions to a web application.
Users report intermittent failures and unexpected responses when calling the endpoint.
You need to identify the appropriate troubleshooting action for each reported issue.
Which troubleshooting action should you perform for each issue? To answer, move the appropriate troubleshooting actions to the correct issues. You may use each troubleshooting action once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Correct Answer:



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68). You manage an Azure Machine Learning workspace.
You need to define an environment from a Docker image by using the Azure Machine Learning Python SDK v2.
Which parameter should you use?

A. image
B. properties
C. conda_file
D. build

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Which parameter should you use?

A. image
B. properties
C. conda_file
D. build

Correct Answer: A

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69). A team plans to deploy a large foundation model in Microsoft Foundry as part of a new enterprise AI capability. Different business units across the team's organization will access the model from various internal applications.
You need to deploy a foundation model by minimizing latency.
Which deployment type should you use?

A. Data Zone Batch
B. Developer
C. Data Zone Standard
D. Global Batch

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Which deployment type should you use?

A. Data Zone Batch
B. Developer
C. Data Zone Standard
D. Global Batch

Correct Answer: C

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70). You create a workspace by using Azure Machine Learning Studio.
You must run a Python SDK v2 notebook in the workspace by using Azure Machine Learning Studio.
You need to reset the state of the notebook.
Which three actions should you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A. Change the compute.
B. stop the current kernel.
C. Reset the compute.
D. Navigate to another section of the workspace.
E. Change the current kernel.

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70). You create a workspace by using Azure Machine Learning Studio.
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NOTE: Each correct selection is worth one point.

A. Change the compute.
B. stop the current kernel.
C. Reset the compute.
D. Navigate to another section of the workspace.
E. Change the current kernel.

Correct Answer: B,C,E

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71). A team is building a Retrieval-Augmented Generation (RAG) system.
The team observes that the retrieved documents are often irrelevant or incomplete.
You need to improve retrieval accuracy.
What should you adjust?

A.Temperature parameter
B.Chunk size and overlap
C.Embedding strategy
D.Token limits

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D. Token limits

Correct Answer: B

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72). A data science team plans to evaluate multiple hyperparameter values automatically while training a model in Azure Machine Learning.
The tuning process must run multiple training trials without manually modifying the training script for each run.
You need to automate hyperparameter tuning for the training job.
What should you do?

A. Run a single training job with fixed hyperparameters.

B. Select hyperparameters based only on default model settings.

C. Adjust hyperparameters after model deployment.

D. Create a tuning job that runs multiple trials with different parameter values.

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Correct Answer: D

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73) A team is working in Microsoft Foundry to test and compare large language model (LLM) prompt variants in a development environment.
The team requires consistent inputs to evaluate prompt variants without relying on live user traffic.
You need to create a controlled evaluation of input data.
Which action should you perform first?

A. Generate synthetic interaction data.

B. Configure content filters.

C. Apply a blocklist.

D. Enable observability metrics.

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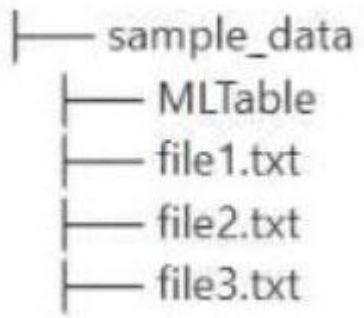
D. Enable observability metrics.

Answer(s): A

Explanation:
In Microsoft Foundry, you can test and compare LLM prompt variants using Prompt Flow. This environment allows you to create a controlled development workflow with consistent inputs, bypassing the need for live user traffic.
1. Generate Synthetic Interaction Data Generating synthetic data is the recommended first step to build a robust, diverse test dataset without manual effort.
2. Compare Prompt Variants Once your synthetic dataset is ready, use Prompt Flow to iterate on your prompts.
3. Controlled Evaluation To measure performance objectively, run automated Eval Runs.

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74) HOTSPOT (Drag and Drop is not supported)
You manage an Azure Machine Learning workspace named workspace1 by using the Python SDK v2.
The default datastore of workspace1 contains a folder named sample_data. The folder structure contains the following content:



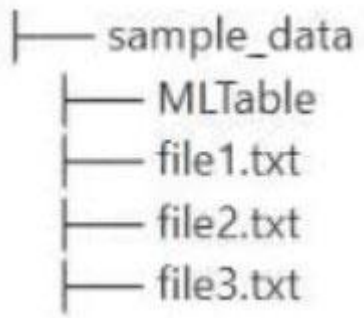
You write Python SDK v2 code to materialize the data from the files in the sample_data folder into a Pandas data frame.
You need to complete the Python SDK v2 code to use the MLTable folder as the materialization blueprint.
How should you complete the code? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point
Hot Area:

Answer Area

```
import mltable
tbl = mltable.[ load | save | take ]("[ ./sample_data | ./sample_data/MLTable | ./sample_data/file*.txt ]")
df = tbl.to_pandas_dataframe()
```

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The default datastore of workspace1 contains a folder named sample_data. The folder structure contains the following content:



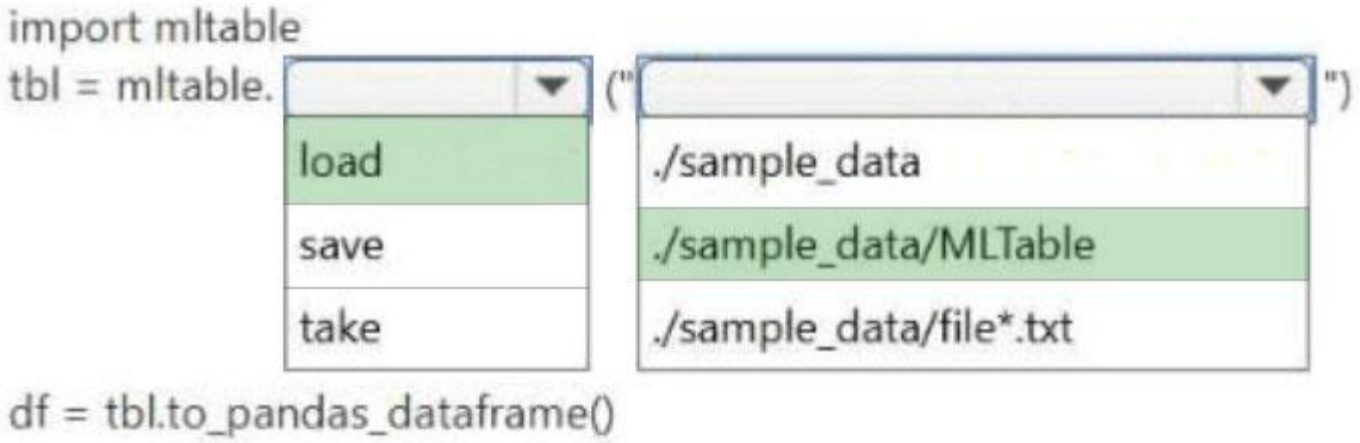
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You need to complete the Python SDK v2 code to use the MLTable folder as the materialization blueprint.
How should you complete the code? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point
Hot Area:

Answer Area

```
import mltable
tbl = mltable.[ load | save | take ]("[ ./sample_data | ./sample_data/MLTable | ./sample_data/file*.txt ]")
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```

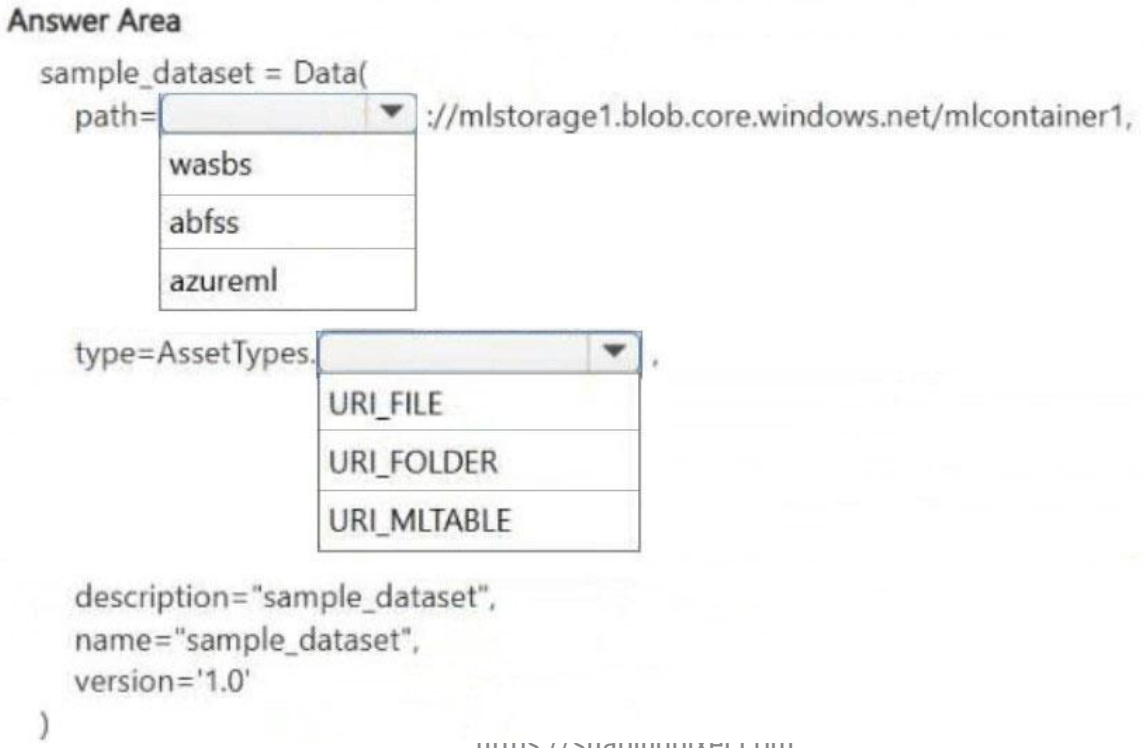
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Answer Area



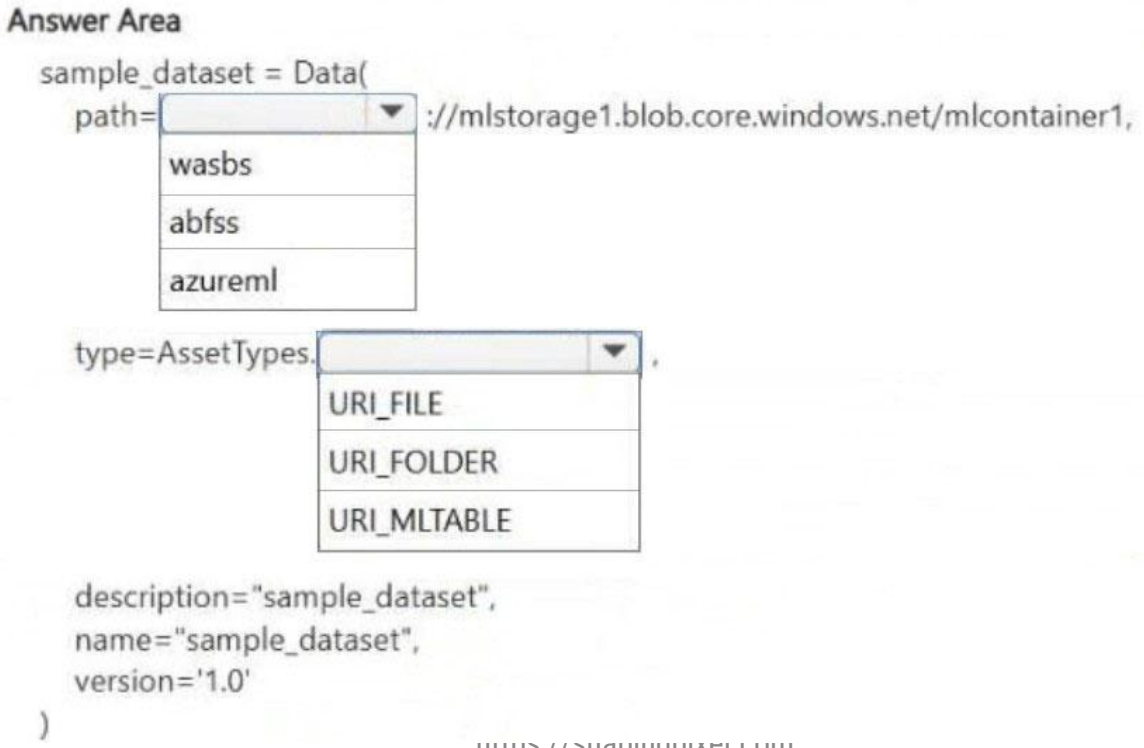
Box 1: load To load your data into a Pandas DataFrame, use the following structure: import mltable # Line 2: Point specifically to the folder containing the 'MLTable' file tbl = mltable.load("./sample_data/MLTable") # Line 3: Materialize the data df = tbl.to_pandas_dataframe()
Box 2: ./sample_data/MLTable
To materialize data using mltable, the path must point to the directory containing the MLTable file (a YAML
metadata file) or directly to the file itself.

75) HOTSPOT (Drag and Drop is not supported)
You manage an Azure Machine Learning workspace named workspace1 by using the Python SDK v2. You create a General Purpose v2 Azure storage account named mlstorage1. The storage account includes a publicly accessible container named mlcontainer1. The container stores 10 blobs with files in the CSV format.
You must develop Python SDK v2 code to create a data asset referencing all blobs in the container named mlcontainer1.
You need to complete the Python SDK v2 code.
How should you complete the code? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



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75) HOTSPOT (Drag and Drop is not supported)
You manage an Azure Machine Learning workspace named workspace1 by using the Python SDK v2. You create a General Purpose v2 Azure storage account named mlstorage1. The storage account includes a publicly accessible container named mlcontainer1. The container stores 10 blobs with files in the CSV format.
You must develop Python SDK v2 code to create a data asset referencing all blobs in the container named mlcontainer1.
You need to complete the Python SDK v2 code.
How should you complete the code? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



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Answer Area

```
sample_dataset = Data(
    path=[wasbs / abfss / azureml] ://mlstorage1.blob.core.windows.net/mlcontainer1,
    type=AssetTypes.[URI_FILE / URI_FOLDER / URI_MLTABLE],
    description="sample_dataset",
    name="sample_dataset",
    version='1.0'
)
```

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Box 1: abfss Supported paths When you provide a data input/output to a Job, you must specify a path parameter that points to the data location. This table shows both the different data locations that Azure Machine Learning supports, and examples for the path parameter:
* A path on Azure Storage https://<account_name>.blob.core.windows.net/<container_name>/<path> abfss://<file_system>@<account_name>.dfs.core.windows.net/<path>
* A path to a Data Asset azureml:<my_data>:<version>
* Etc.
Note: # Supported paths include: # local: './<path>' # blob: 'https://<account_name>.blob.core.windows.net/<container_name>/<path>' # ADLS gen2: 'abfss://<file_system>@<account_name>.dfs.core.windows.net/<path>/' # Datastore: 'azureml://datastores/<data_store_name>/paths/<path>'
Box 2: uri_folder You can create three data asset types:
* Folder, Reference a single folder V2 API: uri_folder V1 API: FileDataset
You must read/write a folder of parquet/CSV files into Pandas/Spark.
Deep-learning with images, text, audio, video files located in a folder.
Incorrect: Not mltable: * Table, Reference a data table V2 API: mltable V1 API: TabularDataset
You have a complex schema subject to frequent changes, or you need a subset of large tabular data.
AutoML with Tables.
Not uri_file: * File, Reference a single file V2 API: uri_file V1 API: FileDataset
Read/write a single file - the file can have any format.

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76) HOTSPOT (Drag and Drop is not supported)
You review the following Azure CLI command and the relevant Bicep excerpt.

```
01 az deployment group create \
02   --resource-group rg-foundry-dev \
03   --template-file ./foundry-environment.bicep \
04   --parameters location=eastus projectName=proj-dev-01
05
06 // foundry-environment.bicep (excerpt)
07 targetScope = 'resourceGroup'
08
09 param location string
10 param projectName string
11
12 resource aiFoundry 'Microsoft.CognitiveServices/accounts@2025-04-01-preview' = {
13  name: aiFoundryName
14  location: location
15  identity: {
16    type: 'SystemAssigned'
17  }
18  sku: {
19    name: 'S0'
20  }
21  kind: 'AIServices'

22  properties: {
24    allowProjectManagement: true
27    customSubDomainName: aiFoundryName
28    disableLocalAuth: true
29  }
30}
```

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(Non-relevant sections are omitted.)
You need to validate what the snippet will do before it is merged. For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Note: Each correct selection is worth one point.
Hot Area:

Microsoft Foundry deployments

Statement	Yes	No
The command deploys the resources into an existing resource group named rg-foundry-dev.	○	○
The system-assigned managed identity defined in the template will automatically be inherited by all Microsoft Foundry projects.	○	○
To deploy this template to a different subscription, you must modify the Bicep file to include a subscriptionId parameter.	○	○

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You review the following Azure CLI command and the relevant Bicep excerpt.

```
01 az deployment group create \
02   --resource-group rg-foundry-dev \
03   --template-file ./foundry-environment.bicep \
04   --parameters location=eastus projectName=proj-dev-01
05
06 // foundry-environment.bicep (excerpt)
07 targetScope = 'resourceGroup'
08
09 param location string
10 param projectName string
11
12 resource aiFoundry 'Microsoft.CognitiveServices/accounts@2025-04-01-preview' = {
13  name: aiFoundryName
14  location: location
15  identity: {
16    type: 'SystemAssigned'
17  }
18  sku: {
19    name: 'S0'
20  }
21  kind: 'AIServices'

22  properties: {
24    allowProjectManagement: true
27    customSubDomainName: aiFoundryName
28    disableLocalAuth: true
29  }
30}
```

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Answer(s):

Microsoft Foundry deployments

Statement	Yes	No
The command deploys the resources into an existing resource group named rg-foundry-dev.	◉	○
The system-assigned managed identity defined in the template will automatically be inherited by all Microsoft Foundry projects.	○	◉
To deploy this template to a different subscription, you must modify the Bicep file to include a subscriptionId parameter.	○	◉

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Box 1: Yes Yes- The command deploys the resources into an existing resource group named rg-foundry-dev.
Box 2: No No- The system-assigned managed identity defined in the template will automatically be inherited by all Microsoft Foundry projects.
In Azure, system-assigned managed identities are strictly bound to the individual resource on which they are enabled. They are not inherited by downstream or associated applications (like Microsoft Foundry projects) in the deployment hierarchy.
Box 3: No No - To deploy the template to a different subscription, you must modify the Bicep file to include a subscriptionID parameter.
A Bicep file does not strictly need to include a subscriptionId parameter to deploy to a different subscription. How you handle the cross-subscription deployment depends entirely on whether you are shifting the entire deployment context using external tooling, or using a multi-scope deployment directly within your code.
Option 1: Handle via External Tooling (No Code Changes)
If your entire Bicep file is designed to deploy to a single subscription, you do not need to alter your Bicep code. Instead, you change the subscription context externally before running the deployment command. The deployment tool will automatically route everything to the active subscription context.
Option 2: Handle via Bicep Modules (Multi-Scope Deployment) If you need a single Bicep deployment to create resources across multiple subscriptions simultaneously, you must use Bicep modules. In this scenario, you do use a subscriptionId parameter or string inside the Bicep template to explicitly route the module to the target scope. You use the subscription() scope function to pass the targeted subscription ID directly to the module definition.

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77) DRAG DROP (Drag and Drop is not supported)
A team performs interactive experimentation during development. The team also runs scalable jobs for model training.
The team must minimize costs while ensuring compute resources scale when needed. Different workloads require different compute behaviors within the same workspace.
You need to configure compute targets that support each workload.
Which compute targets should you use? To answer, move the appropriate compute targets to the correct workload types.
You may use each compute target once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



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Compute targets

- Azure Batch
- Azure Databricks cluster
- Azure Kubernetes Service
- Azure Virtual Machine
- Azure Machine Learning compute cluster
- Azure Machine Learning compute instance

Workload compute targets

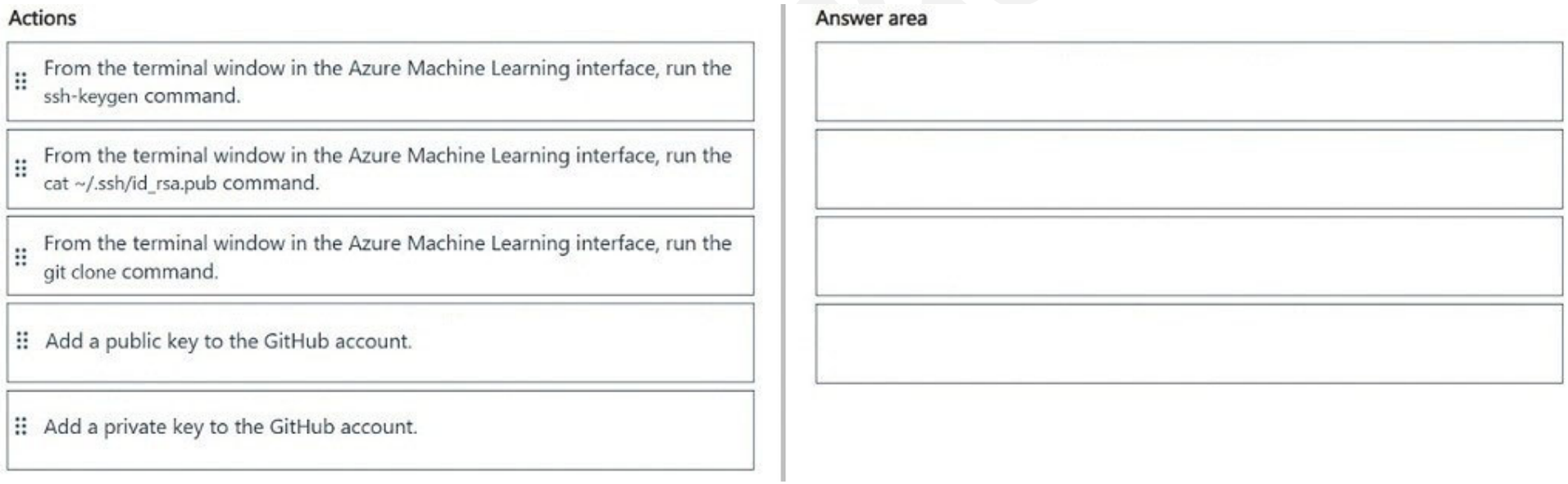
Workload type	Compute target
Interactive experimentation	Azure Machine Learning compute instance
Scalable training jobs	Azure Databricks cluster

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Box 1: Azure Machine Learning Compute Instance Interactive experimentation
The best compute target for interactive experimentation while maintaining low costs and allowing the flexibility to scale is an Azure Machine Learning Compute Instance.
Optimized for Interactive Development: It acts as a managed, cloud-based data science workstation integrated directly with Jupyter Notebooks, JupyterLab, and VS Code.
Cost Controls: It includes built-in cost-saving features like idle shutdown and automated stop/start scheduling. This ensures you only pay for compute hours when the environment is actively being used.
Box 2: Azure Databricks cluster Scalable training jobs
The best compute target for scalable model training jobs in this scenario is an Azure Databricks cluster.
Workload Optimization: Azure Databricks is explicitly optimized for data preparation and large-scale, distributed machine learning model training workloads (such as Apache Spark MLlib or distributed deep learning).
Workload Diversity: Attaching an Azure Databricks cluster as a compute target allows you to handle specific heavy-duty engineering and training workloads while seamlessly transitioning back to your Azure Machine Learning Compute Instance for core interactive development.
Incorrect: Azure Kubernetes Service In Azure Machine Learning architectures, Azure Kubernetes Service (AKS) is primarily designated and optimized as an inference compute target for hosting, scaling, and deploying trained models as real-time production endpoints. It is not the standard choice for general model training pipelines.

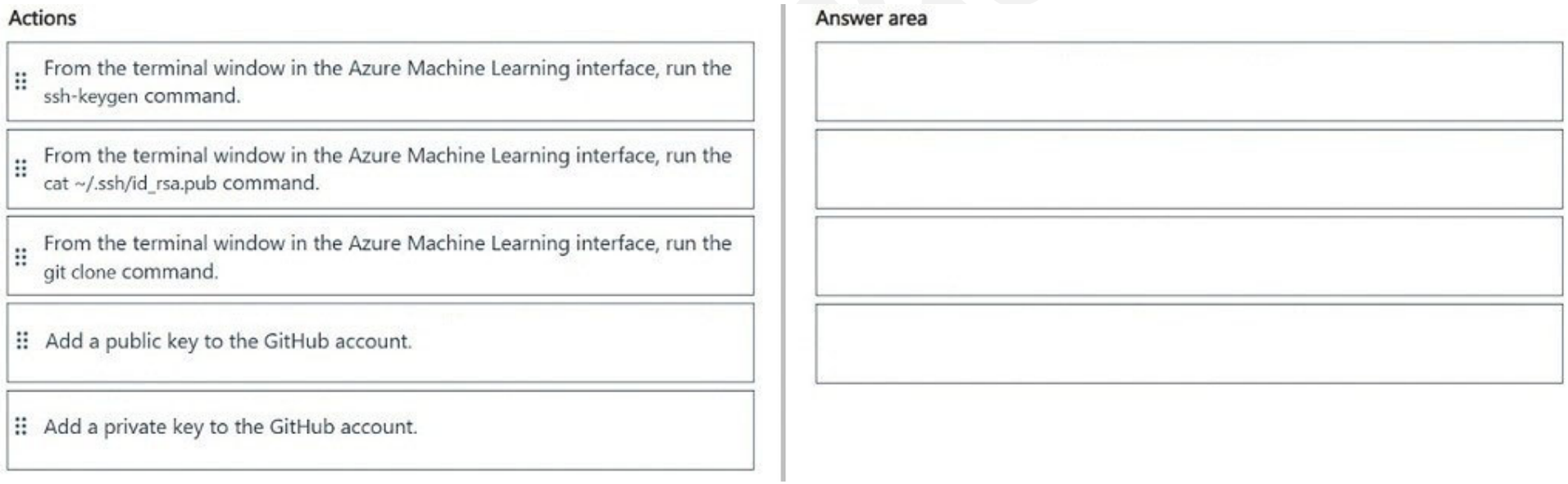
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78) DRAG DROP (Drag and Drop is not supported)
You have an existing GitHub repository containing Azure Machine Learning project files.
You need to clone the repository to your Azure Machine Learning shared workspace file system.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Note: More than one order of answer choices is correct. You will receive credit for any of the correct orders you select.
Select and Place:

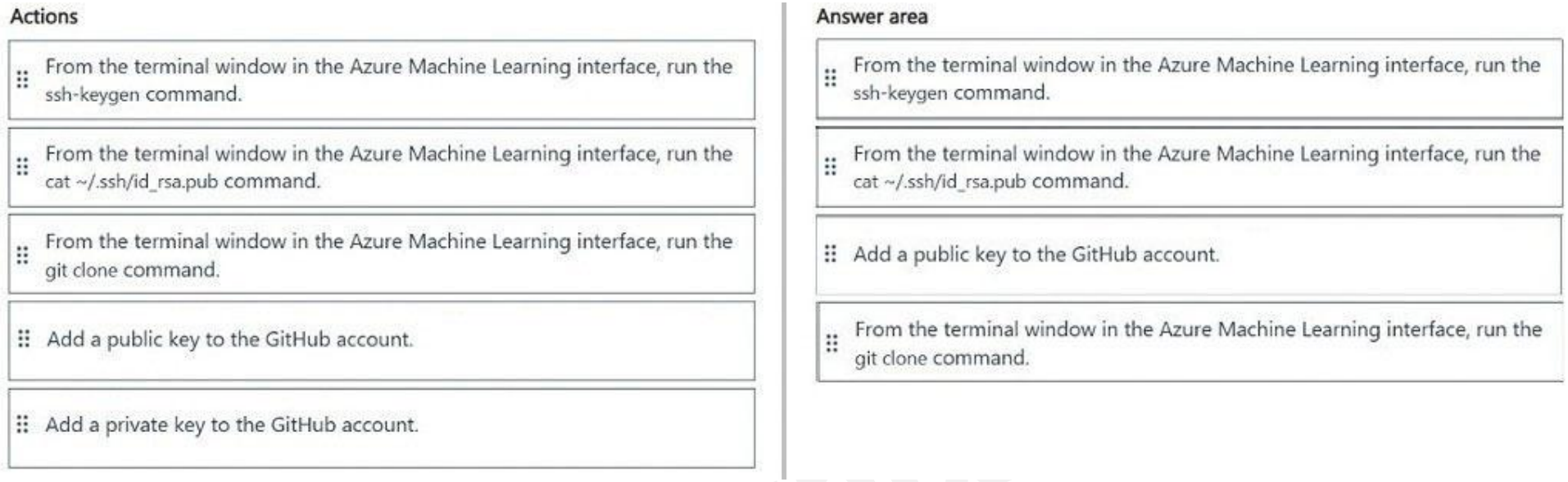


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Select and Place:



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Step 1: From the terminal window in the Azure Machine Learning interface, run the the ssh-keygen command.
1. Generate an SSH key pair: ssh-keygen -t ed25519 -C "your_email@example.com"
Step 2: From the terminal window in the Azure Machine Learning interface, run the cat ~/.ssh/id_rsa.pub command. Retrieve your public key: 2.. Display your public key so you can copy it to your GitHub account settings: cat ~/.ssh/id_ed25519.pub
Step 3: Add a public key to the GitHub Account. Key Type to Add to GitHub 3. You must add the SSH public key (copied from the id_ed25519.pub file in step 2 above) to your GitHub account under Settings > SSH and GPG keys.
Step 4: From the terminal window in the Azure Machine Learning interface, run the git clone command.
Clone the repository: After adding the key to GitHub, navigate to your workspace folder and clone the repository using its SSH URL:
4. Clone the repository using the SSH URL: git clone git@github.com:username/repository-name.git

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79) You manage an Azure Machine Learning workspace. You have an environment for training jobs which uses an existing Docker image.
A new version of the Docker image is available.
You need to use the latest version of the Docker image for the environment configuration by using the Azure Machine Learning SDK v2.
What should you do?

A. Change the description parameter of the environment configuration.
B. Modify the conda_file to specify the new version of the Docker image.
C. Use the create_or_update method to change the tag of the image.
D. Use the Environment class to create a new version of the environment.

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What should you do?

A. Change the description parameter of the environment configuration.
B. Modify the conda_file to specify the new version of the Docker image.
C. Use the create_or_update method to change the tag of the image.
D. Use the Environment class to create a new version of the environment.

Answer(s): C

Explanation:
To use a new version of the Docker image for an environment using the Azure Machine Learning SDK v2, you must instantiate an Environment class object with the new image parameter and then use the ml_client.environments.create_or_update() method.
Required Steps 1. Define the updated Environment: Use the Environment entity from the azure.ai.ml.entities package. Set the image parameter to the URI of the new Docker image version.
2. Register or update the asset: Pass the environment instance into ml_client.environments.create_or_update () to create a new version of that environment asset within your workspace.

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80) A company has multiple data science teams working on separate machine learning projects.
The company requires models to be auditable, reusable, and governed centrally across teams. The models must allow team-level isolation for billing.
You need to establish the foundation for governed machine learning operations.
Which action should you perform first?

A. Register shared datasets in a central storage account.
B. Create a shared hub workspace and project workspaces for each team.
C. Create a resource group for shared machine learning assets.
D. Create a shared Azure Machine Learning workspace.

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C. Create a resource group for shared machine learning assets.
D. Create a shared Azure Machine Learning workspace.

Answer(s): B

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Explanation:
The ideal first step is to implement a Hub-and-Spoke Workspace Architecture combined with Azure Machine Learning Registries. This hybrid blueprint natively decouples your organizational boundaries (billing and access) from your machine learning assets (models, environments, and pipelines).
In enterprise-scale machine learning, an Azure Machine Learning Hub Workspace acts as a central governance plane. It lets IT administrators configure security, shared compute resources, networks, and compliance guidelines centrally once. From this central hub, different teams can spin up isolated project workspaces. This provides strict data and team-level separation, which naturally supports precise cost tracking and billing isolation per team.

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81) A team provisions an Azure Machine Learning environment by triggering pull requests.
Deployments must be automated, auditable, and require approval before running.
You need to select a deployment automation tool.
Which tool should you use?

A. Azure Monitor

B. GitHub Actions

C. MLflow

D. Azure Machine Learning pipelines

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81) A team provisions an Azure Machine Learning environment by triggering pull requests.
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You need to select a deployment automation tool.
Which tool should you use?

A. Azure Monitor
B. GitHub Actions
C. MLflow
D. Azure Machine Learning pipelines

Answer(s): B

Explanation:
GitHub Actions is the best tool for this scenario.
Pull Request Integration: GitHub Actions is a native continuous integration and continuous delivery (CI/CD) platform built directly into GitHub. It can natively listen for repository events—such as triggering a workflow automatically when a pull request is opened, updated, or merged.
Required Approvals: It provides built-in governance through Environments. You can configure specific deployment environments (e.g., Production) to enforce required reviewers, ensuring a deployment cannot run until an authorized user manually approves it.
Auditable Logs: Every workflow execution, event trigger, and manual approval is tracked chronologically within GitHub. This provides a highly transparent, centralized audit trail for compliance requirements.

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82) A team develops and manages a conversational assistant by using Microsoft Foundry.
The team requires generative AI to automatically evaluate every pull request of an agentic application and fail the build if safety thresholds are exceeded.
You need to automate evaluations as part of CI.
What should you configure?

A. Blocklist applied to the model endpoint
B. Content filter configured in warning mode
C. Retrieval chunking strategy
D. GitHub Actions workflow that executes the runs

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You need to automate evaluations as part of CI.
What should you configure?

A. Blocklist applied to the model endpoint
B. Content filter configured in warning mode
C. Retrieval chunking strategy
D. GitHub Actions workflow that executes the runs

Answer(s): D

Explanation:
A GitHub Actions workflow can absolutely be configured to automatically evaluate pull requests using Azure AI Foundry and fail the CI build if safety thresholds are exceeded.By integrating the Azure AI Evaluation SDK into your automated testing suite, you can scan agentic applications for safety risks (like jailbreaks, hate speech, or content harms) during the continuous integration (CI) lifecycle.

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83) A team runs training and inference jobs in Azure Machine Learning.
The team experiences inconsistent runtime dependencies that cause variation in results.
You need to ensure that all jobs use the same execution dependencies.
Which asset should you define?

A. Azure Machine Learning job configuration
B. Azure Machine Learning job input/output bindings
C. Azure Monitor alert
D. Azure Machine Learning environment

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You need to ensure that all jobs use the same execution dependencies.
Which asset should you define?

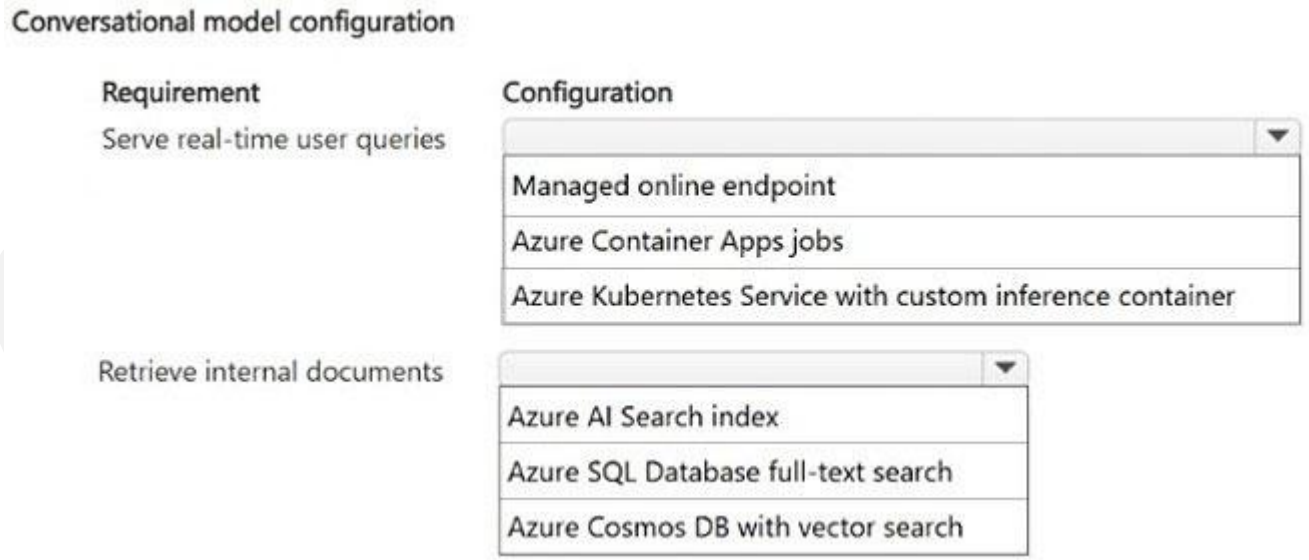
A. Azure Machine Learning job configuration
B. Azure Machine Learning job input/output bindings
C. Azure Monitor alert
D. Azure Machine Learning environment

Answer(s): D

Explanation:
To guarantee consistent runtime dependencies in Azure Machine Learning, you must use Azure ML Environments configured with custom Docker images or pinned Conda dependencies.
An Environment asset encapsulates the exact Python packages, environment variables, and software settings for your training and inference workloads.

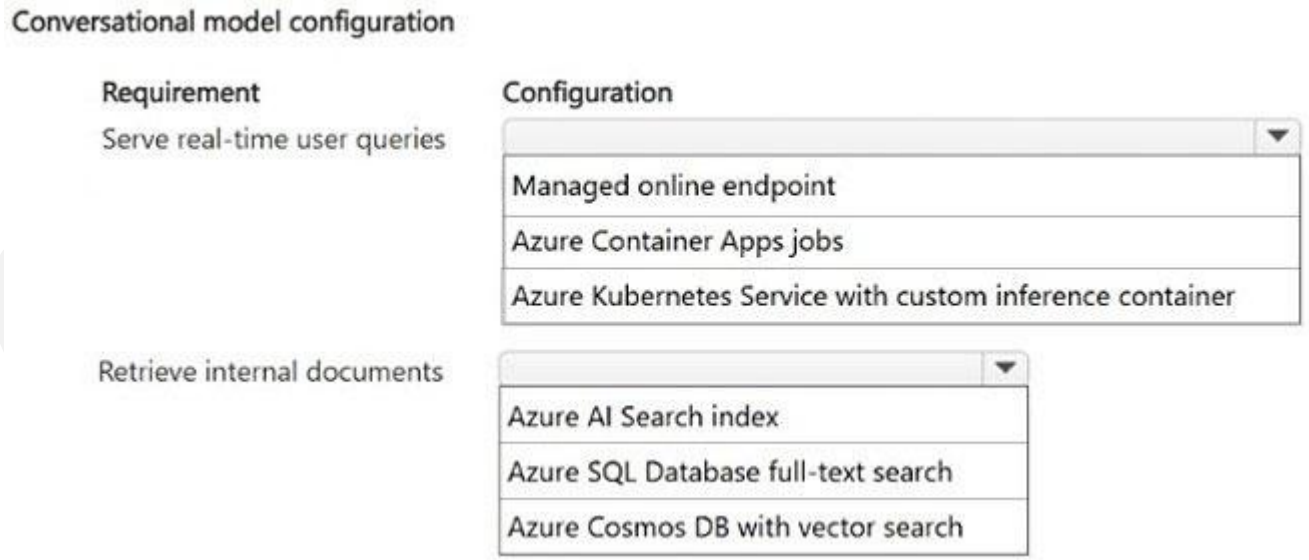
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83) HOTSPOT (Drag and Drop is not supported)
You need to deploy the RAG-based chat application that meets Fabrikam Inc.'s business and technical requirements.
Which configuration should you use for each requirement? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



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83) HOTSPOT (Drag and Drop is not supported)
You need to deploy the RAG-based chat application that meets Fabrikam Inc.'s business and technical requirements.
Which configuration should you use for each requirement? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



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Conversational model configuration

Requirement	Configuration
Serve real-time user queries	Managed online endpoint Azure Container Apps jobs Azure Kubernetes Service with custom inference container
Retrieve internal documents	Azure AI Search index Azure SQL Database full-text search Azure Cosmos DB with vector search

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Scenario: Business Requirements: Provide a conversational interface that answers analytics questions by using internal documents and datasets.
Box 1: Managed Online Endpoint Serve real-time user queries
To serve real-time user queries in your Azure Machine Learning solution, a Managed Online Endpoint is the best deployment type.
Turnkey Infrastructure: Azure manages OS updates, patching, and scaling out-of-the-box.
Low Latency: Built specifically for synchronous, high-throughput, real-time inference.
Native ML Features: Direct integration with Azure ML data drift, monitoring, and security.
Blue/Green Deployment: Safely rollout updates by splitting traffic between models.
Incorrect: Azure Container Apps (ACA) Jobs: Built for batch processing, not real-time APIs. They spin up, run a task, and shut down. This introduces massive cold-start latency.
Azure Kubernetes Service (AKS) with Custom Containers: Offers maximum control but brings high operational overhead. Choose this only if you require complex custom network topologies or strict multi-cloud portability.
Box 2: Azure AI Search index Retrieve internal documents
Azure AI Search is the best choice for retrieving internal documents to power an analytics-focused conversational interface (RAG).While Azure Cosmos DB and Azure SQL Database have specialized retrieval capabilities, Azure AI Search functions as a dedicated enterprise search engine. It outperforms the other options by combining keyword, vector, and semantic ranking out of the box.

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84) You need to recommend a solution to address Fabrikam Inc.’s limited rollback capability.
Which deployment approach should you recommend?

A. VM-hosted REST APIs
B. Azure Kubernetes Service with blue-green switching
C. Managed online endpoints with traffic splitting
D. Batch endpoints

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Which deployment approach should you recommend?

A. VM-hosted REST APIs
B. Azure Kubernetes Service with blue-green switching
C. Managed online endpoints with traffic splitting
D. Batch endpoints

Answer(s): C

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Explanation:
Scenario: Current Environment: Deployment is performed manually by data scientists, with limited rollback capability.
The correct deployment type to address this problem is Managed online endpoints with traffic splitting.
Native traffic control: Managed online endpoints naturally support native blue-green deployments under a single HTTP endpoint.
Granular traffic splitting: You can deploy a new model version (e.g., green) with 0% live traffic, safely test it, and then incrementally shift traffic from the old version (e.g., blue).Instant rollbacks: If the new model shows bugs or degrades performance, you can immediately change the traffic percentage configuration back to 100% for the old deployment. This eliminates the risks of limited rollback capabilities.
No infrastructure overhead: Unlike setting up manual routing, Azure handles the underlying infrastructure and routing mechanisms in a turnkey fashion.
Incorrect: [Not A] VM-hosted REST APIs: This forces you to build, maintain, and configure your own custom load balancers and deployment scripts to handle routing and rollbacks, which increases operational risk and complexity.
[Not B] Azure Kubernetes Service with blue-green switching: While it supports blue-green deployments, it requires you to manage complex Kubernetes infrastructure, service meshes, or ingress controllers manually to handle the traffic switching.
[Not D] Batch endpoints: These are designed for long-running, asynchronous processing of large data batches rather than real-time requests where instant live-traffic rollback is required.

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85) You need to make model training repeatable and auditable to address Fabrikam Inc.'s current environment challenges and technical requirements.
What should you use?

A. Workflow automation by using Azure Logic Apps
B. Training pipelines in Azure Machine Learning
C. Serverless execution by using Azure Functions
D. Scheduled notebook runs by using Azure Machine Learning jobs

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What should you use?

A. Workflow automation by using Azure Logic Apps
B. Training pipelines in Azure Machine Learning
C. Serverless execution by using Azure Functions
D. Scheduled notebook runs by using Azure Machine Learning jobs

Answer(s): B

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Explanation:
Azure Machine Learning (AML) training would be best.
While the other options can execute code, they lack the native machine learning operations (MLOps) capabilities required for enterprise auditing and repeatability.
Azure ML pipelines are purpose-built for machine learning workflows. They natively solve your tracking and auditing requirements.
Built-in Lineage: Automatically tracks data inputs, code versions, environments, and output models.
Native Component Reuse: Steps are modular, containerized, and easily shared across teams.
Data Caching: Skips completed steps if inputs have not changed, saving time and compute costs.
Scenario, Technical Requirements: Implement experiment tracking and model versioning for all training jobs. Orchestrate training and evaluation by using pipelines rather than manually running notebooks.
Problem Statement Fabrikam Inc. must design and implement an Azure-based AI operations solution that enables reliable training, evaluation, deployment, and iteration of generative AI models.

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86) A team manages an Azure Machine Learning workspace and deploys a model to an endpoint.
A deployed online endpoint shows inconsistent response times during periods of high traffic.
You need to identify potential performance degradation.
Which three metrics should you monitor? Each correct answer presents part of the solution. Choose three.
Note: Each correct selection is worth one point.

A. Feature count
B. Requests per minute
C. Connections active
D. Dataset size
E. Request latency

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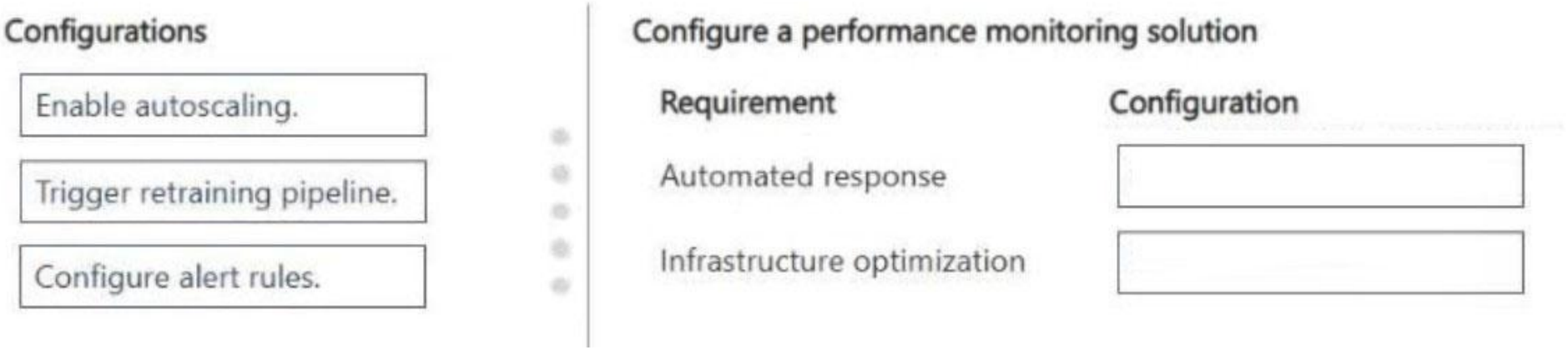
Answer(s): B,C,E

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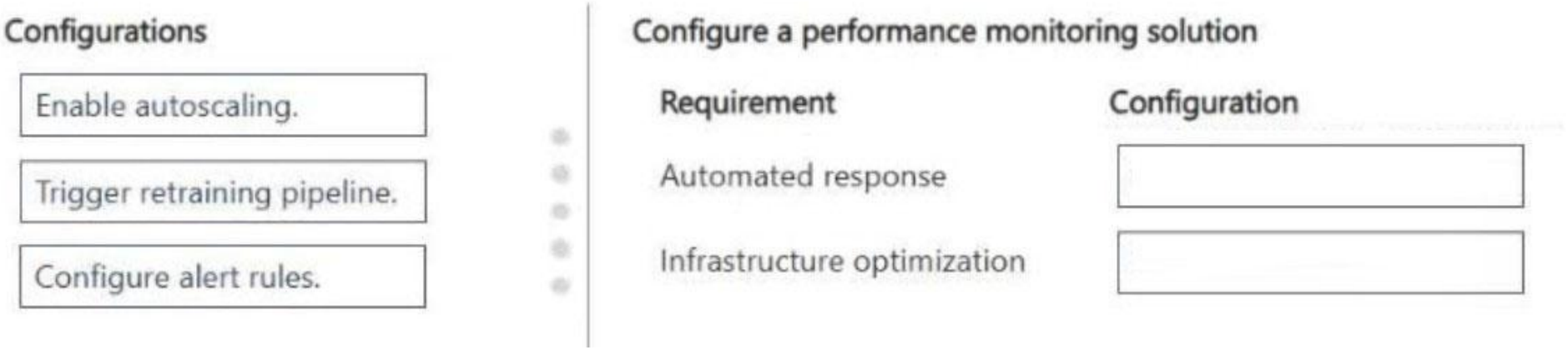
Explanation:
To locate potential performance degradation in an Azure Machine Learning online endpoint during high traffic, you should monitor these three metrics: Requests per minute: This metric tracks the volume of incoming traffic and helps identify if spikes in load correlate with slower response times.
Connections active: This monitors the total number of concurrent TCP connections from clients, which can indicate if the endpoint is reaching its capacity limits during peak periods.
Request latency: This directly measures the time taken to respond to requests, allowing you to observe exactly when and by how much performance is degrading.

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87) DRAG DROP (Drag and Drop is not supported)
A team deploys a machine learning model to a managed online endpoint. The team monitors model performance and data quality metrics in production.
When monitoring thresholds are exceeded, the team requires an automated operational response that notifies downstream systems.
You need to configure the monitoring solution to meet the requirements.
Which configuration should you associate with each requirement as a first step? To answer, move the appropriate configurations to the correct requirements. You may use each configuration once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



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Note: Each correct selection is worth one point.
Select and Place:



Configurations

- Enable autoscaling.
- Trigger retraining pipeline.
- Configure alert rules.

Configure a performance monitoring solution

Requirement	Configuration
Automated response	Configure alert rules.
Infrastructure optimization	Trigger retraining pipeline.

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Box 1: Configure alert rules Automated response
To set up an automated operational response for a deployed machine learning model, you should integrate Azure Machine Learning monitoring with Azure Monitor and Action Groups. This ensures that when performance or data drift thresholds are crossed, downstream systems are notified instantly via webhooks, emails, or automated workflows.
Core Components for Automated Response Monitoring Signal: Tracks Data Drift, Prediction Drift, or Model Performance (Precision/Recall).
*-> Alert Rule: A condition defined in Azure Monitor that triggers when a metric exceeds a threshold.
Action Group: A collection of notification preferences and automated receivers (Webhooks, Azure Functions, Logic Apps).
Box 2: Trigger retraining pipeline Infrastructure optimization
Triggering the Retraining Pipeline Once the threshold is exceeded, the system must optimize the infrastructure or model.
REST API Call: Use the Logic App to call the Azure ML Job/Pipeline endpoint.
SDK Trigger: Use an Azure Function with the azure-ai-ml library to submit a new command or pipeline job.
Data Prep: Ensure the trigger passes the latest data location to the retraining job to address the detected drift.

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88) A data science team completes multiple training runs within an experiment by using MLflow.
The team wants to store a selected model in Azure Machine Learning so that it can be versioned and deployed later.
The model must be versioned centrally for reuse across environments.
You need to version the trained model.
Which two actions should you perform? Each correct answer presents part of the solution. Choose two.
Note: Each correct selection is worth one point.

A. Locate and capture the model artifacts from the outputs of the training run.
B. Register the model in the Azure Machine Learning workspace.
C. Tag the training experiment with a name.
D. Export the model files to local storage.

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B. Register the model in the Azure Machine Learning workspace.
C. Tag the training experiment with a name.
D. Export the model files to local storage.

Answer(s): A,B

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Explanation:

To set up versioning for a trained model in an Azure Machine Learning (Azure ML) workspace using MLflow, you must capture the model artifacts during the training run and then register the model into the centralized registry.

[A] 1. Capture Model Artifacts During each training run, use the MLflow SDK to log the model. This ensures that all necessary files (the model binary, environment dependencies, and the MLmodel metadata) are stored as run outputs in the workspace.

Manual Logging: Use a flavor-specific method like mlflow.sklearn.log_model(model, "model_path") within an active run.

Automatic Logging: Call mlflow.autolog() before starting your training. This automatically captures metrics, parameters, and the model artifacts for supported frameworks.

Artifact Location: Once logged, artifacts are typically found in the outputs/ folder of the specific run, accessible via the Azure Machine Learning Studio.

[B] 2. Register the Model After identifying the best-performing run, you register it to the Model Registry. This creates a named, versioned entity that can be accessed across different environments for deployment.

To set up versioning for an MLflow model in Azure Machine Learning (Azure ML) that is accessible across different environments, you should use a centralized Azure ML Registry.

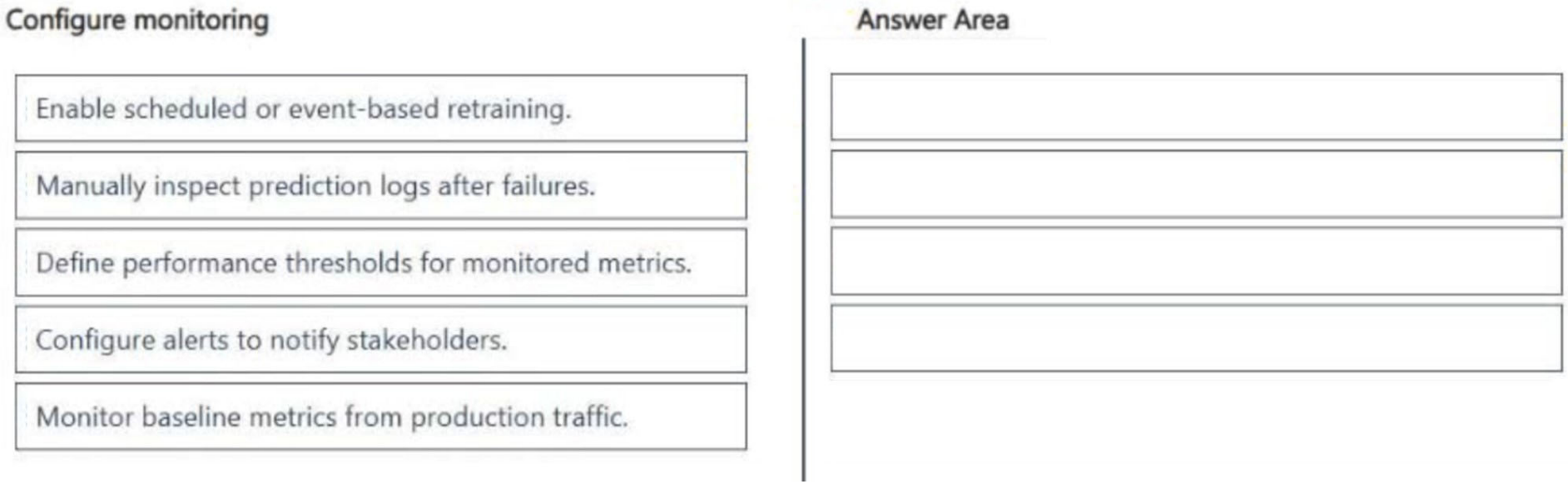
While a standard Azure ML Workspace acts as an MLflow server for individual experiments, an Azure ML Registry is the specifically designed feature for sharing models, environments, and components across multiple workspaces and environments within an Azure tenant.

3. Centralized Reuse By registering the model in the workspace's registry, you establish a single source of truth. You can then load this specific version in any environment (e.g., staging or production) using its registry URI: models:/<model_name>/<version_or_alias>.

Incorrect: [Not D] Must use an Azure ML Registry (Central).

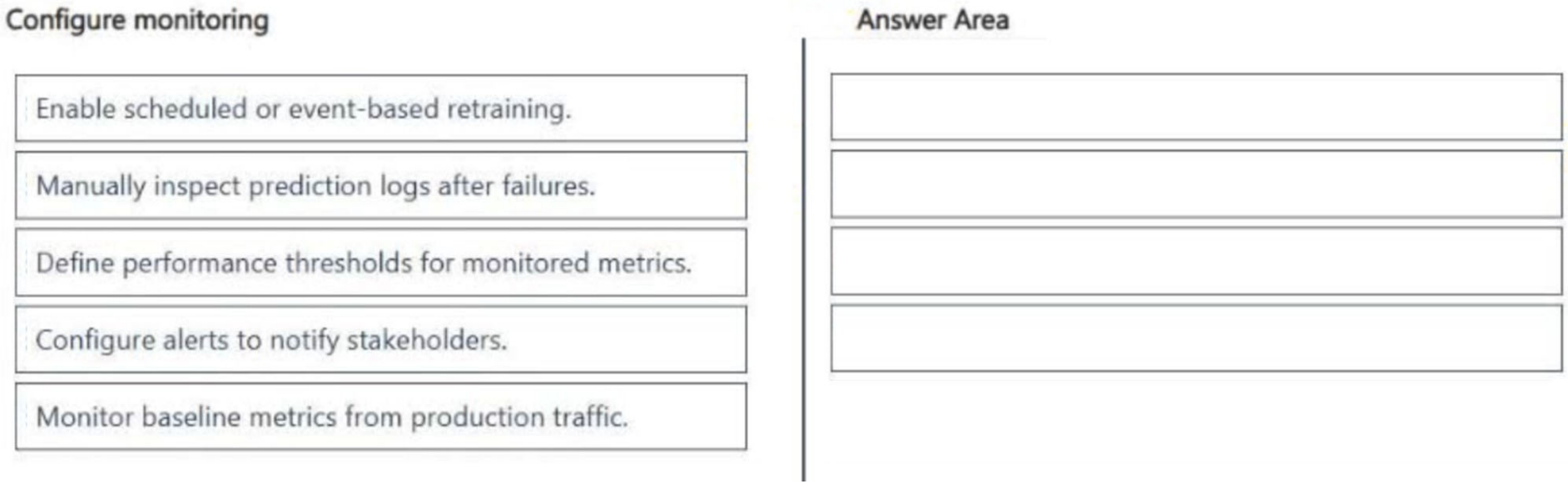
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89) DRAG DROP (Drag and Drop is not supported)
A team deploys a classification model to production and monitors performance and data changes.
The team wants to ensure that significant drops in prediction accuracy automatically trigger the following:
-Stakeholders must be notified of the drops.
-Retraining must be initiated when thresholds are exceeded
You need to configure monitoring to meet the requirements.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:

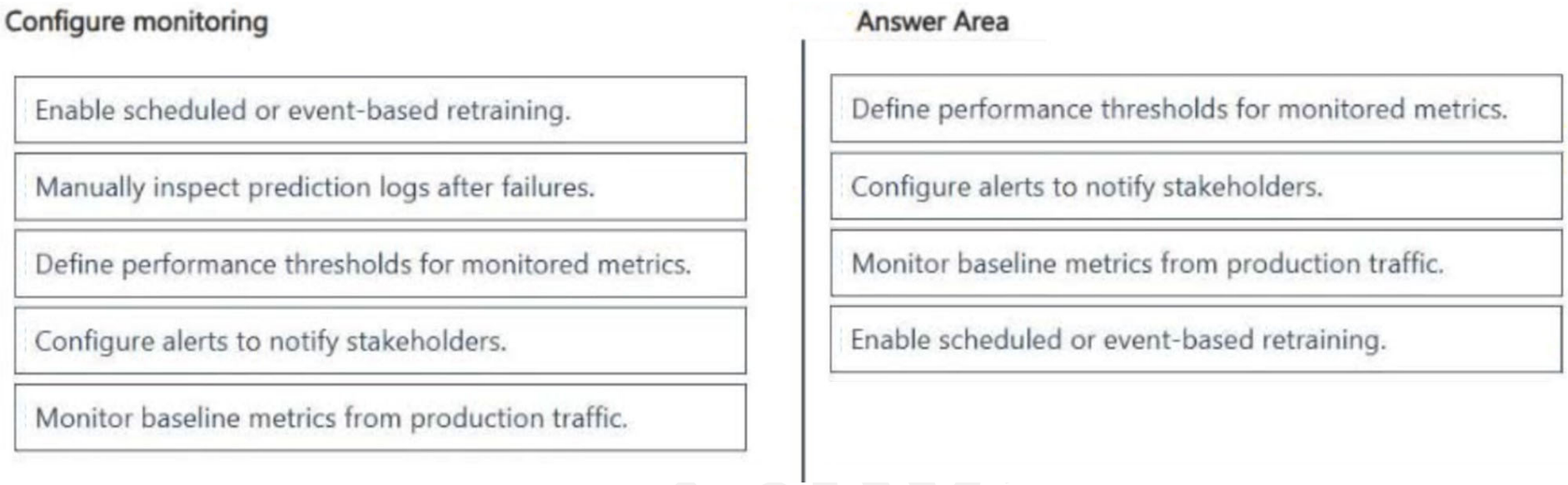


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You need to configure monitoring to meet the requirements.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:



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To meet the requirements of automatic notification and retraining, the steps should follow this sequence:
Step 1: Define performance thresholds for monitored metrics. Define performance thresholds Establish what constitutes a "significant drop." Set specific KPIs (e.g., Accuracy < 90% or Data Drift > 0.5).
Step 2: Configure alerts to notify stakeholders. Configure alerts and triggers Connect thresholds to notification systems (Azure Monitor/Action Groups). Define the "Event" that will kick off the next steps
Step 3: Monitor baseline metrics from production traffic. Monitor baseline and production metrics Collect real-time data from the deployed endpoint. Compare live telemetry against the established baseline.
Step 4: Enable schedule or event-based retraining. Execute the retraining pipeline automatically once the monitor detects a threshold breach.

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90) HOTSPOT (Drag and Drop is not supported)
You are monitoring a fine-tuned large language model deployed in Microsoft Foundry.
You evaluate the model before and after fine-tuning by using the same evaluation dataset.
You review the following evaluation results:

Evaluation metric	Baseline model	Fine-tuned model
Groundedness score	0.82	0.91
Relevance score	0.85	0.88
Hallucination rate	0.06	0.05
Response latency (ms)	420	430

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You need to determine whether the fine-tuned model shows improved performance without introducing regression. For each of the following statements, select Yes if the statement is true. Otherwise, select No.
Note: Each correct selection is worth one point.
Hot Area:

Fine-tuned model evaluation metrics

Statement	Yes	No
The fine-tuned model shows improved grounding compared to the baseline model.	○	○
The fine-tuned model introduced a regression in hallucination behavior.	○	○
The fine-tuned model's response latency indicates a functional failure.	○	○
The evaluation results indicate that fine-tuning improved overall response quality.	○	○

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Statement	Yes	No
The fine-tuned model shows improved grounding compared to the baseline model.	○	○
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The evaluation results indicate that fine-tuning improved overall response quality.	○	○

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Fine-tuned model evaluation metrics

Statement	Yes	No
The fine-tuned model shows improved grounding compared to the baseline model.	◉	○
The fine-tuned model introduced a regression in hallucination behavior.	○	◉
The fine-tuned model's response latency indicates a functional failure.	○	◉
The evaluation results indicate that fine-tuning improved overall response quality.	◉	○

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Box 1: Yes Yes - The fine-tuned model shows improved grounding compared to the baseline model.
The groundness score increased from 0.82 to 0.91 after fine-tuning. In Microsoft Foundry monitoring, a higher groundedness score indicates that the fine-tuned model's answers are better supported by the provided input source (context or database) compared to the baseline model.
Box 2: No No - The fine-tuned model introduced a regression in hallucination behavior.
The hallucination rate decreased from 0.06 to 0.05 after fine-tuning. If the hallucination rate decreases when comparing a fine-tuned model against a baseline model (base model), it indicates that the fine-tuned model is performing better and has introduced a reduction in hallucination behavior, not a regression.
Box 3: No No - The fine-tuned model's response latency indicates a functional failure.
The response latency (ms) increased from 420 to 430 after fine-tuning. In Microsoft AI Foundry, an increase in response latency (ms) for a fine-tuned model compared to its baseline does not necessarily indicate a "functional failure" in the sense of a system crash or broken code. However, it is a critical performance regression that may signal a failure to meet the specific business or technical goals of the fine-tuning project.
Box 4: Yes Yes - The evaluation results indicate that fine-tuning improved overall response quality.
The relevance score increased from 0.85 to 0.88 after fine-tuning. In the context of monitoring a fine-tuned large language model (LLM) in Microsoft (Azure) AI Foundry, an increase in the relevance score when comparing the fine-tuned model against the baseline model generally indicates that the fine-tuning process has improved the overall response quality.

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91) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear in the review screen.
You have an Azure Machine Learning workspace. You connect to a terminal session from the Notebooks page in Azure Machine Learning studio.
You plan to add a new Jupyter kernel that will be accessible from the same terminal session.
You need to perform the task that must be completed before you can add the new kernel.
Solution: Delete the Python 3.6 - AzureML kernel.
Does the solution meet the goal?

A. Yes

B. No

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Solution: Delete the Python 3.6 - AzureML kernel.
Does the solution meet the goal?

A. Yes

B. No

Answer(s): B

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Explanation:
Correct: * Create an environment.
Incorrect: * Delete the Python 3.6 - AzureML kernel. * Delete the Python 3.8 - AzureML kernel.
Note: Before you can add a new Jupyter kernel on an Azure Machine Learning compute instance terminal, you must create a Conda environment.
Required Workflow To officially provision and expose the new kernel to your Azure Machine Learning studio Notebooks, you need to execute the following full process from your terminal session: 1. Create the environment: Provision a new isolated environment (e.g., using conda create -n newenv python=3.10). 2. Activate the environment: Run conda activate newenv.
3. Install dependencies: Add the required ipykernel package using conda install ipykernel or pip install ipykernel. 4. Register the kernel: Bind the new environment configuration to the global Jupyter directory by running: python -m ipykernel install --user --name newenv --display-name "My New Kernel"

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Solution: Delete the Python 3.8 - AzureML kernel.
Does the solution meet the goal?

A. Yes

B. No

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Solution: Delete the Python 3.8 - AzureML kernel.
Does the solution meet the goal?

A. Yes

B. No

Answer(s): B

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You need to perform the task that must be completed before you can add the new kernel.
Solution: Create an environment.
Does the solution meet the goal?

A. Yes

B. No

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Solution: Create an environment.
Does the solution meet the goal?

A. Yes

B. No

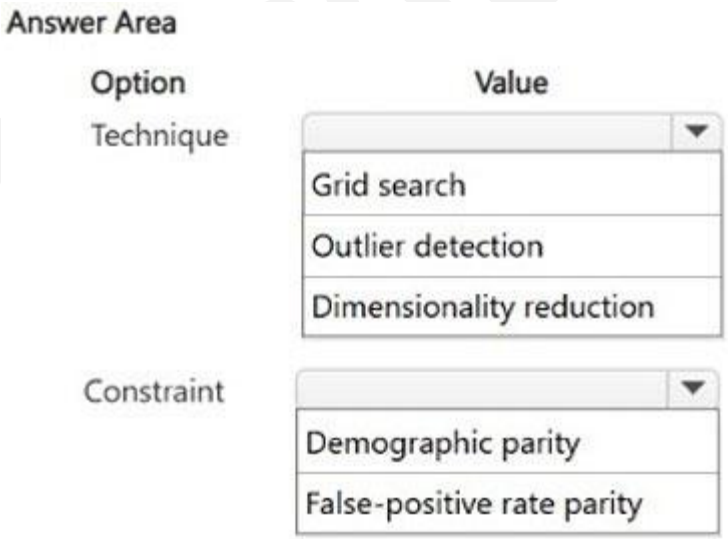
Answer(s): A

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Note: Before you can add a new Jupyter kernel on an Azure Machine Learning compute instance terminal, you must create a Conda environment.
Required Workflow To officially provision and expose the new kernel to your Azure Machine Learning studio Notebooks, you need to execute the following full process from your terminal session: 1. Create the environment: Provision a new isolated environment (e.g., using conda create -n newenv python=3.10). 2. Activate the environment: Run conda activate newenv. 3. Install dependencies: Add the required ipykernel package using conda install ipykernel or pip install ipykernel. 4. Register the kernel: Bind the new environment configuration to the global Jupyter directory by running: python -m ipykernel install --user --name newenv --display-name "My New Kernel"

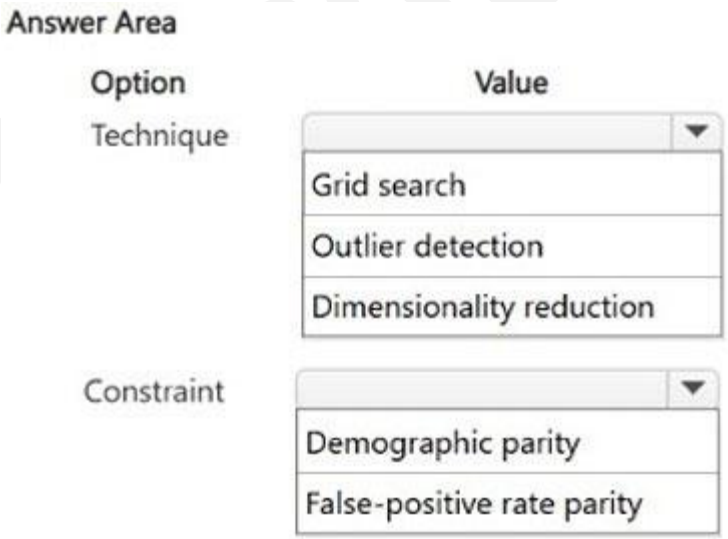
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94) HOTSPOT (Drag and Drop is not supported)
A biomedical research company plans to enroll people in an experimental medical treatment trial.
You create and train a binary classification model to support selection and admission of patients to the trial.
The model includes the following features: Age, Gender, and Ethnicity.
The model returns different performance metrics for people from different ethnic groups.
You need to use Fairlearn to mitigate and minimize disparities for each category in the Ethnicity feature.
Which technique and constraint should you use? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:

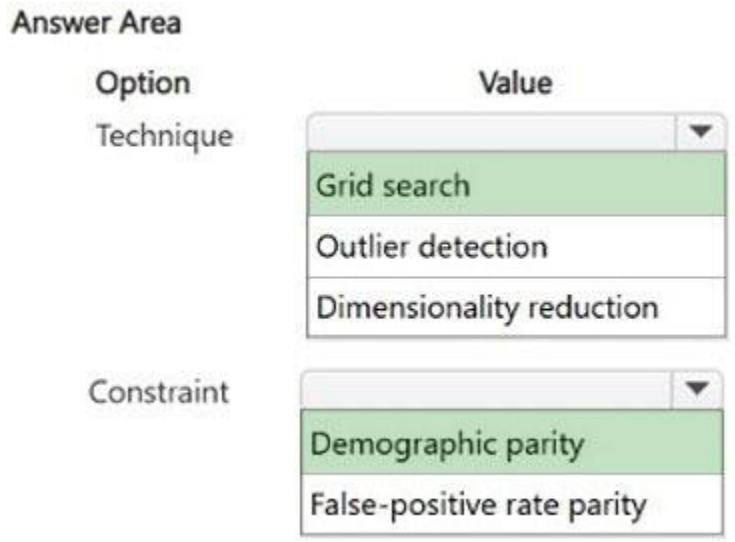


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Box 1: Grid search To minimize disparities in a binary classification model across the categories of a sensitive feature (such as Ethnicity), the best approach within the Azure Machine Learning and Fairlearn framework is to use Grid Search combined with a Demographic Parity constraint.

Technique: Grid Search (fairlearn.reductions.GridSearch) This is a reduction-based, in-processing technique. It takes your standard estimator (e.g., a classification algorithm) and optimizes it across a grid of systematically reweighted training datasets.

It generates multiple candidate models, allowing you to select the model that provides the best trade-off between predictive accuracy and fairness metrics.

Box 2: Demographic Parity Constraint: Demographic Parity (DemographicParity) This constraint ensures selection-rate equity.It forces the model's positive outcome prediction (e.g., admission to the medical trial) to be statistically independent of the sensitive attribute.

As a result, the probability of being selected is balanced evenly across all ethnic groups.

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95) You are authoring a notebook in Azure Machine Learning studio.
You must install packages from the notebook into the currently running kernel. The installation must be limited to the currently running kernel only.
You need to install the packages.
Which magic function should you use?

A. !pip
B. !conda
C. %load
D. %pip

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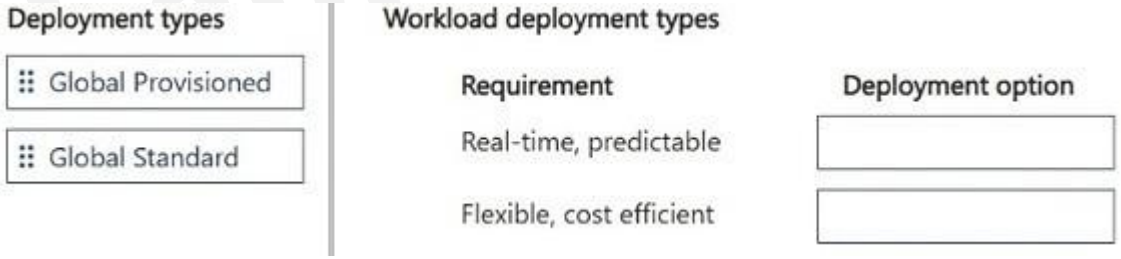
A. !pip
B. !conda
C. %load
D. %pip

Answer(s): D

Explanation:
To target and install Python packages into the currently running kernel only, you should use the %pip line magic function.
Kernel Isolation: Unlike using the shell command !pip install, which installs the package into the underlying compute instance's base environment, the %pip magic explicitly targets the specific Python executable assigned to your active notebook kernel.
No Restart Needed: It automatically syncs the newly installed packages with the current session so you can import them immediately without restarting the kernel.
Example Usage Simply type and run the following python line in a notebook cell:
%pip install <package_name>

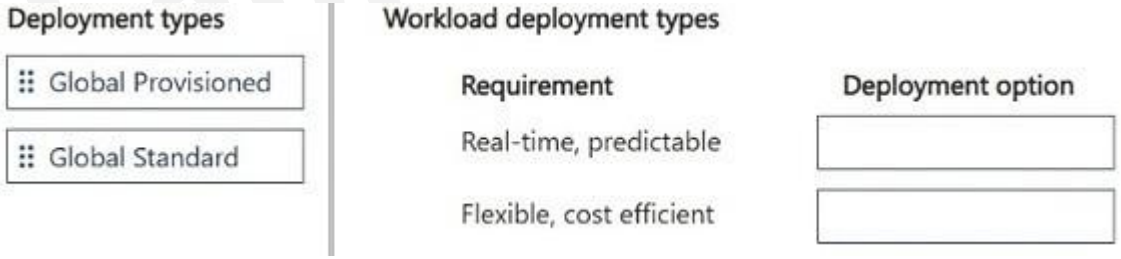
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96) DRAG DROP (Drag and Drop is not supported)
An organization is deploying generative AI solutions by using Microsoft Foundry to support multiple production workloads.
The organization has the following workload requirements:
-One workload must be real-time, latency-sensitive, and have predictable global usage patterns that demand consistent performance.
-One workload must have variable performance and be optimized for cost-efficient operation.
You need to select a global deployment type for each workload.
Which type of deployment should you use for each workload requirement? To answer, move the appropriate deployment types to the correct requirements. You may use each deployment type once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



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Select and Place:



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Deployment types

- Global Provisioned
- Global Standard

Workload deployment types

Requirement	Deployment option
Real-time, predictable	Global Provisioned
Flexible, cost efficient	Global Standard

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Box 1: Global Provisioned. Real-time, predictable
The best deployment type for this workload is Global Provisioned.
Dedicated Resources for Real-Time Needs: This deployment option uses Provisioned Throughput Units (PTUs) to reserve dedicated compute capacity exclusively for your application. It eliminates "noisy neighbor" issues found in shared capacity options, ensuring the low and stable latency needed for interactive user experiences.
Predictable Performance: Because the processing capacity is fixed and fully guaranteed, you avoid queueing delays and unexpected latency spikes, matching your strict requirement for consistent performance.
Global Infrastructure Routing: It uses Microsoft’s global infrastructure to intelligently handle traffic without region-specific quota constraints. This provides maximum resilience and optimizes performance for predictable global usage patterns.
Box 2: Global Standard Flexible, cost efficient
The Standard (or Global Standard) deployment type is best for this workload.
Standard Deployment: This type utilizes a pay-per-token (consumption-based) billing model. It is specifically designed to handle variable or bursty traffic patterns cost-effectively. Because you only pay for the tokens you actively consume, you avoid the high fixed costs associated with unused capacity during low-demand periods.
Performance Impact: Since standard deployments utilize multi-tenant shared infrastructure, they feature variable performance and latency fluctuations based on regional demand. This perfectly matches your requirement for a workload that must allow for variable performance.

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97) You create an Azure Machine Learning workspace. You train an MLflow-formatted regression model by using tabular structured data.
You must use a Responsible AI dashboard to assess the model.
You need to use the Azure Machine Learning studio UI to generate the Responsible AI dashboard.
What should you do first?

A. Register the model with the workspace.
B. Create the model explanations.
C. Convert the model from the MLflow format to a custom format.
D. Deploy the model to a managed online endpoint.

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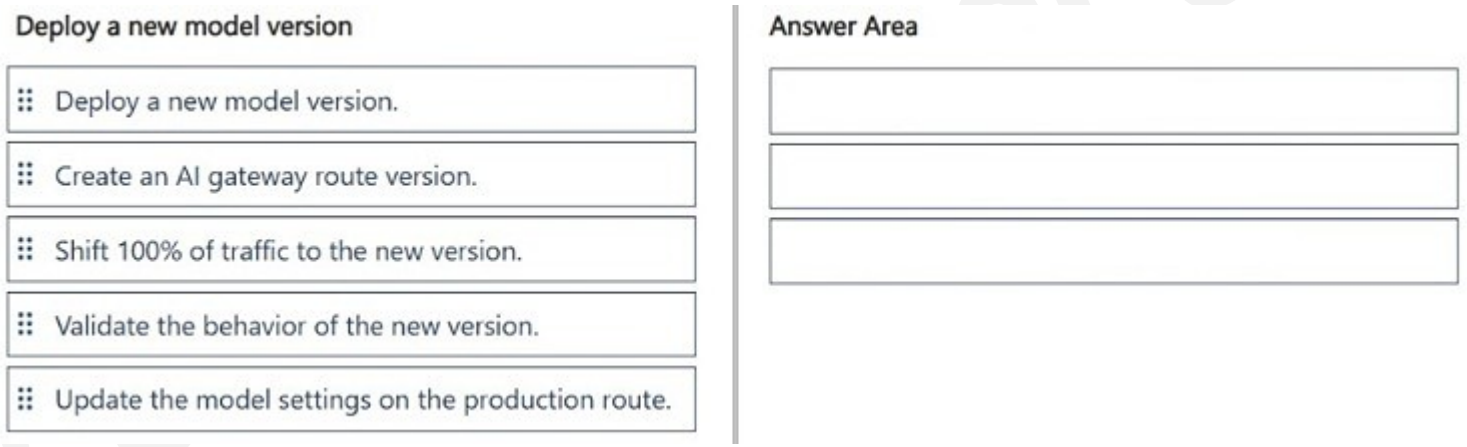
A. Register the model with the workspace.
B. Create the model explanations.
C. Convert the model from the MLflow format to a custom format.
D. Deploy the model to a managed online endpoint.

Answer(s): A

Explanation:
The first step you must take is to register the model with the workspace.
To access the no-code, guided wizard for generating a Responsible AI dashboard directly within the Azure Machine Learning studio UI, the trained model must first exist as a recognized asset inside your workspace's model registry.

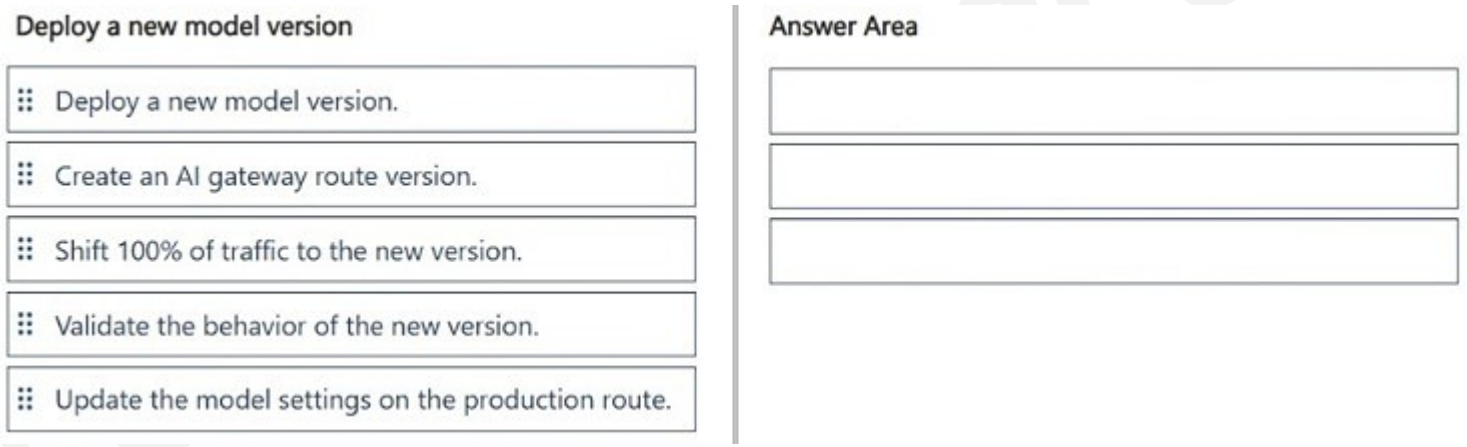
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98) DRAG DROP (Drag and Drop is not supported)
A customer-facing web application uses a foundational model deployed through Microsoft Foundry.
A new model version must be introduced and validated without disrupting production traffic.
You need to deploy the new version by using a safe promotion strategy.
Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:

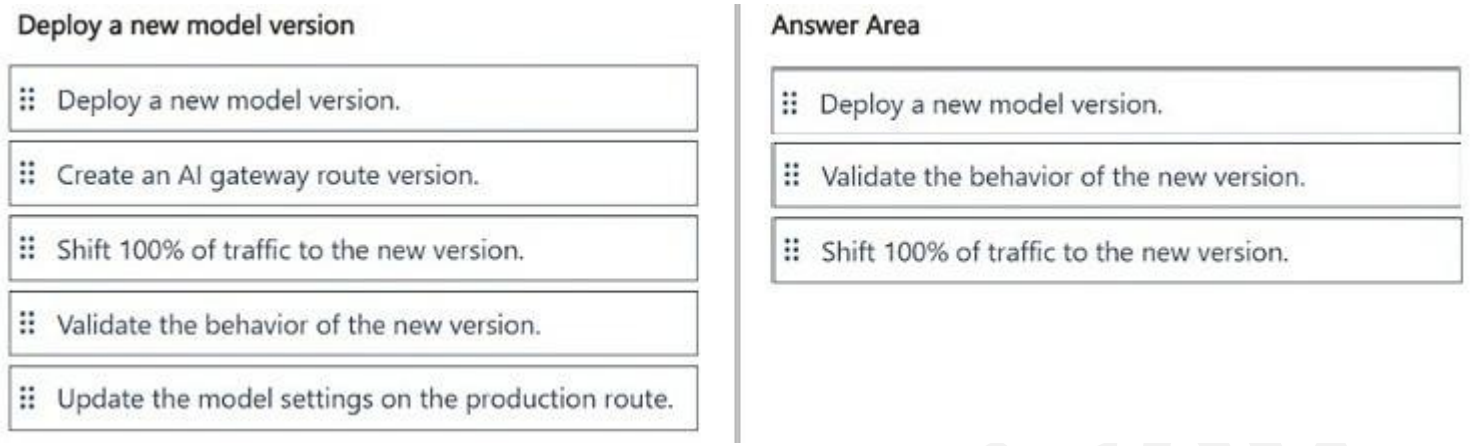


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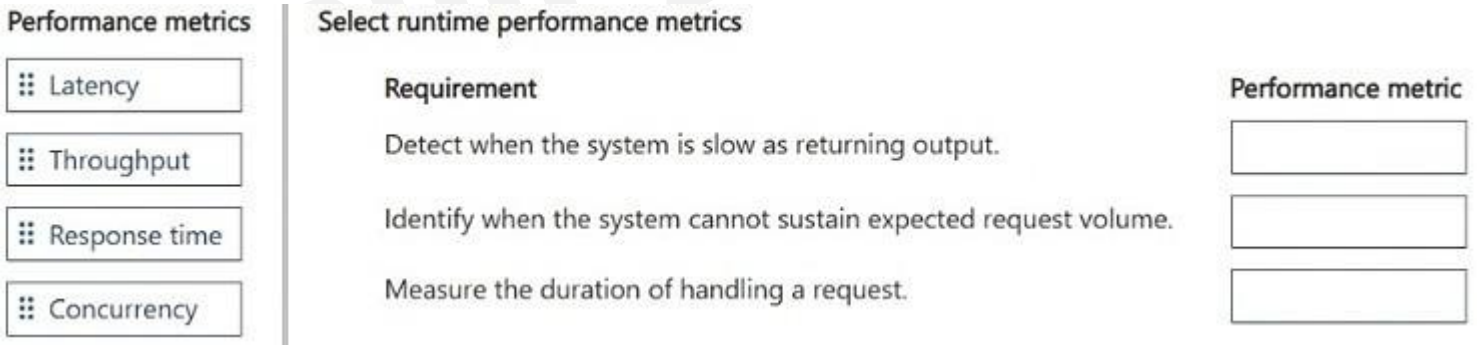
To update the foundation model version safely in Microsoft Foundry without affecting live production traffic, you must implement a structured Blue-Green Deployment and continuous evaluation pipeline. This decouples the deployment of the new model from its public exposure, ensuring a zero-downtime transition.
Here are the precise chronological steps you should take: 1. Disable Platform Auto-Upgrades [Skip] Before introducing any new updates, prevent the system from shifting beneath you.
Pin Your Current Baseline: Modify your current deployment configurations to "Stay on this version" or "No Auto Upgrade". This ensures that Microsoft Foundry does not automatically update your production models to a new default version before you have validated it.
Step 1: Deploy a new model version 2. Deploy the New Model Version in Isolation (Green Deployment) Deploy the updated model version as an entirely separate instance within the same endpoint.

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Create a Green Instance: Leave your original model (Blue) running at 100% traffic.
Set Initial Traffic to Zero: Deploy the new model version (Green) to your Microsoft Foundry resource or online endpoint with a traffic weight of 0%.This provides a live, isolated endpoint for the new model version without routing customer traffic to it.
Step 2: Validate the behavior of the new version. 3. Conduct Isolated Testing and Evaluations Validate the Green deployment's outputs against your original performance baselines.
Run Evaluation Pipelines: Use Microsoft Foundry's built-in Evaluation tools or Prompt Flow to test the new model. Test against benchmark datasets to evaluate response quality, accuracy, and safety filters.
Targeted Request Verification: Send targeted testing requests directly to the Green deployment using specific HTTP headers (e.g., azureml-model-deployment: green) to bypass global traffic routing rules.
Step 3: Shift 100% of traffic to the new version 4. Implement Shadow Testing or Canary Routing Gradually expose the new model version to production traffic patterns.
Canary Release / Traffic Splitting: Update your endpoint configurations to route a minimal portion (e.g., 90% Blue / 10% Green) of live customer requests to the new model.
Shadow Mirroring (Optional): If supported by your API architecture, mirror 100% of live traffic to the Green deployment concurrently. Discard its responses to the user but log and evaluate how it handles real-world requests in real time.
Incorrect: * Create an AI gateway route version Once the new model is deployed and vetted, you can then implement routing mechanisms (such as creating an AI gateway route version or configuring Azure API Management) to perform A/B testing, shadowing, or safe traffic shifting to production.

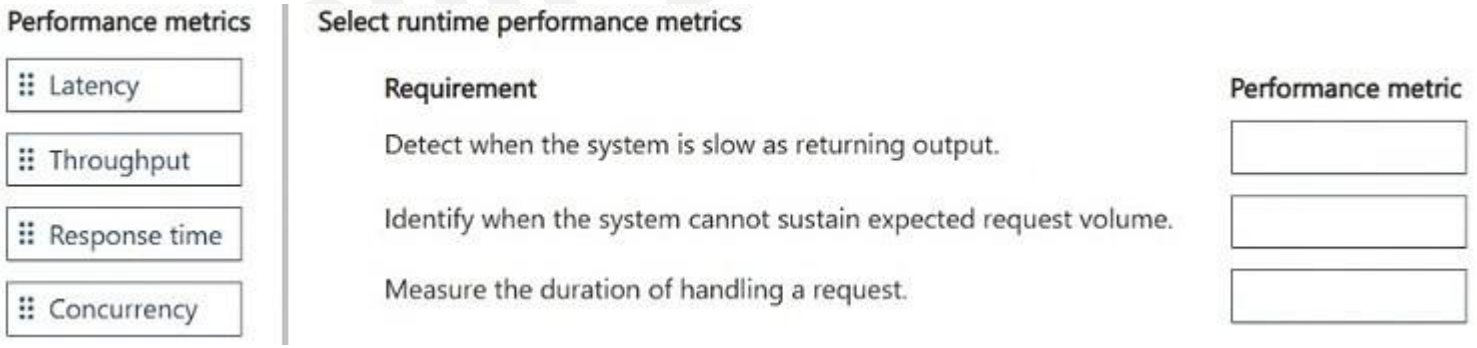
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99) DRAG DROP (Drag and Drop is not supported)
A team operates a generative AI-powered customer support assistant built on Microsoft Foundry. The application serves users globally and supports both real-time chat interactions and batch summarization jobs.
The team must ensure that the application continues to meet defined service-level objectives (SLO) as usage increases.
The team requires visibility into runtime behavior to identify performance regressions that affect the user experience and system capacity.
You need to select the performance metrics that meet the requirements.
Which performance metric should you monitor for each requirement? To answer, move the appropriate performance metrics to the correct requirements. You may use each performance metric once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



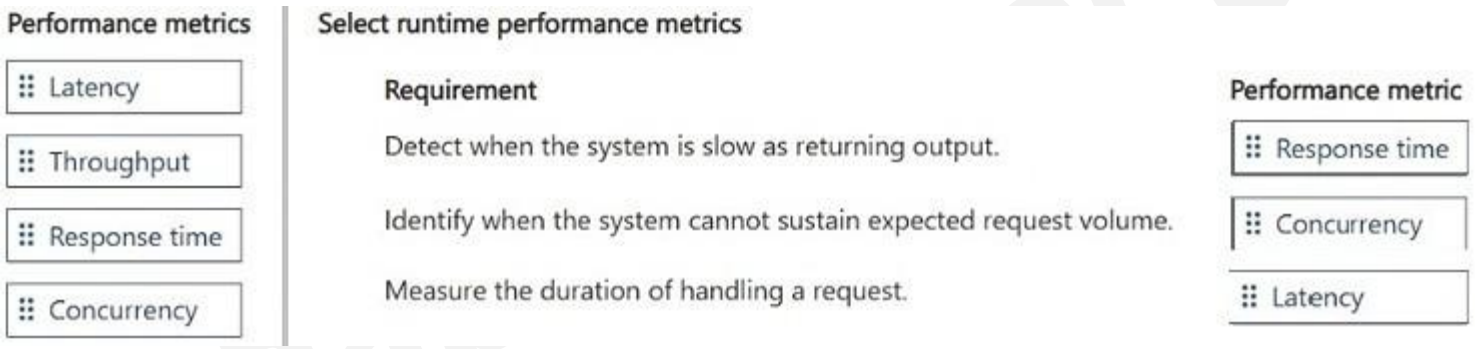
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Answer(s):

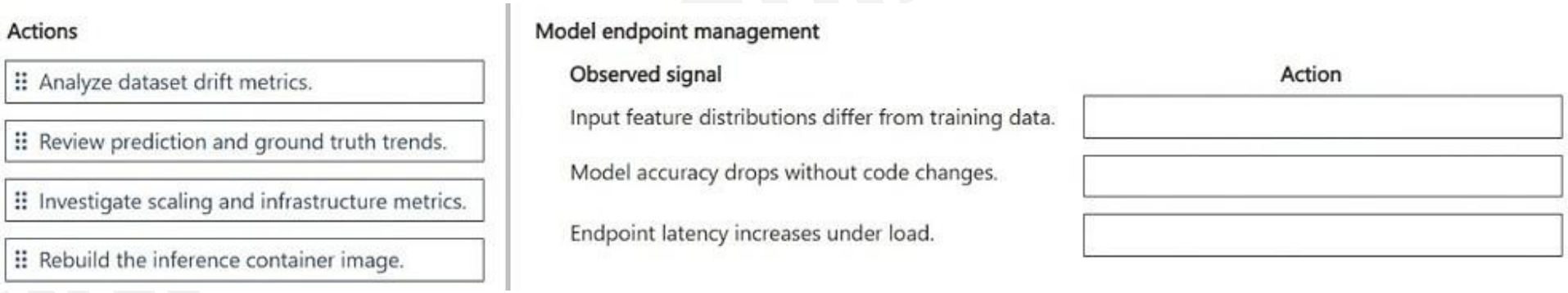


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Box 1: Response time Detect when the system is slow as returning output.
Response time is the best single metric to detect when the system is slow at returning output to the user.
Response time measures the total duration from the moment a user sends a request until they receive the final output.
Because this specific application handles real-time chat interactions, response time directly captures the exact slowness a user experiences. For generative AI applications, this often tracks critical user-centric metrics like Time to First Token (TTFT) and overall end-to-end generation time.
Incorrect: Latency focuses strictly on the time spent on server-side processing or network transit. It often excludes components of the user experience, such as data rendering time or client-side delays, making it less comprehensive than response time for measuring end-to-end user-perceived slowness.
Throughput measures the volume of data or number of requests processed over a specific period (e.g., tokens per second or requests per minute).
While throughput is crucial for evaluating the capacity of your batch summarization jobs, a system can have high throughput while individual real-time users still suffer from slow, high-response-time delays.
Box 2: Concurrency Identify when the system cannot sustain expect request volume.
When a system cannot sustain expected request volumes, Concurrency acts as the single best leading metric to identify infrastructure bottlenecks.
Active Load Exposure: Concurrency tracks the exact number of active, parallel requests currently being processed by the system at any single moment.
Resource Exhaustion Warning: In generative AI applications, real-time chat interactions hold long-lived connections, while batch summarization jobs consume massive computational threads. If concurrency spikes or plateaus at your system's maximum thread/hosting limit, it serves as a direct indicator that the application cannot handle more parallel volume and will begin queueing or dropping incoming requests.

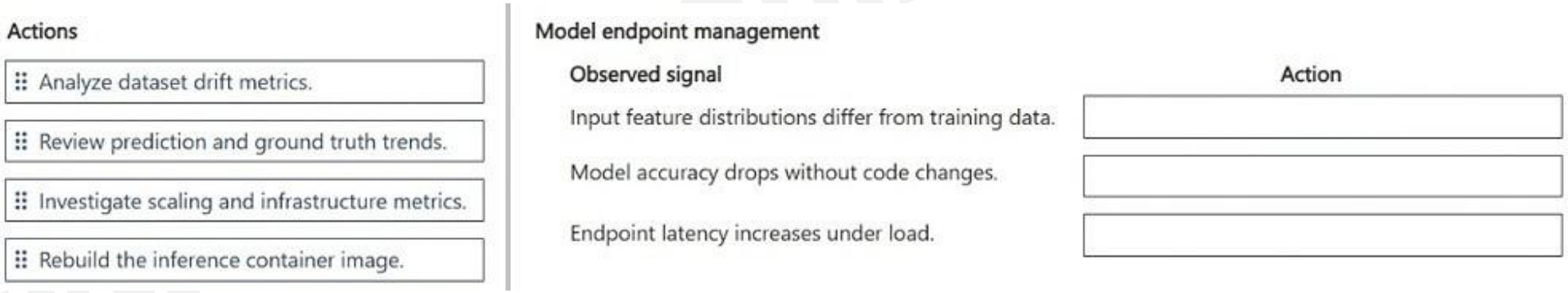
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100) DRAG DROP (Drag and Drop is not supported)
A team deploys a classification model to production and scores incoming customer data daily.
After several weeks, business stakeholders report unexpected changes in prediction behavior, even though the endpoint remains healthy.
You need to determine whether data drift is occurring and if it is, identify the appropriate actions.
Which action should you perform for each observed signal? To answer, move the appropriate actions to the correct observed signals. You may use each action once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
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Select and Place:



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Actions

- Analyze dataset drift metrics.
- Review prediction and ground truth trends.
- Investigate scaling and infrastructure metrics.
- Rebuild the inference container image.

Model endpoint management

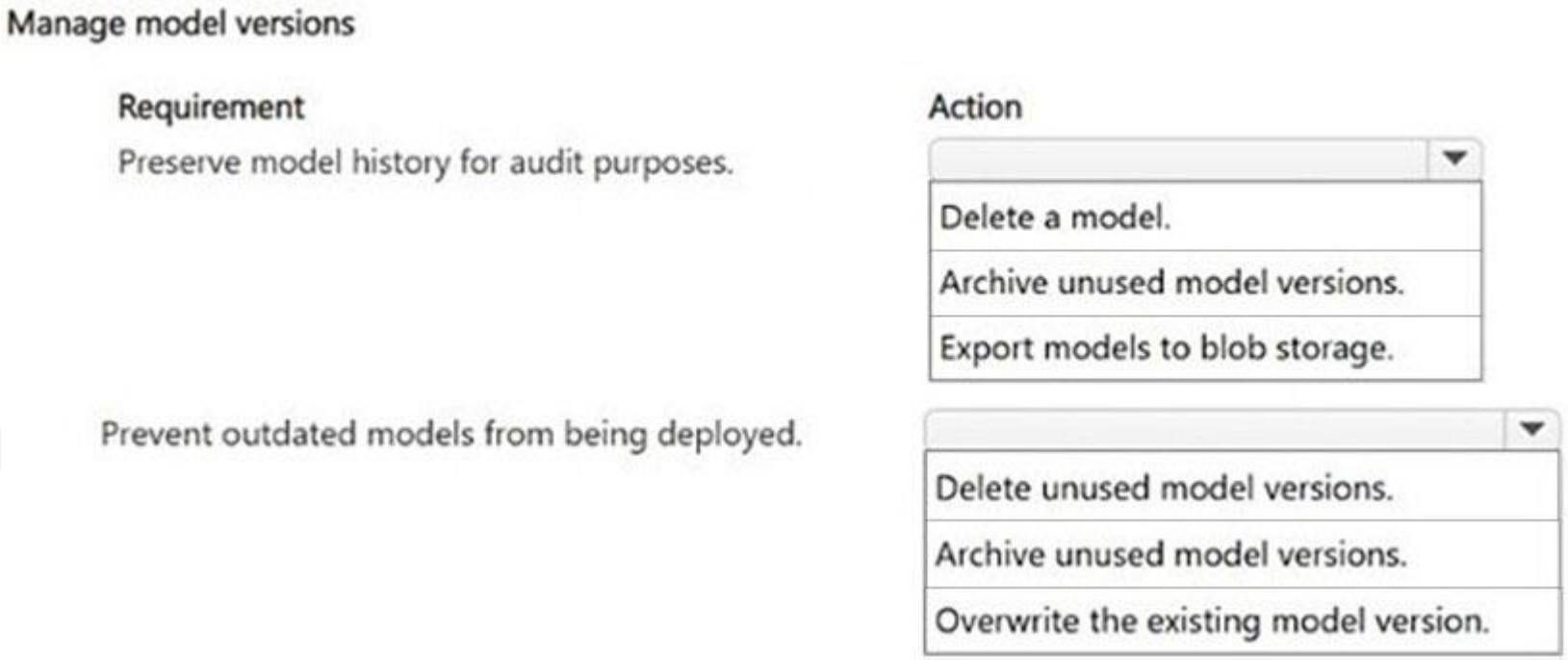
Observed signal	Action
Input feature distributions differ from training data.	Analyze dataset drift metrics.
Model accuracy drops without code changes.	Analyze dataset drift metrics.
Endpoint latency increases under load.	Investigate scaling and infrastructure metrics.

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Box 1: Analyze dataset drift metrics Input feature distributions differ from training data.
The direct action you must take when input feature distributions differ from the training data is to retrain the model using the updated data distribution.
This scenario describes a classic case of data drift (also known as covariate shift).
While the endpoint infrastructure remains perfectly healthy, the statistical properties of the production data have shifted away from the baseline training data, leading to a degradation in predictive behavior.
Box 2: Analyze dataset drift metrics
Model accuracy drops without code changes.
You should analyze dataset drift metrics to identify why the model accuracy dropped without any changes to the code.
In production machine learning, dataset drift occurs when the statistical properties of the input data change over time. Even if your Azure Machine Learning endpoint remains perfectly healthy and functional, changes in the real-world data distribution will cause the model's predictive performance to degrade. This phenomenon is often referred to as data drift or concept drift.
Box 3: Investigate scaling and infrastructure metrics Endpoint latency increases under load.
You should immediately investigate infrastructure metrics, as latency spikes under load typically indicate resource exhaustion.

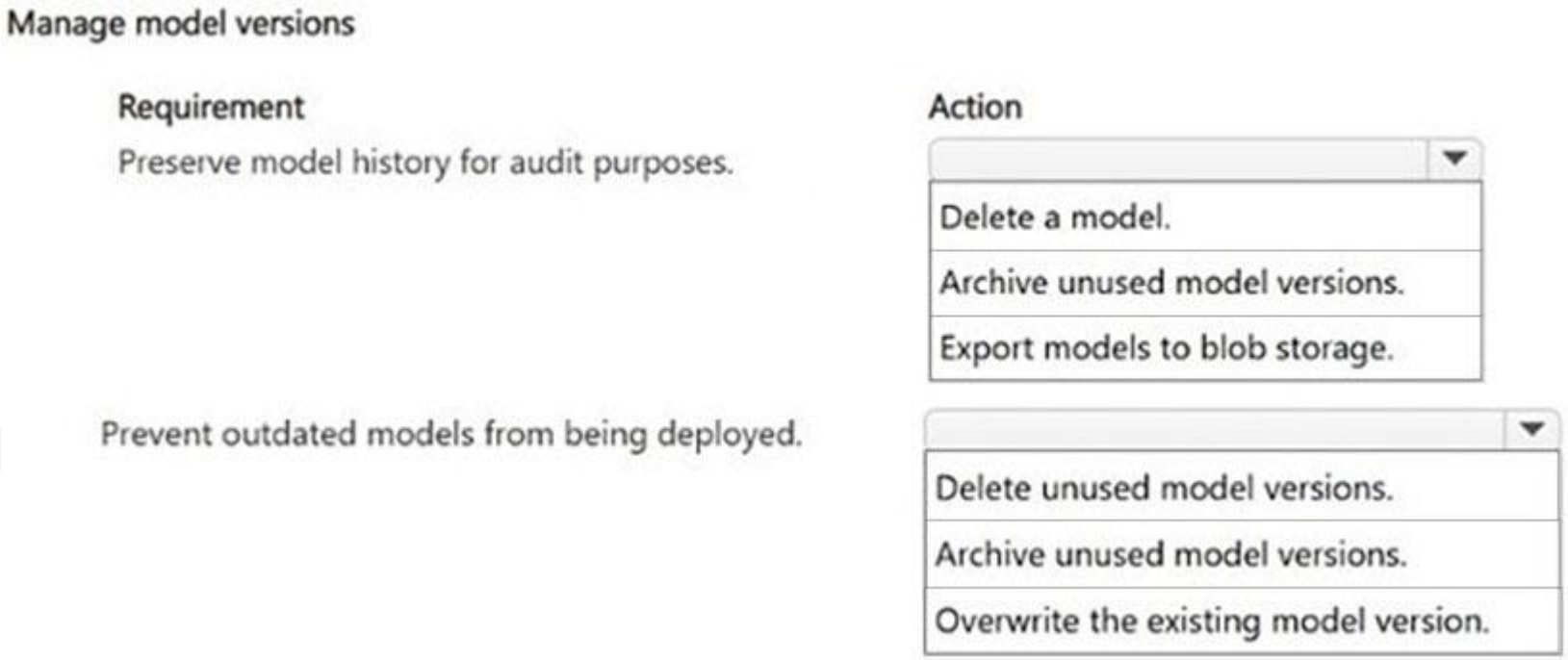
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102) HOTSPOT (Drag and Drop is not supported)
A team manages an Azure Machine Learning workspace to train and register machine learning models.
Previous model versions must be retained for audit and rollback purposes but must not be used for new deployments.
You need to manage model versions.
What should you do for each requirement? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



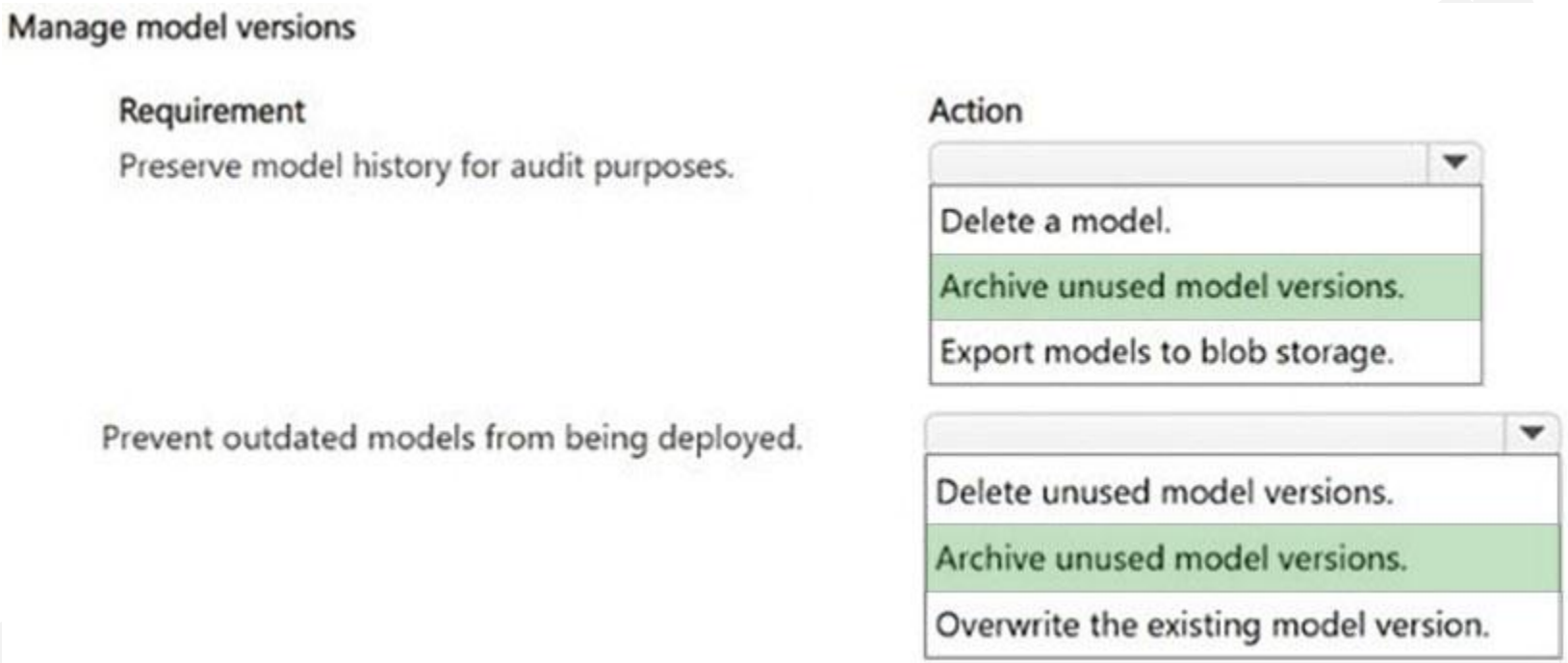
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Answer(s):

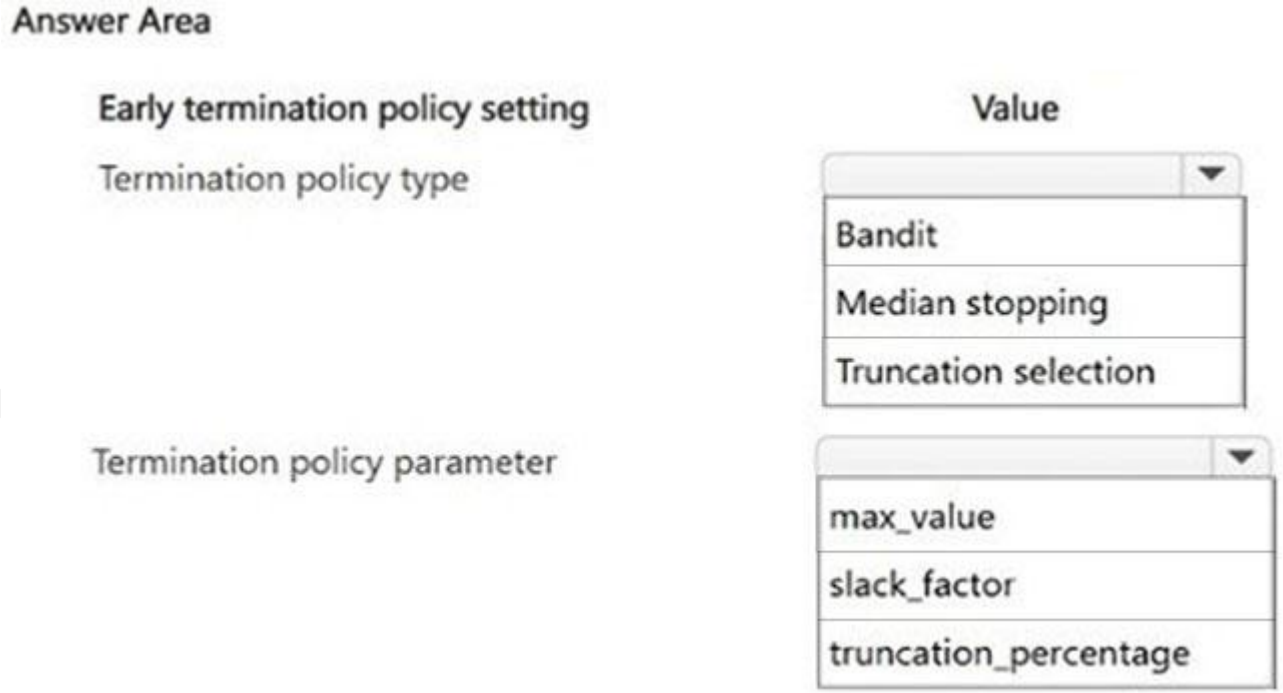


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Box 1: Archive unused model versions Preserve Model history for audit purposes.
Archiving unused model versions is highly appropriate and recommended for this scenario.
Archiving hides unused versions from standard queries and clean user interfaces, while fully preserving the underlying artifacts and metadata required for compliance audits and production rollbacks.
Preserves Audit Trails: Archiving does not delete the model. It perfectly retains the model files, lineage, run histories, and regulatory logs required for compliance audits.
Enables Seamless Rollbacks: An archived model version remains active in your workflows. If a rollback becomes necessary, you can directly reference or restore the archived model to standard views via the Azure
ML CLI.
Declutters the Registry: Standard commands like az ml model list ignore archived components. This simplifies your view to display only active models without disrupting underlying production pipelines.
Box 2: Archive unused model versions Prevent outdate models from being deployed.
Archiving hides unused versions from standard queries and clean user interfaces, while fully preserving the underlying artifacts and metadata required for compliance audits and production rollbacks.

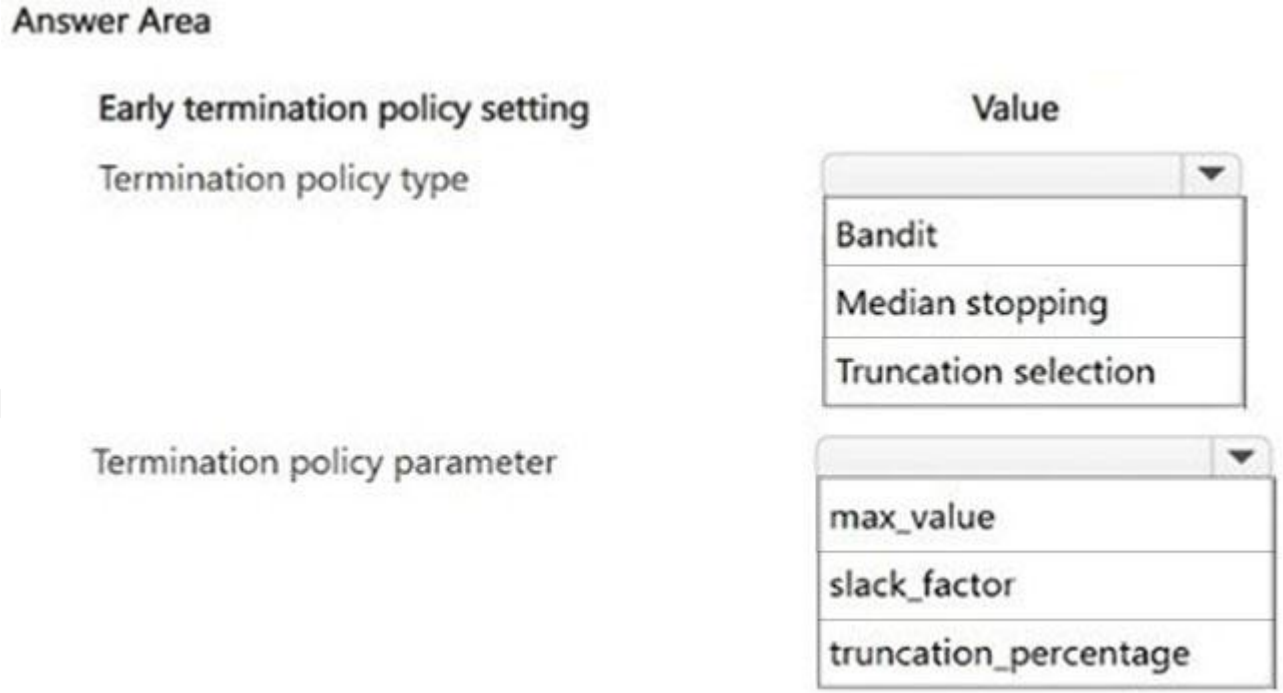
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103) HOTSPOT (Drag and Drop is not supported)
You use Azure Machine Learning to implement hyperparameter tuning for an Azure ML Python SDK v2-based model training.
Training runs must terminate when the primary metric is lowered by 25 percent or more compared to the best performing run.
You need to configure an early termination policy to terminate training jobs.
Which values should you use? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:



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ivel.com

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Hot Area:



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Answer(s):

Answer Area

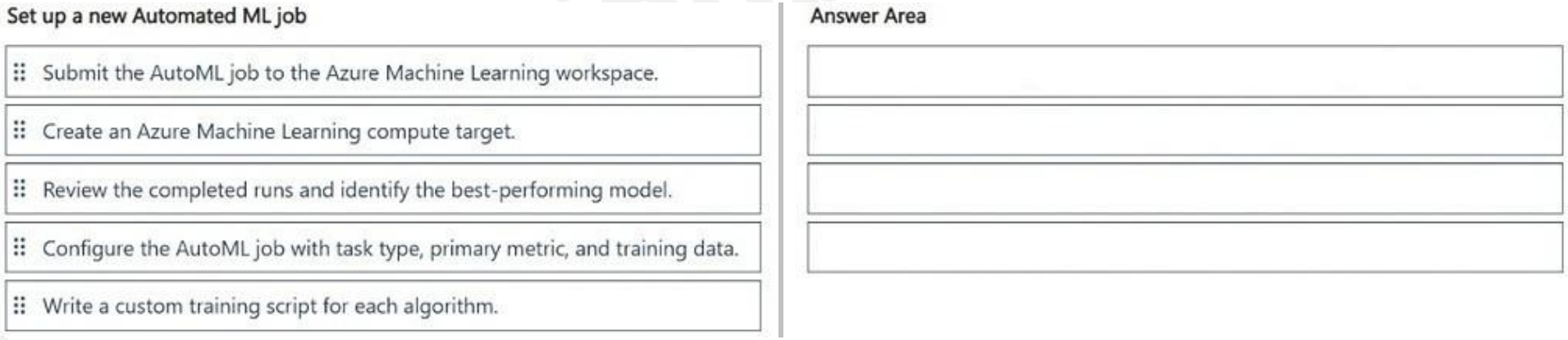
Early termination policy setting	Value
Termination policy type	Bandit
	Median stopping
	Truncation selection
Termination policy parameter	max_value
	slack_factor
	truncation_percentage

www

Box 1: Bandit The Bandit policy is the best termination policy type for this requirement.
Within Azure Machine Learning (Python SDK v2), a Bandit Policy terminates ongoing training runs if the primary performance metric drops below a specified threshold relative to the top-performing run to date.
Box 2: slack_factor To achieve this configuration, you should define the slack_factor parameter.
In the Azure Machine Learning Python SDK v2, the BanditPolicy class allows you to specify a drop-off threshold using either a relative ratio or an absolute value: slack_factor: Specifies the allowable slack as a ratio or percentage relative to the best performing run.
slack_amount: Specifies the allowable slack as an absolute distance or amount.Because your threshold requirement states that the primary metric cannot drop by a relative percentage (25% or more compared to the best run), you must use the ratio-based parameter.To configure a 25% allowable drop, you would set the slack_factor to 0.25.

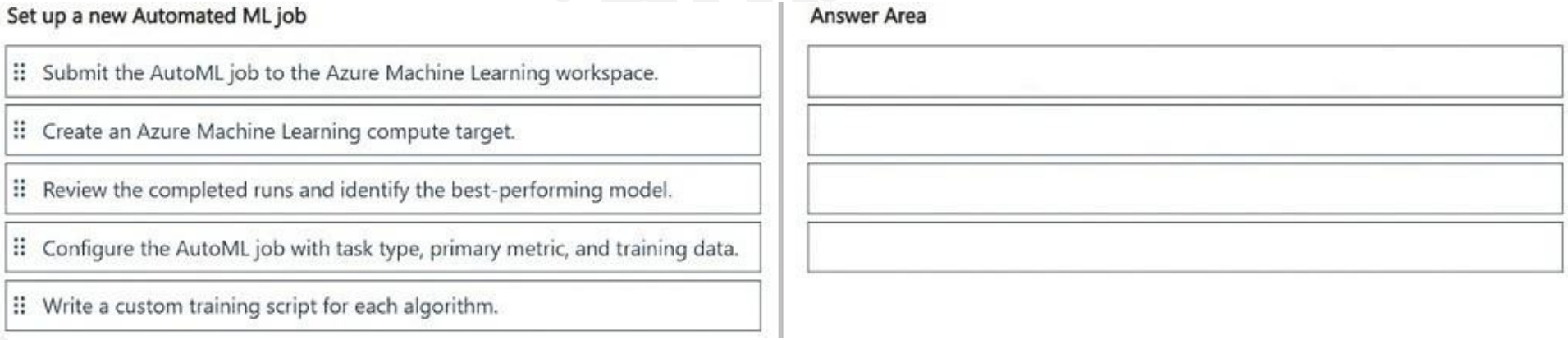
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104) DRAG DROP (Drag and Drop is not supported)
A data science team plans to evaluate multiple algorithms and hyperparameters for a classification problem without manually authoring separate training scripts.
The team must run an automated machine learning (AutoML) job that compares models and identifies the best-performing configuration.
You need to configure and run an AutoML training job.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:



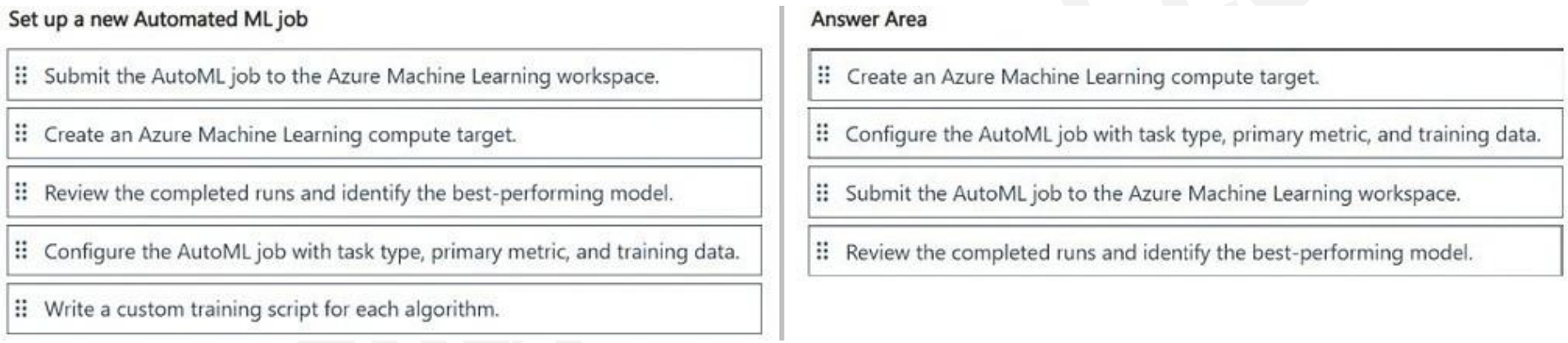
https://shapingpixel.com

104) DRAG DROP (Drag and Drop is not supported)
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The team must run an automated machine learning (AutoML) job that compares models and identifies the best-performing configuration.
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Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.
Select and Place:



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Answer(s):



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To run an automated Azure Machine Learning (AutoML) job for classification without manual scripting, you must use the Azure ML Python SDK v2 to configure an automl.classification() job, submit it via the MLClient, and retrieve the best model.

1. Set Up Your Workspace Connection [skip] Connect to your Azure Machine Learning workspace using your credentials.

Step 1: Create an Azure Machine Learning compute target. 2. Prepare Data and Compute Resources Define your training data input and specify the compute cluster where the parallelized training iterations will execute.

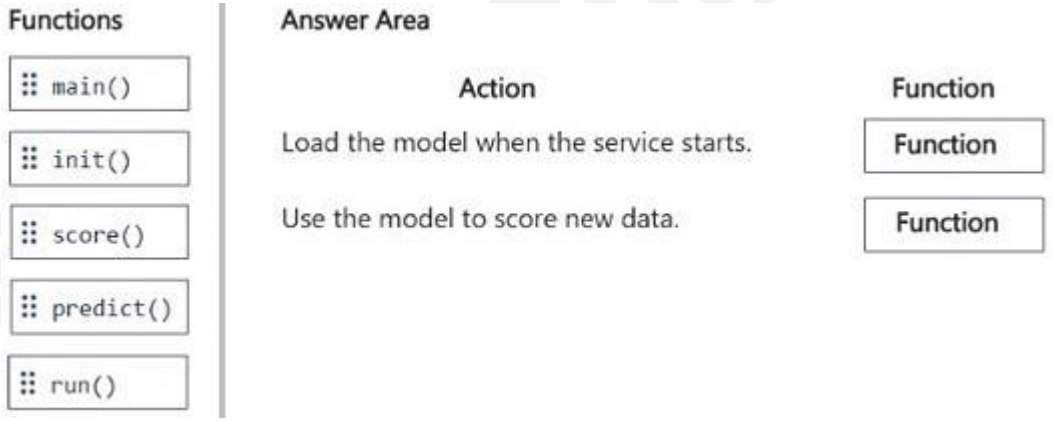
Step 2: Configure the AutoML Job with task type, primary metric, and training data. 3. Configure the AutoML Classification Job Use the automl.classification() function to define the target column, primary metric, limits, and training constraints.

Step 3: Submit the AutoML Job to the Azure Machine Learning workspace. 4. Submit and Track the Job Submit the configured job to your workspace and wait for its execution to complete.

Step 4: Review the completed runs and identify the best-performing model 5. Retrieve the Best Model Query the finished job properties to identify the pipeline execution that yielded the highest primary metric score.

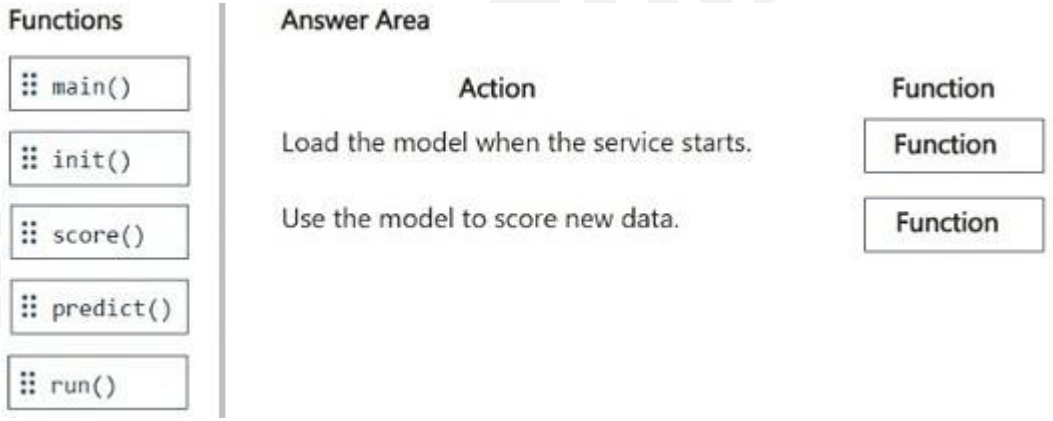
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106) DRAG DROP (Drag and Drop is not supported)
You use Azure Machine Learning to deploy a model as a real-time web service.
You need to create an entry script for the service that ensures that the model is loaded when the service starts and is used to score new data as it is received.
Which functions should you include in the script? To answer, drag the appropriate functions to the correct actions. Each function may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
Note: Each correct selection is worth one point.
Select and Place:



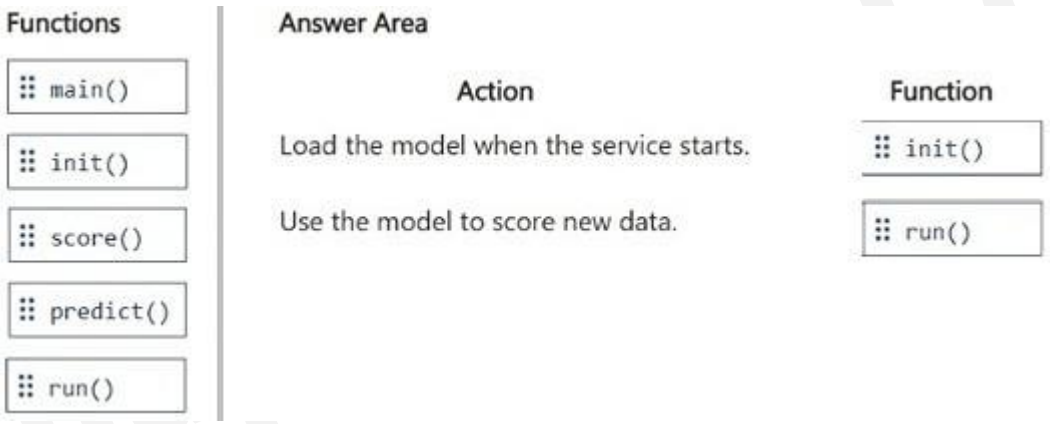
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Select and Place:



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Answer(s):



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Box 1: init() Load the model when the service starts.
To load an Azure Machine Learning model when a real-time web service starts, you must use the init() function in your entry script.
Key Components of an Entry Script An Azure Machine Learning entry script (typically named score.py) requires two core functions to manage the web service lifecycle: init(): This function runs exactly once when the Docker container initializes. You use it to load the model from the registry into a global variable so it is cached and ready for incoming
requests.run()/run(): This function executes every time the service receives a prediction request. It uses the model loaded by init() to process the input data and return a prediction.
Box 2: run() Use the model to score new data.
Use the run(data) function in your entry script to score new data as it is submitted to the deployed web service.

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107) HOTSPOT (Drag and Drop is not supported)
A regression model is deployed to a production online endpoint and is actively serving predictions.
Business stakeholders report that the model's prediction quality has changed.
The endpoint remains available and responsive.
You need to determine which metric to examine to validate each quality concern.
Which metric should you review for each concern? To answer, select the appropriate options in the answer area.
Note: Each correct selection is worth one point.
Hot Area:

Evaluate deployed model metrics

Concern	Metric
Prediction accuracy declines over time.	Endpoint latency metrics Model performance metrics Prediction output distributions trends Request throughput
Response times increase during peak usage.	Endpoint latency metrics Model performance metrics Prediction output distribution trends Feature importance
Predictions remain accurate, but business outcomes worsen.	Endpoint latency metrics Model performance metrics Prediction output distribution trends Training dataset size

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Evaluate deployed model metrics

Concern	Metric
Prediction accuracy declines over time.	Model performance metrics
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Endpoint latency metrics
Model performance metrics
Prediction output distributions trends
Request throughput

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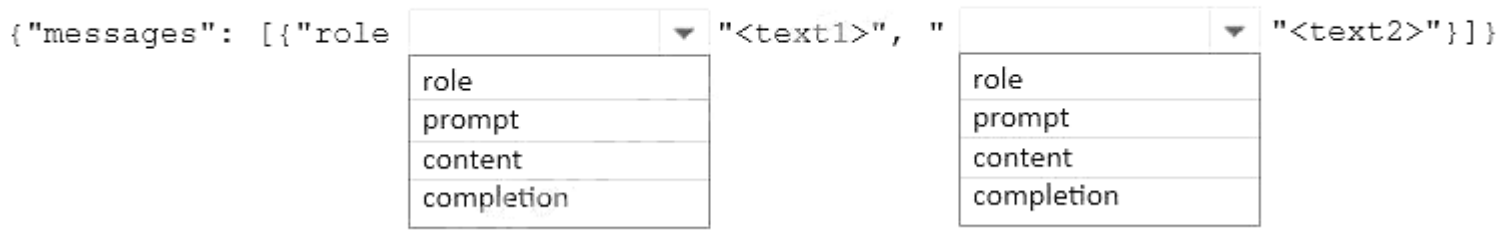
Box 1: Model performance metrics Prediction accuracy declines over time.
To address a decline in model accuracy while the endpoint remains available, you should investigate model performance metrics, such as Data Drift and Prediction Drift.
When a model's prediction quality declines over time without API or network failure, it is typically due to shifts in the real-world data (data drift) or concept drift.
Box 2: Endpoint latency metrics Response times increase during peak usage.
Use Endpoint latency metrics to investigate the increased response times during peak usage.
Box 3: Prediction output distribution trends Predictions remain accurate, but business outcomes worsen.
The most appropriate metric to look into further is Prediction output distribution trends (often analyzed as part of data drift or concept drift).
When predictions remain statistically accurate (e.g., low mean squared error against actual target variables) but business outcomes degrade, it usually indicates a shift in the real-world environment.
Monitoring the distribution of prediction outputs over time helps identify downstream concept drift or changes in user behavior.
For example, if a pricing model accurately predicts lower market values during a recession, the model is "accurate," but the business loses revenue because the market itself has shifted.

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108) Hotspot Question
You manage an Azure OpenAI deployment of the gpt-4o base model.
You plan to fine-tune the deployed model.
You need to prepare a file that contains training data.
Which keys should you include in each line of the training data file? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Training data keys

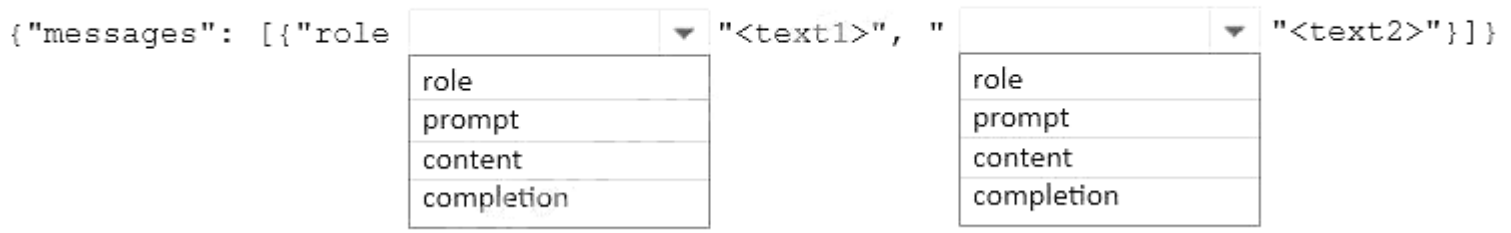


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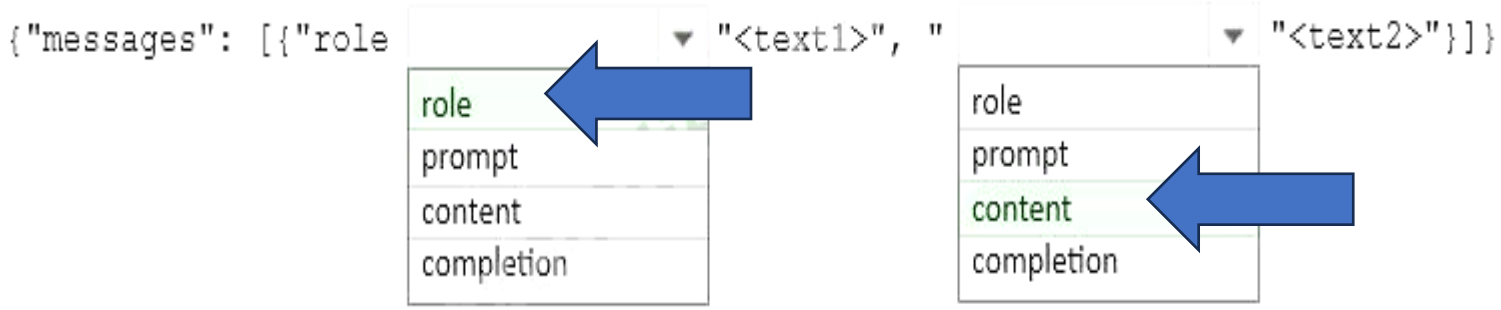
Training data keys



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Answer Area

Training data keys



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Explanation:
For fine-tuning conversational models like gpt-4o on Azure OpenAI, the training file must be formatted in a JSON Lines (.jsonl) file, where each line represents a single training conversation.
Each line must strictly contain a messages key wrapping an array of message objects.
Box 1: role
role: Defines the sender of the message. The accepted strings for fine-tuning are "system", "user", or "assistant".
Box 2: content
content: Contains the actual text string or multi-modal payload of the message.
Reference:
https://learn.microsoft.com/en-us/azure/foundry/openai/how-to/reinforcement-fine-tuning

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109) You have a Microsoft Foundry project and a CSV file stored in Azure Blob Storage.
You plan to add the CSV file as the grounding data to the project for RAG by using the file data type.
You need to specify the URL schema to designate the blob location.
Which URL schema should you specify?

A. hdfs

B. nfs

C. file

D. wasbs

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Which URL schema should you specify?

A. hdfs

B. nfs

C. file

D. wasbs

Answer: D

wasbs (Windows Azure Storage Blob Secure) is the officially supported URL scheme used by the underlying Azure Machine Learning and Microsoft Foundry data infrastructure to securely access and locate storage files directly from an Azure Blob Storage account. A typical URI format looks like:
wasbs://<containername>@<accountname>.blob.core.windows.net/<path_to_data>.csv.
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-create-data-assets

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110) Hotspot Question
You use Azure Machine Learning to implement hyperparameter tuning with a Bandit early termination policy for an Azure ML Python SDK v2-based model training.
The policy uses a slack_factor set to 0.1, an evaluation interval set to 1, and an evaluation delay set to 5.
You need to evaluate the outcome of the early termination policy.
What should you evaluate? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Scenario	Value
Percentage of worst performing runs to be terminated	1 percent 91 percent 99 percent
Run termination interval	Every interval when metrics are reported, starting at evaluation interval 5. Every 5th interval when metrics are reported. Every 6th interval when metrics are reported.

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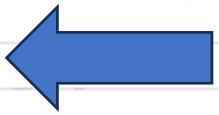
Answer Area

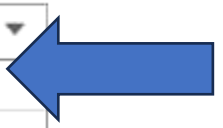
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Scenario	Value
Percentage of worst performing runs to be terminated	1 percent 91 percent 99 percent
Run termination interval	Every interval when metrics are reported, starting at evaluation interval 5. Every 5th interval when metrics are reported. Every 6th interval when metrics are reported.





Explanation:
Box 1: 91%
Box 2: Every interval when metrics are reported, starting at evaluation interval 5 Every interval when metrics are reported, starting at evaluation interval 5 is the best description for the run termination schedule in this scenario.
evaluation_interval = 1: The policy checks for potential termination every time the training script logs the primary metric.
evaluation_delay = 5: Policy evaluation is suspended for the first 5 intervals to prevent the premature termination of training runs before they have time to stabilize.
Combined Behavior: Once the evaluation_delay threshold of 5 is met, the policy applies at every subsequent multiple of the evaluation_interval (Intervals 5, 6, 7, etc.).
Reference:
https://azure.github.io/azureml-sdk-for-r/reference/bandit_policy.html

111) You need to refine a GPT-5 model so that its performance and behavior align with the technical and business requirements of Fabrikam Inc.
Which two Foundry strategies should you apply? Each correct answer presents a complete solution. Choose two.
NOTE: Each correct selection is worth one point.

A. Supervised fine-tuning

B. Evaluations

C. Guardrails

D. Synthetic data generation

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Answer: A,B

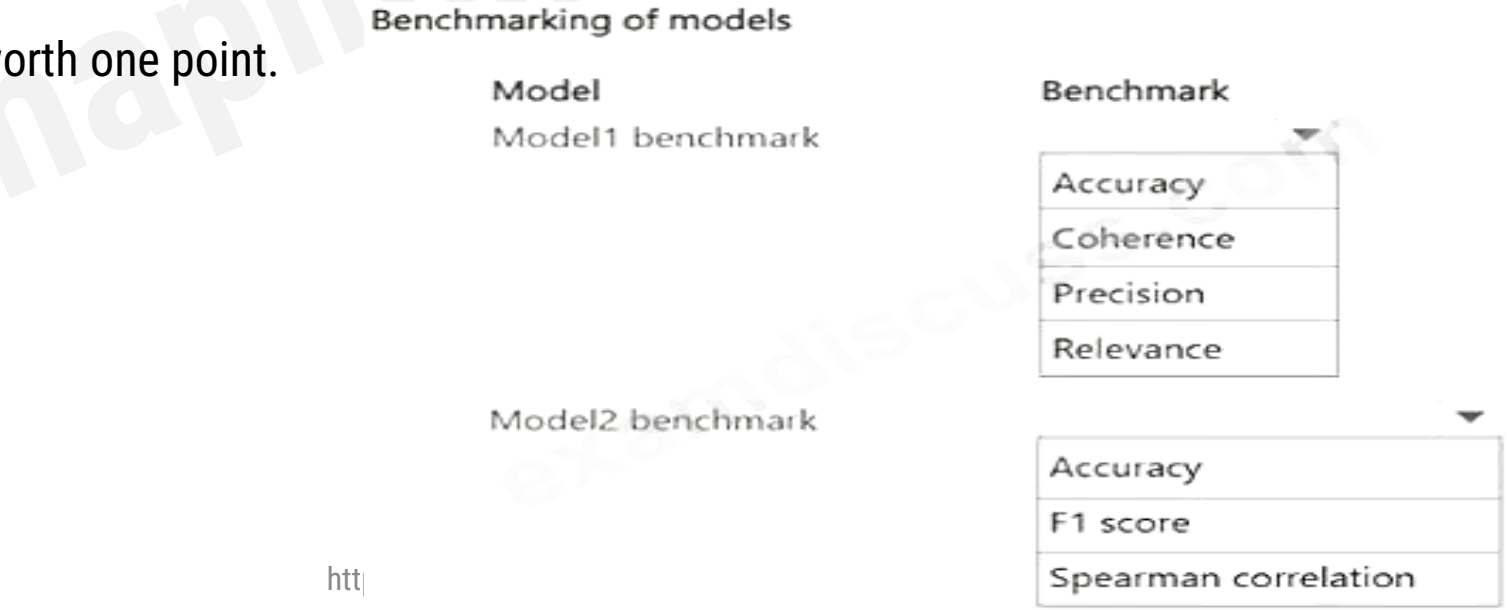
Scenario, Technical Requirements:
Improve RAG-based solution output quality.
Apply advanced fine-tuning techniques only when prompt engineering is insufficient Business Requirements:
Provide a conversational interface that answers analytics questions by using internal documents and datasets.
The two Foundry strategies that would be most useful in this scenario are Evaluations and Supervised fine-tuning.
Evaluations: This strategy is critical for improving RAG-based solution output quality. It provides systemic measurement to test how changes to your data, prompts, or retrieval chunks impact the accuracy, relevance, and groundedness of the model's answers before moving to more complex methods.

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Supervised fine-tuning: This strategy directly satisfies the requirement to apply advanced fine- tuning techniques only when prompt engineering is insufficient. It allows you to deeply customize the model's behavior and conversational tone to fit your specific analytics interface requirements.
Incorrect:
Guardrails: While valuable for safety and compliance, guardrails focus on enforcing hard operational boundaries (e.g., blocking toxic content or preventing data leaks). They do not optimize RAG output quality or provide advanced behavioral updates when prompt engineering fails.
Synthetic data generation: While helpful for bootstrapping training datasets when real-world data is scarce, it is an data-preparation step rather than a core model refinement strategy designed to iteratively solve prompt engineering limitations or build conversational systems.
Reference:
https://medium.com/@hugoparreao/context-engineering-what-really-improves-the-performance-of-llms-0c0e3ed45c98

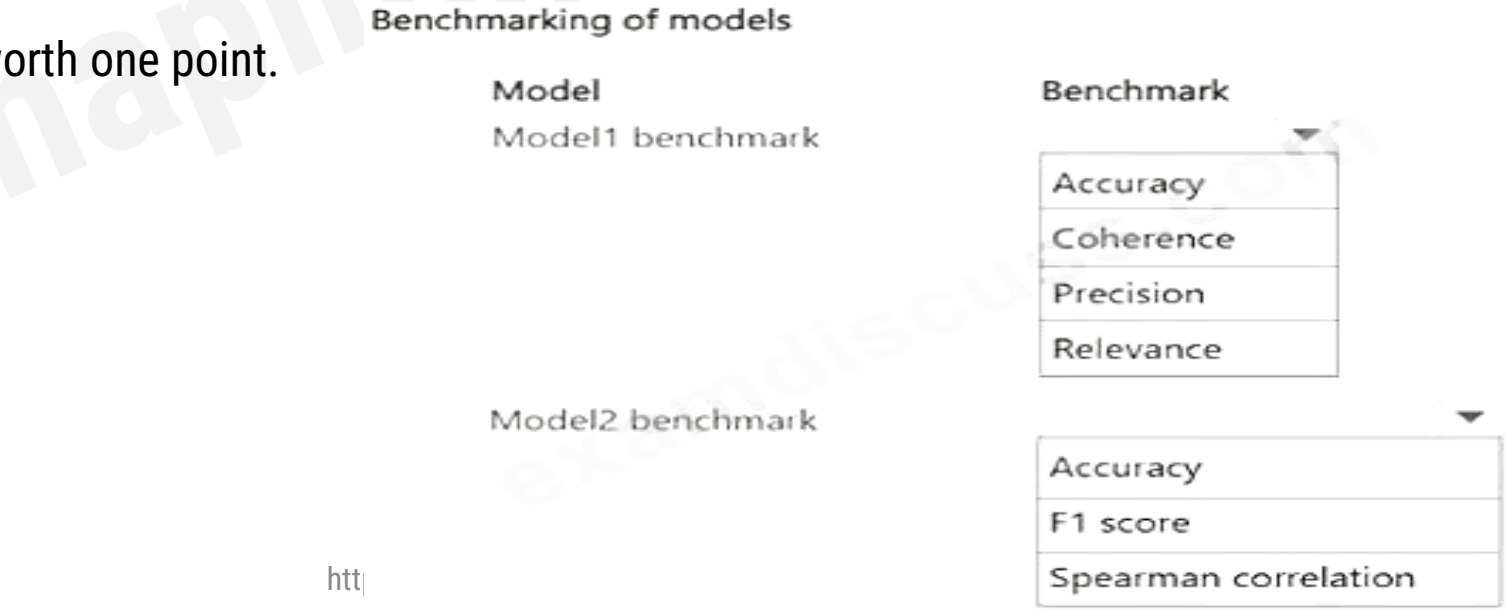
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112) Hotspot Question
You manage a Microsoft Foundry project.
You plan to build a RAG solution.
The solution must include two models:
- One for text output, named Model1. This model must resemble human language and read naturally.
- One for creating embeddings, named Model2. This model must maximize the retrieval of relevant results (high recall) while minimizing irrelevant or incorrect matches (high precision).
You need to compare different models by using benchmarking metrics to select the appropriate models for Model1 and Model2.
Which benchmarking metric should you select for each model? To answer, select the appropriate options in the answer area.
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Answer:

Benchmarking of models

Model	Benchmark
Model1 benchmark	Accuracy Coherence Precision Relevance
Model2 benchmark	Accuracy F1 score Spearman correlation

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113) Hotspot Question
You have an Azure Machine Learning workspace.
You plan to set up logging and tracking experiments by using MLflow Tracking.
You need to log the accuracy as a numerical value and the training loss as a plot.
How should you complete the commands? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Logging experiments with MLflow

```
import mlflow
from matplotlib import pyplot as plt

mlflow.set_experiment("mlflow-experiment")

plt.savefig('plot.png')

num1 = 82
img1 = "plot.png"

with mlflow.start_run() as run:
    mlflow.
```

- log_artifact("log", num1)
- log_metric("log", num1)
- log_metrics("log", num1)
- log_params("log", num1)

```
    mlflow.
```

- log_artifact(img1)
- log_artifacts(img1)
- log_figure(fig, img1)
- log_image(img, img1)

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Answer:

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- log_metric("log", num1) ← **Answer**
- log_metrics("log", num1)
- log_params("log", num1)

```
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```

- log_artifact(img1) ← **Answer**
- log_artifacts(img1)
- log_figure(fig, img1)
- log_image(img, img1)

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Explanation:
Box 1: log_metric("log", num1)
Log Numerical Metrics
To track a single numerical value that changes over time (like accuracy or evaluation scores), you use the mlflow.log_metric() function. This enables Azure Machine Learning to plot the value on a performance graph automatically.
Box 2: log_artifact(img1)
Log Plots and FilesTo save external files, images, or plots generated during training, you use the mlflow.log_artifact() function. This uploads the specified file (in this case, your saved Matplotlib image) to the run's artifact storage.
Reference:
https://levelup.gitconnected.com/mlflow-made-easy-your-beginners-guide-bf63f8fed915

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114) You manage an Azure Machine Learning workspace. You have a folder that contains a CSV file.
The folder is registered as a folder data asset.
You plan to use the folder data asset for data wrangling during interactive development.
You need to access and load the folder data asset into a Pandas data frame.
Which method should you use to achieve this goal?

A. mltable.load()

B. mltable.from_delimited_files()

C. mltable.from_parquet_files()

D. mltable.from_delta_lake()

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Answer: B

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Azure Machine Learning supports a Table type (mltable). This allows for the creation of a blueprint that defines how to load data files into memory as a Pandas or Spark data frame.
Authoring MLTable Files
To directly create the MLTable file, we recommend that you use the mltable Python SDK to author your MLTable files instead of a text editor.
Supported file types
You can create an MLTable using a range of different file types:
* Delimited Text
(for example, CSV files)
from_delimited_files(paths=[path])
Incorrect:
* Parquet
from_parquet_files(paths=[path])
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-mltable

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115) You are preparing training data for a fine-tuning job in Microsoft Foundry.
Real production conversations cannot be used due to compliance requirements.
You need to generate synthetic interaction data that can be used for fine-tuning a generative model.
What should you do?

A. Export model evaluation logs and use them directly as training data.

B. Use a simulator to generate prompt-response interaction data that matches the target task.

C. Enable A/B testing and capture live user traffic for data generation.

D. Run a simulator to produce telemetry logs and trace data from user interactions.

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Answer: B

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You can use a simulator or an LLM-as-a-judge pipeline to generate synthetic interaction data for fine-tuning. This technique is standard practice for maintaining strict data privacy while training models on specific business tasks.

Here is how to effectively structure and execute a synthetic data generation pipeline for Microsoft Azure AI Foundry (formerly Azure AI Studio).

Generation Methods

Persona-Driven Simulation: Prompt one LLM to act as a customer and another as your support agent to generate multi-turn dialogues.

Seed Data Expansion: Feed 10-20 hand-written, compliant examples into an LLM and instruct it to generate hundreds of diverse variations.

Schema-Based Evolution: Define variables (e.g., product types, user intents, sentiment levels) and programmatically combine them into prompt templates for an LLM to flesh out.

Reference:

https://www.digitaldividedata.com/blog/synthetic-data-generation-in-gen-ai

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116) A data science team plans to evaluate multiple hyperparameter values automatically while training a model in Azure Machine Learning.
The tuning process must run multiple training trials without manually modifying the training script for each run.
You need to automate hyperparameter tuning for the training job.
What should you do?

A. Manually change hyperparameter values between training runs.

B. Create a tuning job that runs multiple trials with different parameter values.

C. Duplicate the training script for each parameter combination.

D. Adjust hyperparameters after model deployment.

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Answer: B

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Correct:
* Create a tuning job that runs multiple trials with different parameter values To best automate hyperparameter tuning in Azure Machine Learning, you should create a Sweep Job (in SDK v2) or a HyperDrive experiment (in SDK v1).
This creates a tuning job that automatically launches multiple training trials (child runs) using your single base training script without requiring manual code modifications for each run.
Key Components to Automate the Job
To set up this job successfully using the Azure Machine Learning Python SDK v2, you will define:
Parameterized Training Script: Write your code to accept hyperparameters as command-line arguments (e.g., using Python's argparse), allowing the tuning job to pass different values to each trial.
Search Space: Define the range or specific discrete/continuous choices for the values you want to test (e.g., learning rates, batch sizes).Sampling Algorithm: Choose how Azure ML should navigate your search space. Options include Random sampling, Grid sampling (testing every possible combination), or Bayesian sampling (using previous trial results to pick the next best values).Primary Metric: Specify the performance metric your script logs (like accuracy or loss) so Azure ML knows which target to optimize.
Early Termination Policy: Optional policy (like a Bandit Policy) to automatically cancel poorly performing trials early, saving you compute time and cost.

Incorrect:
* Adjust hyperparameters after model deployment.
* Duplicate the training script for each parameter combination.
* Manually change hyperparameter values between training runs.
* Run a single training job with fixed hyperparameters.
* Select hyperparameters based only on default model settings.

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You need to evaluate whether a proposed change improves RAG answer quality by using supported and measurable techniques.
Solution: Compare embedding vector dimensions used by the retrieval pipeline before and after the change.
Does the solution meet the goal?

A. Yes

B. No

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Does the solution meet the goal?

A. Yes

B. No

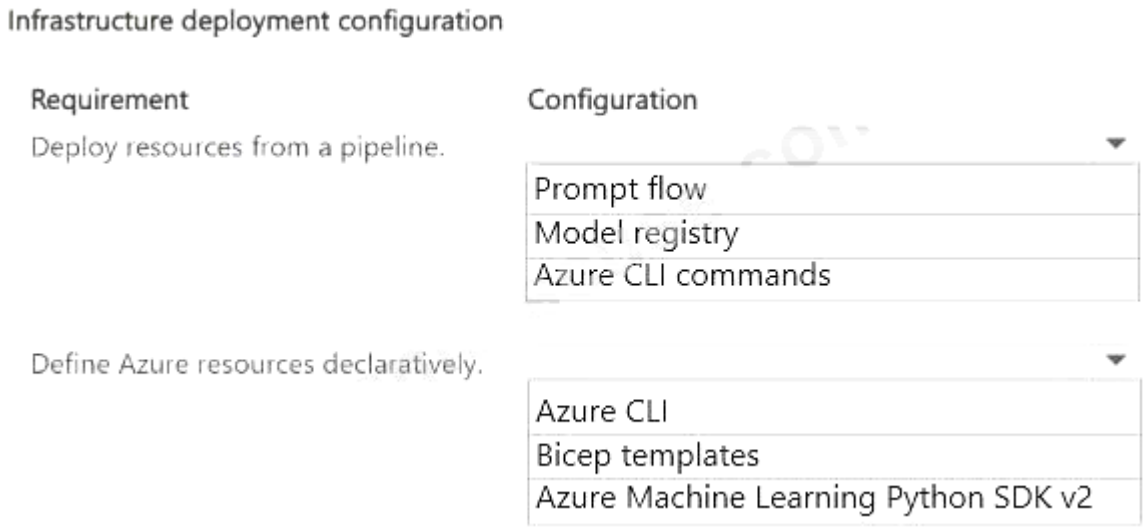
Answer: B

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Correct:
* Review user feedback comments collected after deployment to determine whether answers appear more accurate.
Relying solely on user feedback comments is not the proper immediate action for evaluating this specific issue but it is still the best choice.
Evaluate whether your proposed change improves the Retrieval-Augmented Generation (RAG) system by reducing hallucinations (unsupported information), you need to measure faithfulness and context relevance using automated, quantifiable metrics.
The proper course of action is to implement an LLM-as-a-Judge framework using an open-source evaluation library like Ragas or TruLens.
Recommended Evaluation Plan
Establish a baseline: Run your current RAG pipeline through a test dataset of 50-100 representative user queries.
Capture the outputs: Save the user query, the exact retrieved document snippets, and the generated response for every test.
Apply the change: Deploy your proposed modification (e.g., altered prompt, different temperature, or re-ranking algorithm).Run the evaluation: Pass the test dataset through the updated pipeline to generate a new set of responses.
Compare the metrics: Use the framework to score both sets of data and mathematically verify if the change reduced unsupported claims.
Incorrect:
* Compare embedding vector dimensions used by the retrieval pipeline before and after the change.
Comparing embedding vector dimensions is not a valid or effective method for measuring RAG answer quality. Vector dimensions (e.g., 1536 or 3072) are static architectural properties of your embedding model. They do not reflect factual accuracy, semantic grounding, or the rate of hallucinations in your text generation.
* Measure token throughput and average response latency before and after applying the proposed change.
Measuring token throughput and latency is not the correct action to solve this specific problem.

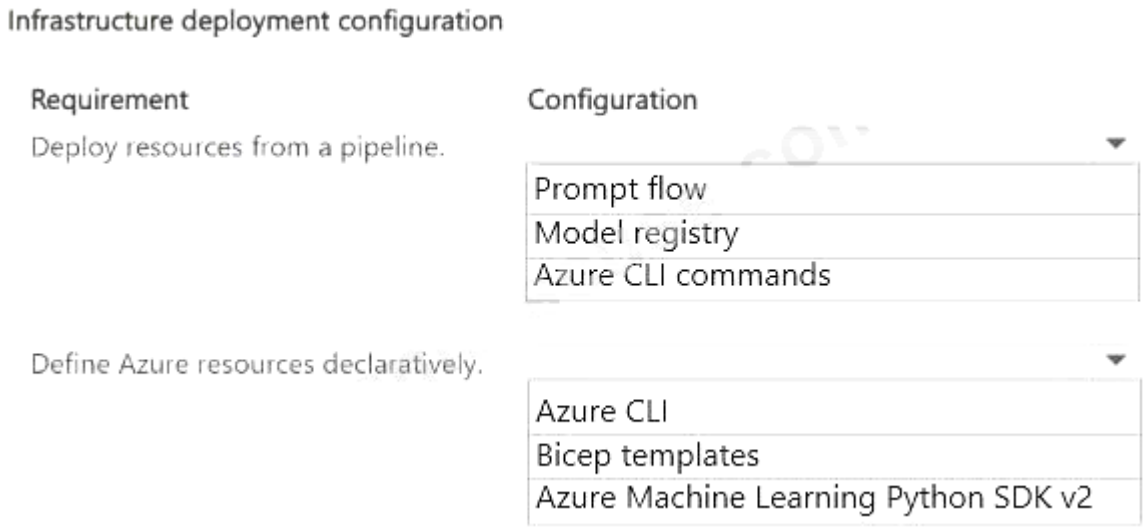
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118) Hotspot Question
A team is standardizing MLOps practices by using automated deployments.
The team requires infrastructure to be defined declaratively and deployed through automation pipelines.
You need to configure infrastructure deployment.
What should you configure for each requirement? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



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Infrastructure deployment configuration

Requirement	Configuration
Deploy resources from a pipeline.	Prompt flow Model registry Azure CLI commands
Define Azure resources declaratively.	Azure CLI Bicep templates Azure Machine Learning Python SDK v2

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Explanation:
Box 1: Model Registry
Deploy resources from a pipeline.
To standardize MLOps practices with declarative infrastructure and automated pipelines, the best choice is to use a Model Registry via Infrastructure as Code (IaC) for your core setup, supplemented by Azure CLI inside your pipelines for operational tasks. Prompt Flow is an application development tool and should not be used to deploy infrastructure.
A Model Registry (like Azure Machine Learning registry or MLflow) is the standard architectural pattern for managing ML artifacts declaratively.
Declarative Tracking: It stores model versions, lineages, and environments as code-like configurations.
Pipeline Integration: CI/CD pipelines can natively fetch from or push to a registry using configuration files.
Environment Isolation: It allows you to promote the exact same model artifact across Dev, Staging, and Production environments without rebuilding.

Box 2: Bicep templates
Define Azure resources declaratively.
Bicep templates are an excellent choice for defining Azure resources declaratively in an automated MLOps pipeline.
Native Azure Integration: Bicep supports all Azure Machine Learning (Azure ML) resources immediately upon release.
Declarative Syntax: You define the desired end-state of your infrastructure without writing complex deployment scripts.
No State Management: Azure manages the state automatically, unlike Terraform which requires a remote state file.
Tooling Support: Deep integration with Azure Pipelines and GitHub Actions allows easy deployment validation.

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119) Hotspot Question
You download a .csv file from a notebook in an Azure Machine Learning workspace to a data/sample.csv folder on a compute instance. The file contains 10,000 records.
You must generate the summary statistics for the data in the file. The statistics must include the following for each numerical column:
- number of non-empty values
- average value
- standard deviation
- minimum and maximum values
- 25th, 50th, and 75th percentiles
You need to complete the Python code that will generate the summary statistics.
Which code segments should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
import [ ▼ ] as lib
```
- csv
- numpy
- pandas

```
datasample = lib.read csv(data/sample.csv)
datasample.[ ▼ ]()
```
- describe
- query
- rank

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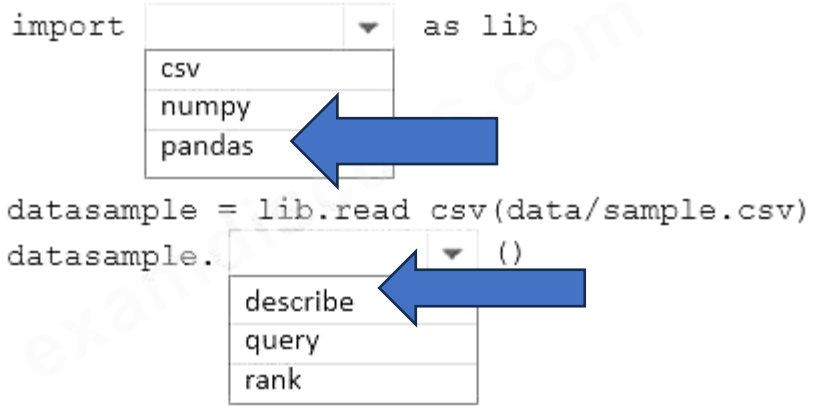
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Answer:

Answer Area



Explanation:
Box 1: pandas
Pandas read_csv() - How to read a csv file in Python
Most of the data is available in a tabular format of CSV files. It is very popular. You can convert them to a pandas DataFrame using the read_csv function. The pandas.read_csv is used to load a CSV file as a pandas dataframe.
Step 1: pandas
import pandas as pd
Step 2: Read the CSV
Read the csv file
df = pd.read_csv("data1.csv")

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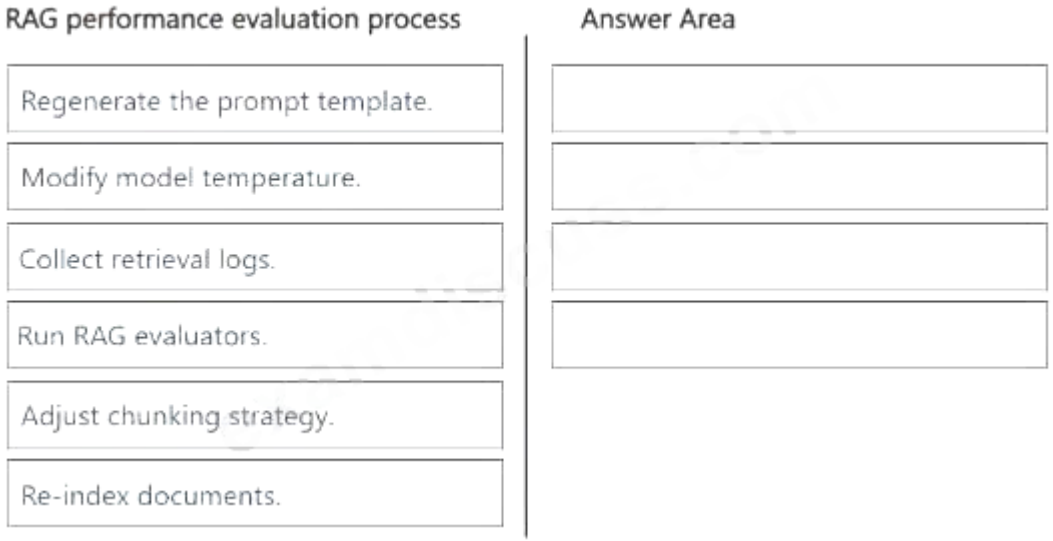
Box 2: describe
The pandas.describe function is used to get a descriptive statistics summary of a given dataframe. This includes mean, count, std deviation, percentiles, and min-max values of all the features.
Pandas Describe function on DataFrame
On applying pandas describe function to a dataframe, the result is also returned as a dataframe .
This dataframe will consist of a statistics summary for all the numeric features of the dataframe.

```
# Create a dataframe
df = pd.DataFrame({
'Subject_1_Marks': [14, 42, 21, 12, 45],
'Subject_2_Marks': [32, 43, 23, 50, 21],
'Subject_3_Marks': [45.0, 34.0, 23.0, 8.0, 21.0],
'Names': ['Saksham', 'Ayushi', 'Abhishek', 'Saksham', 'Saumya']
}
)
# Apply describe function
df.describe()
```

Incorrect:
* rank
Reference:
https://www.machinelearningplus.com/pandas/pandas-read_csv-completed/
https://www.machinelearningplus.com/pandas/pandas-describe/

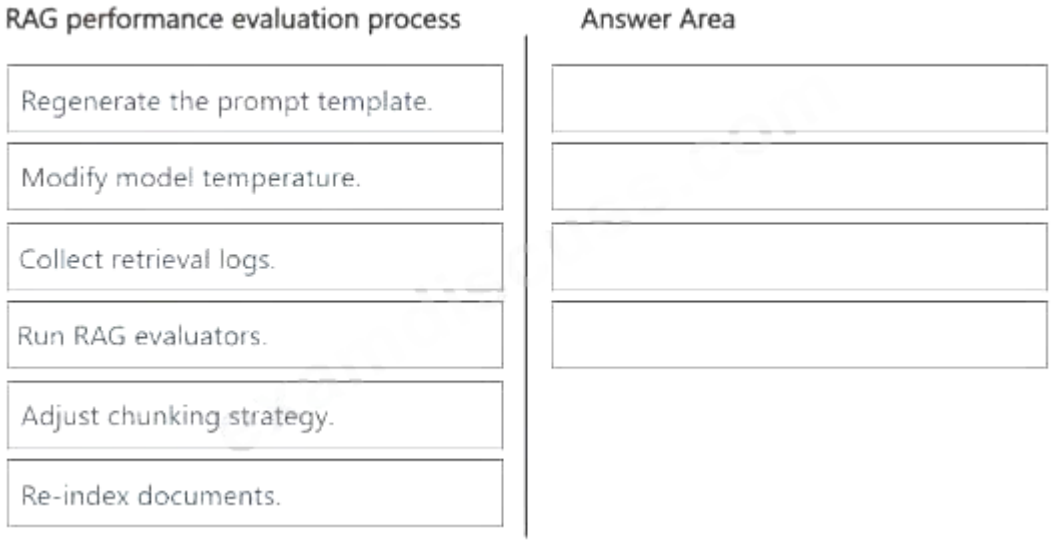
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120) Drag and Drop Question
A team is developing a Retrieval-Augmented Generation (RAG) system.
The team requires improvements to the system's retrieval quality to ensure accurate, grounded responses.
You need to assess RAG performance before you can suggest an improvement strategy.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



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Answer:

RAG performance evaluation process	Answer Area
Regenerate the prompt template.	Run RAG evaluators.
Modify model temperature.	Collect retrieval logs.
Collect retrieval logs.	Modify model temperature.
Run RAG evaluators.	Regenerate the prompt template.
Adjust chunking strategy.	
Re-index documents.	

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Explanation:
To properly assess your Azure Retrieval-Augmented Generation (RAG) system's performance before implementing an improvement strategy, you should include the following four steps: Run RAG evaluators, Collect retrieval logs, Modify model temperature, and Regenerate the prompt template.
Step 1: Run RAG evaluators
Run RAG evaluators is the primary method to objectively measure system performance.
Evaluators calculate data-driven metrics like groundedness, relevance, and retrieval precision using tools like Azure AI Studio Evaluators.
Step 2: Collect retrieval logs
Collect retrieval logs provides the raw operational data needed for assessment. Analyzing these logs helps you identify exactly which documents were retrieved, their relevance scores, and where the retrieval pipeline failed to fetch the correct context.
Step 3: Modify model temperature
Modify model temperature: Adjusting the temperature during assessment helps isolate whether poor responses are caused by bad retrieval or by the LLM being too creative (high temperature) or too rigid (low temperature). Testing variations helps establish a performance baseline.

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121) Hotspot Question
You manage an Azure Machine Learning workspace by using the Python SDK v2.
You must create a compute cluster in the workspace. The compute cluster must run workloads and properly handle interruptions. You start by calculating the maximum amount of compute resources required by the workloads and size the cluster to match the calculations.
The cluster definition includes the following properties and values:
- name="mlcluster1"
- size="STANDARD_DS3_v2"
- min_instances=1
- max_instances=4
- tier="dedicated"
The cost of the compute resources must be minimized when a workload is active or idle. Cluster property changes must not affect the maximum amount of compute resources available to the workloads run on the cluster.
You need to modify the cluster properties to minimize the cost of compute resources.
Which properties should you modify? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Workload status	Property
active	size / tier / max_instances
idle	size / min_instances / max_instances

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Answer Area

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active	size / tier / max_instances
idle	size / min_instances / max_instances

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Answer:

Answer Area

Workload status	Property
active	size **tier** max_instances
idle	size **min_instances** max_instances

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Explanation:
Box 1: tier
Set tier to low_priority.
Lower your compute cluster cost with low priority VMs
Use any of these ways to specify a low-priority VM:

```
from azure.ai.ml.entities import AmlCompute
cluster_low_pri = AmlCompute(
name="low-pri-example",
size="STANDARD_DS3_v2",
min_instances=0,
max_instances=2,
idle_time_before_scale_down=120,
tier="low_priority",
)
ml_client.begin_create_or_update(cluster_low_pri).result()
```

Note: You may also to use low-priority VMs to run some or all of your workloads. These VMs don't have guaranteed availability and may be preempted while in use. You'll have to restart a preempted job.
Using Azure Low Priority Virtual Machines allows you to take advantage of Azure's unused capacity at a significant cost savings. At any point in time when Azure needs the capacity back, the Azure infrastructure will evict Azure Low Priority Virtual Machines. Therefore, Azure Low Priority Virtual Machines are great for workloads that can handle interruptions. The amount of available capacity can vary based on size, region, time of day, and more. When deploying Azure Low Priority Virtual Machines, Azure will allocate the VMs if there's capacity available, but there's no SLA for these VMs. An Azure Low Priority Virtual Machine offers no high availability guarantees. At any point in time when Azure needs the capacity back, the Azure infrastructure will evict Azure Low Priority Virtual Machines.

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122) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
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You need to evaluate whether a proposed change improves RAG answer quality by using supported and measurable techniques.
Solution: Review user feedback comments collected after deployment to determine whether answers appear more accurate.
Does the solution meet the goal?

A. Yes

B. No

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Solution: Review user feedback comments collected after deployment to determine whether answers appear more accurate.
Does the solution meet the goal?

A. Yes

B. No

Answer: A

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Correct:
* Review user feedback comments collected after deployment to determine whether answers appear more accurate.
Relying solely on user feedback comments is not the proper immediate action for evaluating this specific issue but it is still the best choice.
Evaluate whether your proposed change improves the Retrieval-Augmented Generation (RAG) system by reducing hallucinations (unsupported information), you need to measure faithfulness and context relevance using automated, quantifiable metrics.
The proper course of action is to implement an LLM-as-a-Judge framework using an open-source evaluation library like Ragas or TruLens.
Recommended Evaluation Plan
Establish a baseline: Run your current RAG pipeline through a test dataset of 50-100 representative user queries.
Capture the outputs: Save the user query, the exact retrieved document snippets, and the generated response for every test.
Apply the change: Deploy your proposed modification (e.g., altered prompt, different temperature, or re-ranking algorithm).Run the evaluation: Pass the test dataset through the updated pipeline to generate a new set of responses.
Compare the metrics: Use the framework to score both sets of data and mathematically verify if the change reduced unsupported claims.

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123) An organization operates a customer-facing generative AI chat service deployed by using Microsoft Foundry. The service processes a predictable, sustained volume of requests. The service must meet strict response time service-level agreements (SLAs) during peak business hours.
The organization requires that:
- Model responses remain consistent during sustained high traffic.
- Latency does not degrade during peak usage periods.
- Capacity planning avoids throttling and unpredictable performance.
You need to ensure that the deployed foundation model can reliably handle sustained, high- volume traffic while meeting performance and availability requirements.
What should you do?

A. Implement spillover traffic management for excess demand.

B. Increase batch size for inference requests.

C. Optimize prompts to reduce token usage.

D. Deploy the model by using serverless API endpoints.

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B. Increase batch size for inference requests.

C. Optimize prompts to reduce token usage.

D. Deploy the model by using serverless API endpoints.

Answer: A

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Implement spillover traffic management for excess demand is the best action among the provided options to ensure the system gracefully maintains availability and handles sustained, high-volume traffic without degradation or unpredictable throttling under strict SLAs.

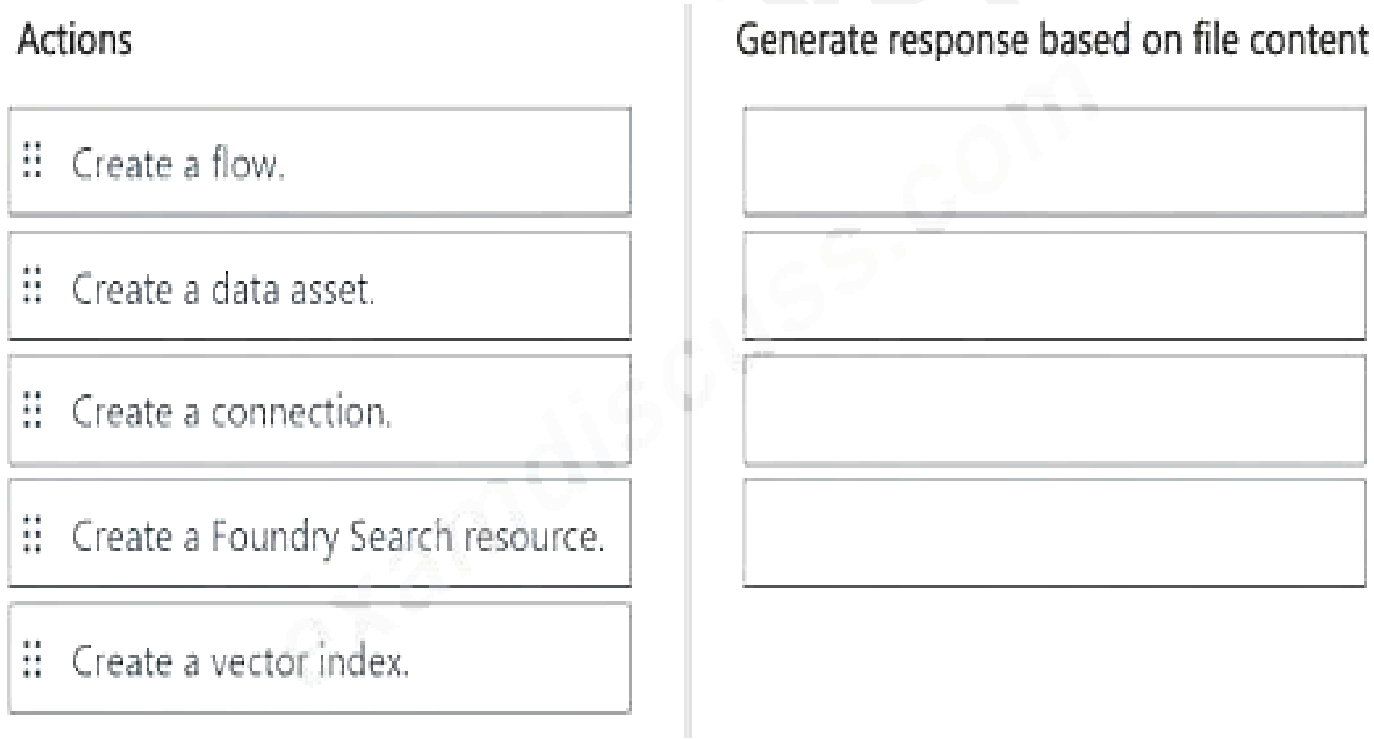
Under strict SLAs and predictable, sustained peak volumes, Azure AI Foundry utilizes provisioned throughput (PTUs) to guarantee latency and throughput. However, if traffic spikes beyond predictable levels, a spillover strategy redirects excess demand to standard pay-as-you- go slots or alternative regions. This prevents the primary deployment from throttling, dropping requests, or suffering from latency degradation.

Reference:

https://smartbridge.com/azure-openai-service-guide/

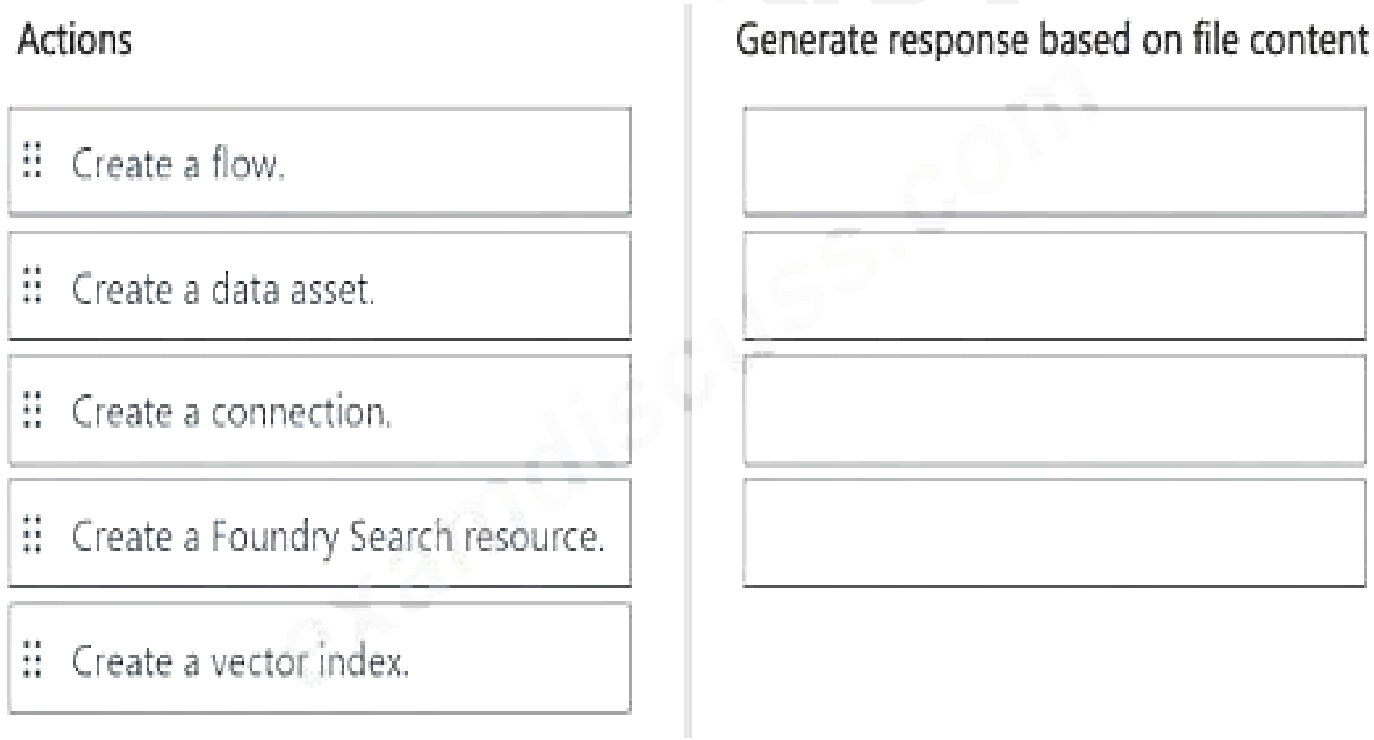
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124) Drag and Drop Question
You have a Microsoft Foundry project with a connected Azure OpenAI Service model.
You have a set of text files stored locally on your computer.
You must set up a flow that will generate responses based on the content of your local files.
You need to implement a solution.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



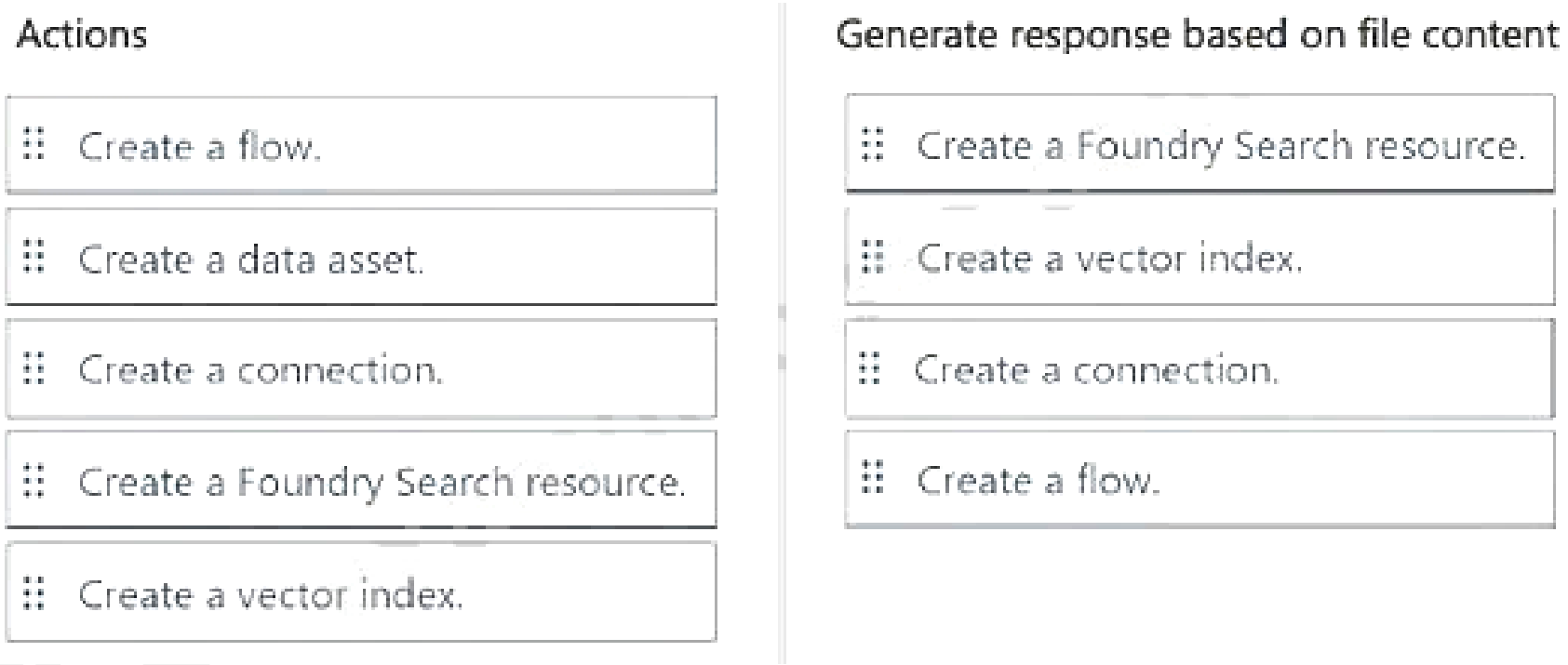
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Answer:



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125) Hotspot Question

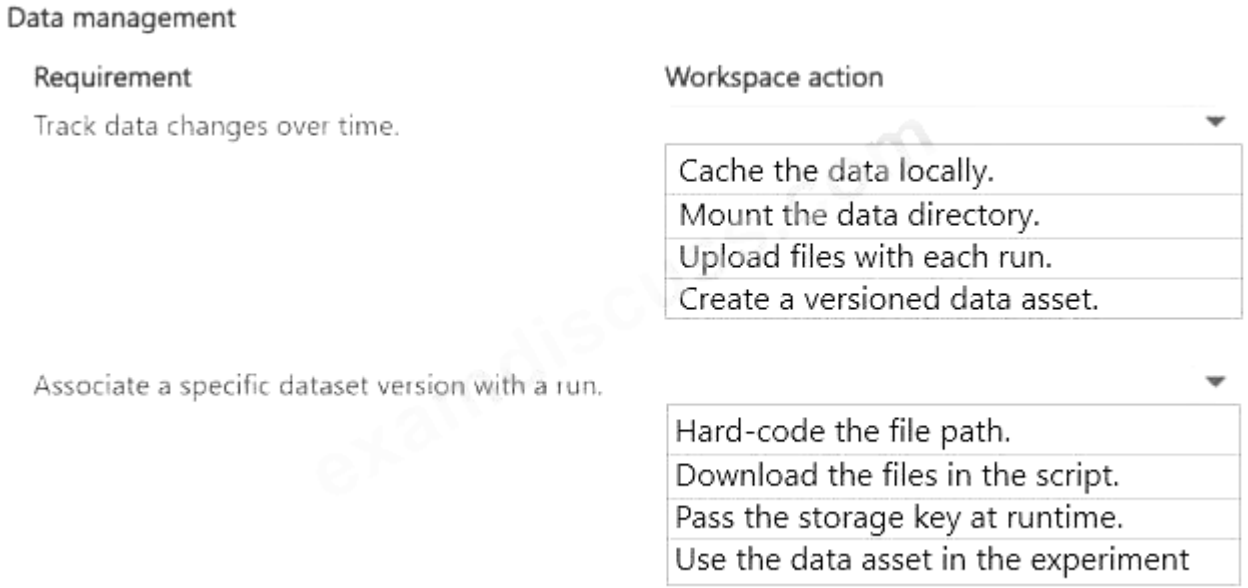
A team retrains a machine learning model on a weekly basis by using updated training data.

The team must be able to rerun any previous experiment by using the exact data that was available at the time it was originally run. The solution must preserve historical versions of the data without duplicating training scripts.

You need to manage the data so that experiments can be reproduced reliably.

Which workspace actions should you perform? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



125) Hotspot Question

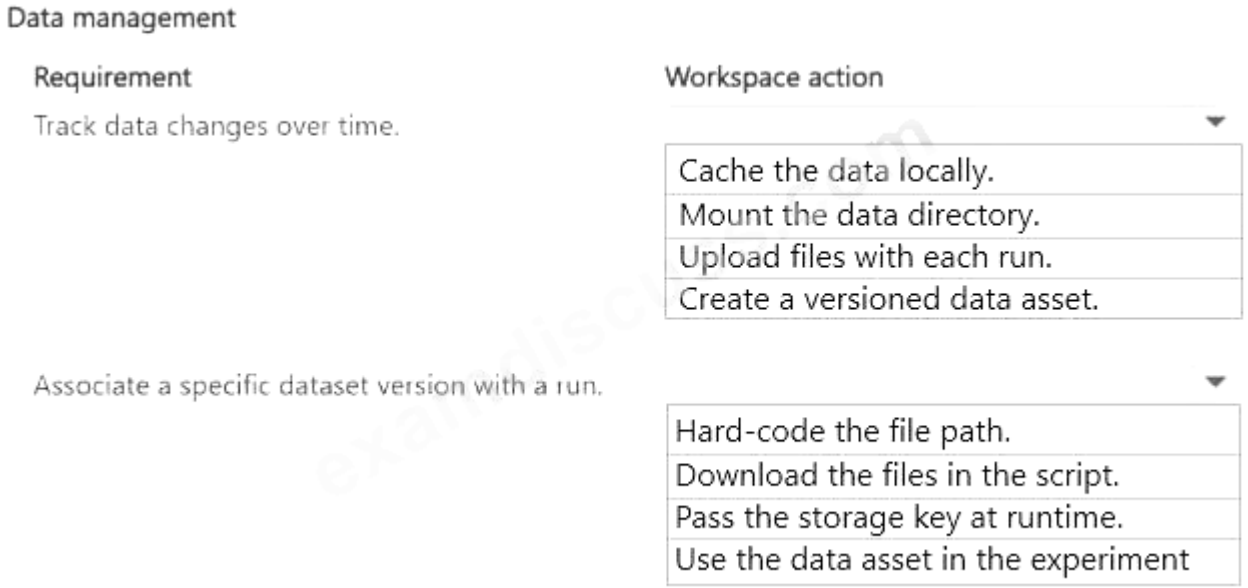
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answers

Requirement	Workspace action
Track data changes over time	✅ Create a versioned data asset.
Associate a specific dataset version with a run	✅ Use the data asset in the experiment.

Why

- **Create a versioned data asset** — Azure Machine Learning data assets support **versioning**, allowing each weekly update to be stored as a distinct version without duplicating the training scripts.
- **Use the data asset in the experiment** — When the experiment references a specific data asset version, the run can be reproduced using the **exact same dataset version** that was available at that time.

❌ Caching, mounting, uploading files each run, hard-coding paths, or passing storage keys do not provide reliable historical data versioning.

Final:

1. **Create a versioned data asset**
2. **Use the data asset in the experiment**

Explanation:
Box 1: Create a versioned dataset
Track data changes over time.
Creating a versioned dataset in your machine learning workspace configuration is the correct and industry-standard way to solve this.
Versioned datasets allow you to decouple your training scripts from your data state. This approach satisfies all your constraints without duplicating code or creating messy infrastructure.
Enables Reproducibility: Every time you run a training job, you log the specific dataset version used (e.g., v1, v2, v3). To rerun a historical experiment, you simply call that specific version identifier.
Prevents Code Duplication: Your training scripts remain identical. They accept a dataset version as an input parameter rather than pointing to hardcoded file paths.
Tracks Weekly Updates: When the weekly data refresh occurs, you register the new data as v+1 in the workspace registry, keeping a clean chronological ledger.
Box 2: Use the data asset in the experiment
Associate a specific dataset version with a run.
To best associate a specific dataset version with a run in the workspace, you should use the data asset in the experiment.
Automatic Lineage: Registering data as a versioned data asset automatically tracks which exact version was used in which specific run.

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126) You manage an Azure Machine Learning workspace. You build a model for which you must configure a Responsible Al dashboard.
Based on what you learn from the dashboard, you must perform the following activities:
- Determine what must be done to get a desirable outcome from the model.
- Identify the features that have the most direct effect on your outcome of interest.
You need to select the components to use for the Responsible Al dashboard configuration.
Which two components should you add? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. counterfactuals

B. error analysis

C. causal

D. explanation

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A. counterfactuals

B. error analysis

C. causal

D. explanation

Answer: A,C

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To meet your requirements, you should configure the Responsible AI (RAI) dashboard with the Counterfactuals (What-If) and Causal Analysis components.

1. Counterfactuals (What-If)

Purpose: This component helps you figure out how to change the model's output to a target result for specific instances.

Mechanism: It generates counterfactual examples (the closest possible data points) that yield a different, desirable outcome. For example, it can answer: "What is the minimum amount this user's income needs to increase for their loan application to be approved?"

2. Causal Analysis

Purpose: This component estimates the direct causal effect of specific "treatment" features on your ultimate outcome of interest.

Mechanism: Unlike standard feature importance (which only shows correlation), causal inference answers deep prescription questions. It separates pure correlation from true causation to tell you exactly how changing a real-world policy or feature directly moves your target metric.

Reference:

https://docs.azure.cn/en-us/machine-learning/concept-responsible-ai-dashboard?view=azureml-api-2

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127) You are a data scientist working for a hotel booking website company. You use the Azure Machine Learning service to train a model that identifies fraudulent transactions.
You must deploy the model to an Azure Machine Learning online endpoint by using the Azure Machine Learning Python SDK v2. The deployed model must return real-time predictions of fraud based on transaction data input.
You need to create the script that is specified as the scoring_script parameter for the CodeConfiguration class used to deploy the model.
What should the entry script do?

A. Create a Conda environment for the online endpoint compute and install the necessary Python packages.

B. Register the model with appropriate tags and properties.

C. Load the model and use it to predict labels from input data.

D. Start a node on the inference cluster where the model is deployed.

E. Specify the number of cores and the amount of memory required for the online endpoint compute.

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D. Start a node on the inference cluster where the model is deployed.

E. Specify the number of cores and the amount of memory required for the online endpoint compute.

Answer: C

The entry script (scoring script) for an Azure Machine Learning online endpoint must initialize the model when the container starts and process incoming transaction data to return real-time fraud predictions.
init() function: Runs once when the container is initialized. It must locate and load the trained model into memory (typically using a global variable) from the path specified by the AZUREML_MODEL_DIR environment variable.
run(data) function: Executes every time the endpoint receives a real-time HTTP request. It accepts the raw transaction payload, deserializes it, processes the features, passes them to the loaded model for prediction, and returns a JSON-serializable response.

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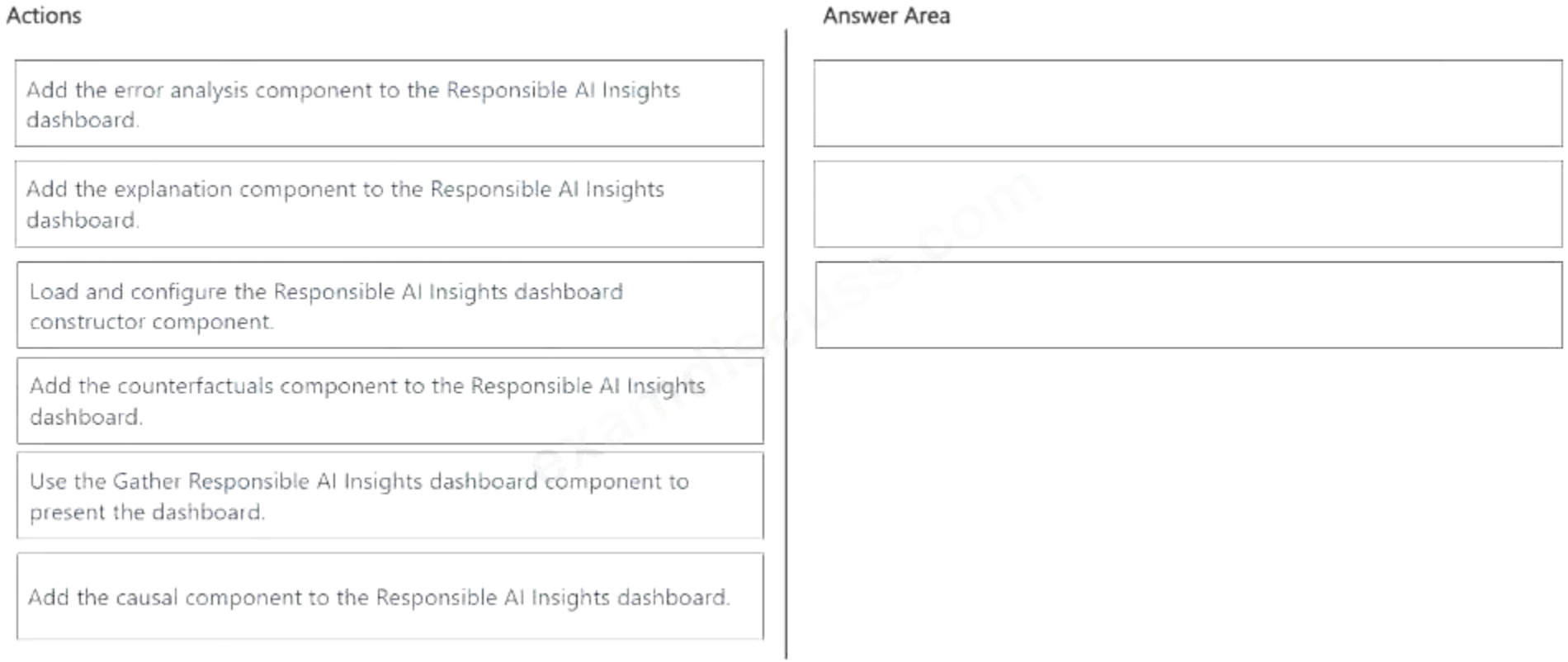
128) Drag and Drop Question

You manage an Azure Machine Learning workspace. You train a model named model1.

You must identify the features to modify for a differing model prediction result.

You need to configure the Responsible AI (RAI) dashboard for model1.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



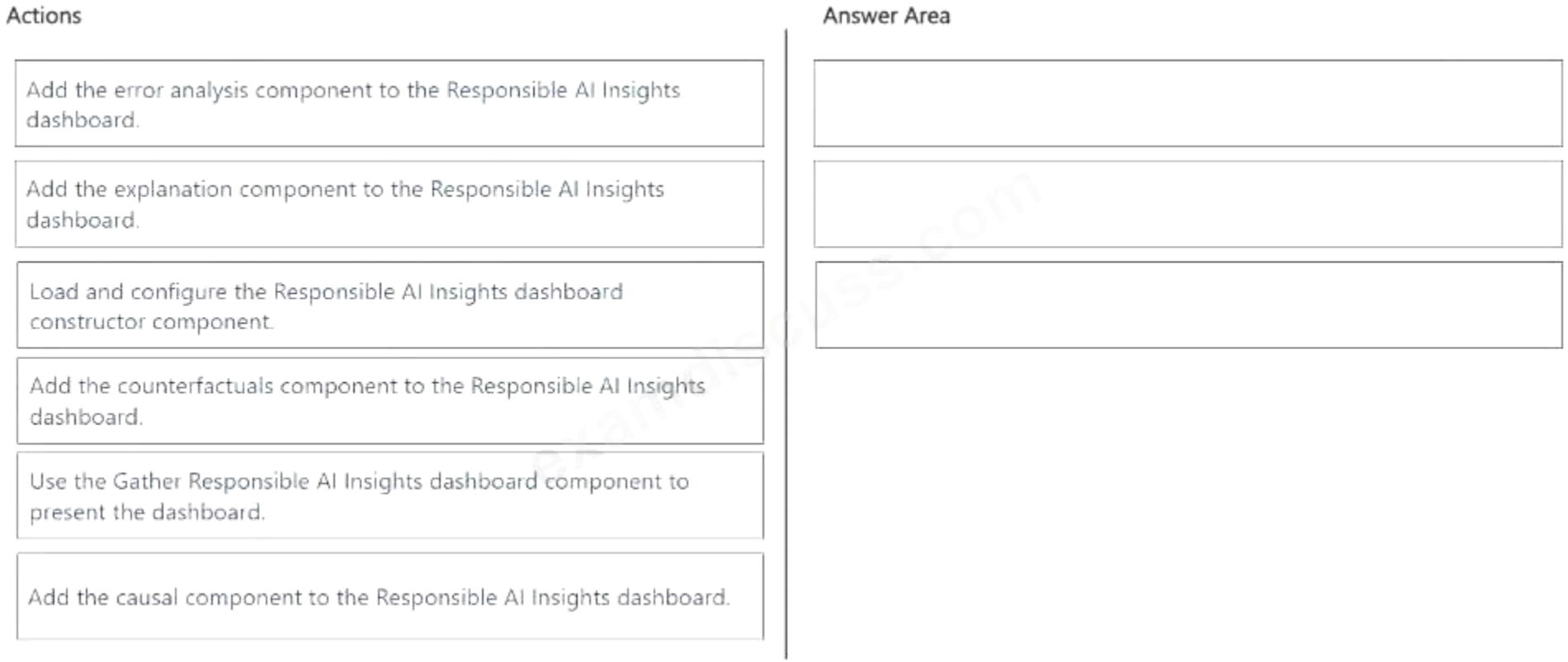
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Answer:

Actions	Answer Area
Add the error analysis component to the Responsible AI Insights dashboard.	Load and configure the Responsible AI Insights dashboard constructor component.
Add the explanation component to the Responsible AI Insights dashboard.	Add the counterfactuals component to the Responsible AI Insights dashboard.
Load and configure the Responsible AI Insights dashboard constructor component.	Use the Gather Responsible AI Insights dashboard component to present the dashboard.
Add the counterfactuals component to the Responsible AI Insights dashboard.	
Use the Gather Responsible AI Insights dashboard component to present the dashboard.	
Add the causal component to the Responsible AI Insights dashboard.	

Explanation:
To setup the Responsible AI (RAI) dashboard in Azure Machine Learning and specifically analyze minimal feature modifications needed to change a prediction result, you must use Counterfactual Analysis.
Here are the specific sequential steps you need to take:
Step 1: Load and configure the Responsible AI Insights dashboard constructor component.
Initialize the RAI Insights Dashboard Constructor
1. Create the root pipeline component that acts as the container for your tools.
2. Call the RAI Insights dashboard constructor component.
3. Pass your registered model and your test dataset as the mandatory inputs to this component.
Step 2: Add the Counterfactuals to Responsible AI Insights dashboard
Configure Counterfactual Analysis
1. Add the specific component required to identify what features to modify for a differing prediction.
2. Call the Add Counterfactuals to RAI Insights dashboard component.
3. Link its input to the output of the constructor component initialized in Step 1 above.
4. Configure the parameter variables, including the number of counterfactual examples you want to generate per data point.
Step 3: Use the Gather Responsible AI Insights dashboard component to present the dashboard.
Assemble and Submit the Dashboard Pipeline
1. Gather the components into a cohesive Azure ML pipeline job to execute them.
-> 2. Call the Gather RAI Insights dashboard component to aggregate the constructor and the counterfactual tool outputs.
3. Submit the pipeline job to your Azure Machine Learning workspace compute cluster for processing.
Reference:
https://oneuptime.com/blog/post/2026-02-16-how-to-implement-responsible-ai-dashboards-in-azure-machine-learning/view

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129) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear on the review screen.
An organization provisions Azure Machine Learning workspaces for development, test, and production environments.
Each environment must be deployed consistently and updated through source control. The deployment process must be automated, repeatable, and auditable.
You need to deploy Azure Machine Learning resources in a consistent and controlled manner.
Solution: Clone an existing Azure Machine Learning workspace to create additional environments.
Does the solution meet the goal?

A. Yes

B. No

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Solution: Clone an existing Azure Machine Learning workspace to create additional environments.
Does the solution meet the goal?

A. Yes

B. No

Answer: B

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Correct:
* Define Azure Machine Learning resources in a Bicep template and deploy them within a GitHub Action.
This action is best because it fulfills all of your operational requirements:
Consistency & Controlled Manner:
Infrastructure as Code (IaC) via Bicep ensures that dev, test, and prod environments are configured identically, eliminating configuration drift.
Source Control: Storing the Bicep template in a Git repository satisfies the requirement that each environment must be updated through source control.
Automated & Repeatable: Orchestrating the deployment using GitHub Actions fully automates the workflow, allowing it to be executed reliably every time a change is merged.
Auditable: Git commit histories combined with GitHub deployment logs provide a comprehensive, compliant audit trail of exactly who modified the infrastructure and when.
Incorrect:
* Clone an existing Azure Machine Learning workspace to create additional environments.
* Create Azure Machine Learning workspaces manually in the Azure portal for each environment.
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-github-actions-machine-learning

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130) Drag and Drop Question
You build and manage a model by using Azure Machine Learning workspace.
Before you deploy the model, you must create a Responsible AI dashboard in Azure Machine Learning studio. The dashboard must provide observation of the following:
- metrics that show real-world impact on an outcome of interest due to taking a treatment policy
- examples with minimal changes to a particular data point such that the model's prediction changes
You need to implement the components for the Responsible AI dashboard.
Which components should you implement? To answer, move the appropriate components to the correct observations.
You may use each component once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.

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Answer Area

Components

- Explanation
- Error analysis
- Causal analysis
- Counterfactual analysis

Create a Responsible AI dashboard

Observation	Component
Impact of treatment policy	
Effect of change in data	

130) Drag and Drop Question
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Answer Area

Components

- Explanation
- Error analysis
- Causal analysis
- Counterfactual analysis

Create a Responsible AI dashboard

Observation	Component
Impact of treatment policy	
Effect of change in data	

Answer:

Answer Area

Components

- Explanation
- Error analysis
- Causal analysis
- Counterfactual analysis

Create a Responsible AI dashboard

Observation	Component
Impact of treatment policy	Causal analysis
Effect of change in data	Counterfactual analysis

https://shapingpixel.com

Explanation:
Box 1: Causal analysis
To observe metrics that show the real-world impact on an outcome of interest due to taking a treatment policy, you must implement the Causal analysis component.
Real-World Intervention: Causal analysis specifically uses historical data to evaluate how changing a particular feature (the treatment) directly impacts a real-world outcome.
Policy Making: It generates policy insights by showing which treatment policy is most effective for a specific cohort to optimize your outcome of interest.
Box 2: Counterfactual analysis
You must implement Counterfactual analysis.
The Counterfactual what-if component of the Azure Machine Learning Responsible AI dashboard is designed specifically to "interrogate" a model. It calculates the exact feature tweaks necessary to reverse or modify a specific forecast:
Minimizing Changes: It generates a set of examples demonstrating the smallest possible adjustments to a data point that cause the model's prediction to flip.
Actionable Scenarios: It surfaces the closest data points with opposing or alternative predictions, allowing you to easily test "what-if" scenarios.Why Other Components Do Not Match Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/concept-responsible-ai-dashboard

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131) Hotspot Question
You create an Azure Machine Learning workspace.
You must use the Python SDK v2 to implement an experiment from a Jupyter notebook in the workspace. The experiment must log a table in the following format:

```
table = {
    "col1" : [1, 2, 3],
    "col2" : [4, 5, 6]
}
```

You need to complete the Python code to log the table.
How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
import json
with open("table.json", 'w') as f:
   json. [ ▼ ] (table, f)
```

- load
- dump
- encode

```
[ ▼ ] ("table.json")
```

- mlflow.log_text
- mlflow.log_artifact
- MlflowClient().log_batch

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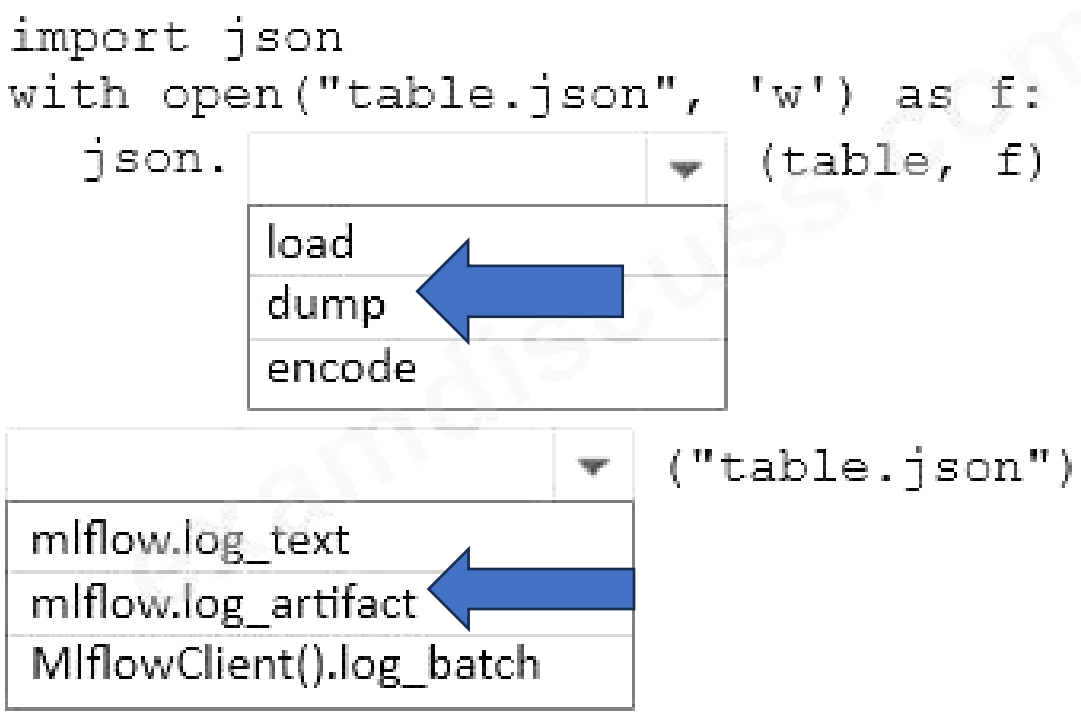
```
[ ▼ ] ("table.json")
```

- mlflow.log_text
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Answer:

Answer Area



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Explanation:
Box 1: dump
Log a table, SDK v2 with MLflow
Add a metric for each column prefixed by metric name. Similar to log_row row1 = {"table.col1": 5, "table.col2": 10}
To be done for each row in the table
mlflow.log_metrics(row1)
Using mlflow.log_artifact
import json
with open("table.json", 'w') as f:
json.dump(table, f)
mlflow.log_artifact("table.json")
Box 2: mlflow.log_artifact
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/reference-migrate-sdk-v1-mlflow-tracking

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You need to deploy Azure Machine Learning resources in a consistent and controlled manner.
Solution: Create Azure Machine Learning workspaces manually in the Azure portal for each environment.
Does the solution meet the goal?

A. Yes

B. No

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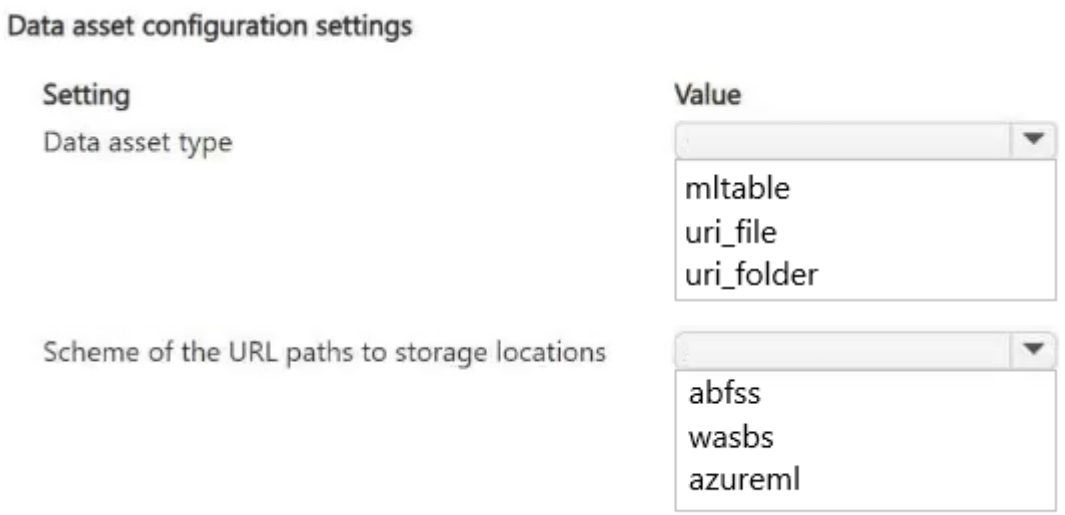
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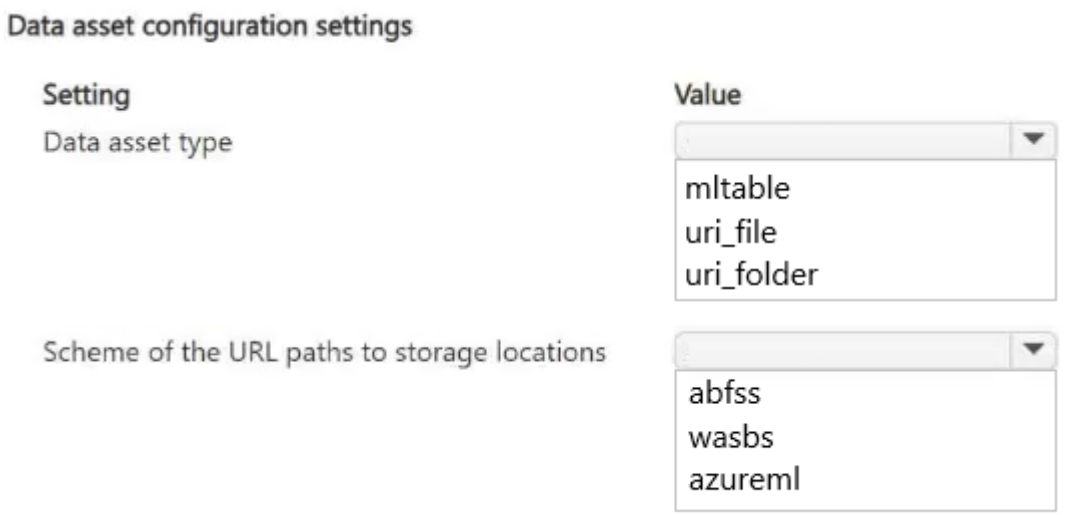
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133) Hotspot Question
You have an Azure Machine Learning workspace and a collection of image files stored in two Azure Blob Storage accounts.
You need to configure data asset properties.
Which values should you use in your configuration? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



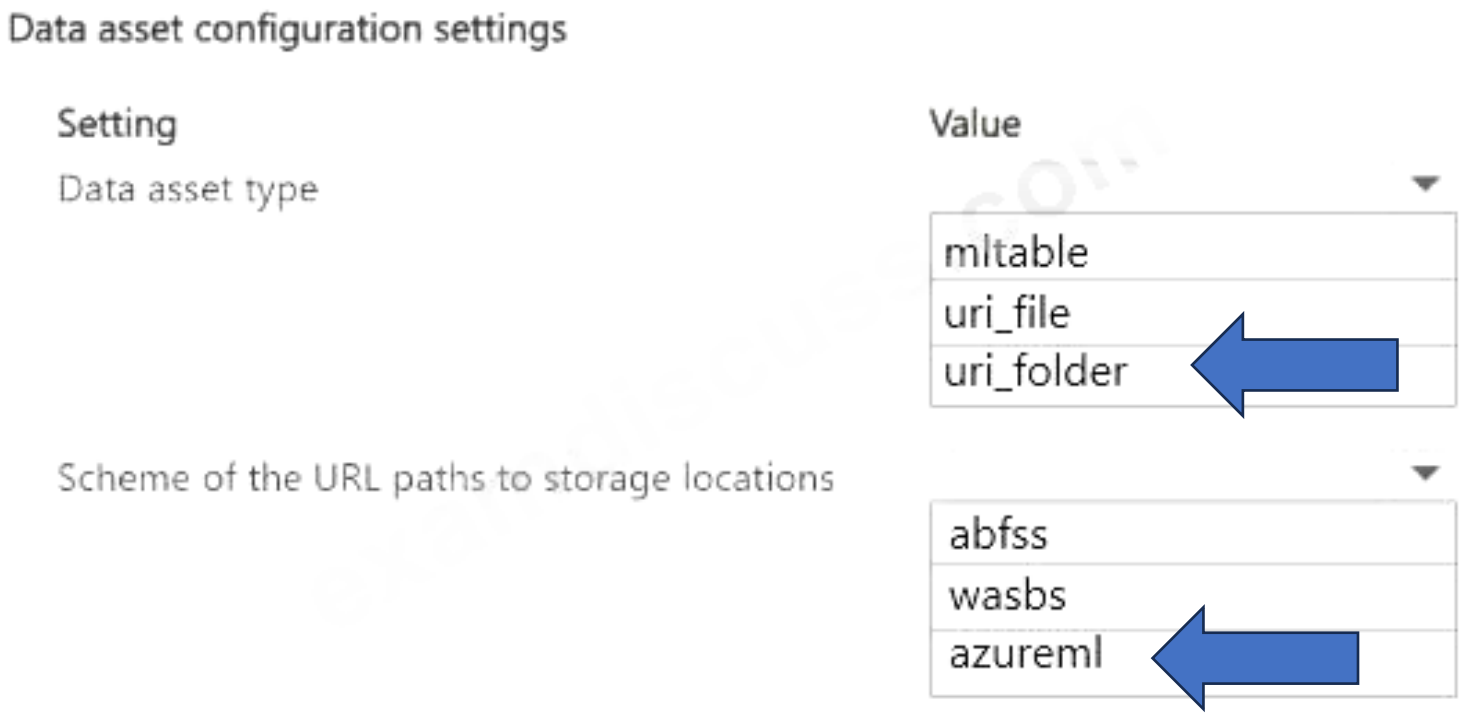
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Answer:



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Explanation:
Box 1: uri_folder
The best data asset type for this scenario is a File data asset (uri_file or uri_folder).
In Azure Machine Learning, image files used for training computer vision models are best managed as file-based data assets because machine learning frameworks read individual images directly from storage paths rather than tabular rows.
Preserves Formats: Keeps images in their native formats (PNG, JPEG).Direct Access: Allows training scripts to mount or download files easily.
*-> Folder Mapping: A uri_folder references the entire directory containing your images.
Efficiency: Avoids the overhead of parsing unstructured binary data into a table.
Box 2: azureml
To point a uri_folder data asset to your Azure Blob Storage locations in Azure Machine Learning, you should use the azureml:// URI scheme.
This is the recommended, modern standard that leverages Azure ML datastores for secure tokenless access.
Alternatively, you can use direct Azure Storage URI schemes depending on your configuration Recommended Scheme: Azure ML Datastore This scheme abstracts the storage credentials by referencing an Azure ML Datastore.
Format: azureml://datastores/<datastore_name>/paths/<path_to_folder>/
Example: azureml://datastores/myblobdatastore/paths/images/training_set/ Incorrect:
[not wasbs]
Azure Blob Storage (wasbs)
Used when connecting via the legacy Windows Azure Storage Blob driver.
Format:
wasbs://<container_name>@<storage_account_name>.blob.core.windows.net/<path_to_folder>/
[Not abfss]
Azure Data Lake Storage Gen2 (abfss)
Used if your Blob Storage accounts have the hierarchical namespace enabled.
Format:
abfss://<container_name>@<storage_account_name>.dfs.core.windows.net/<path_to_folder>/ Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-create-data-assets

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134) You are designing a new machine learning solution to predict customer churn by using Azure Machine Learning. You have raw data in CSV format stored in Azure Data Lake.
You need to design the solution so that it can efficiently handle large-scale model training and iterative development.
Which two actions should you perform? Each correct answer presents part of the solution.
Choose two.
NOTE: Each correct selection is worth one point.

A. Convert the data into the JSONL format and upload into Blob Storage.

B. Schedule training using an Azure Data Factory pipeline.

C. Configure an Azure Machine Learning compute cluster for model training.

D. Register the data as a tabular dataset in the Azure Machine Learning workspace.

E. Configure an Azure Machine Learning compute instance for model training.

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D. Register the data as a tabular dataset in the Azure Machine Learning workspace.

E. Configure an Azure Machine Learning compute instance for model training.

Answer: C,D

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Here is the correct and optimized architectural design for the solution.
1. Data Engineering & Registration
Connect Data Lake: Create an Azure ML Datastore pointing to your Azure Data Lake Storage (ADLS) Gen2.
[D] -> Modern Data Asset: Register the CSV data as an MLtable (the modern replacement for Tabular Datasets).
Schema Enforcement: Define types (e.g., CustomerID as string, Churn as boolean) in the MLTable YAML file.
Lazy Loading: MLtable loads data lazily during training to prevent out-of-memory errors on massive datasets.
2. Compute Strategy
[C] -> Compute Cluster (Scale-Out): Use an Azure ML Compute Cluster (AmlCompute) for actual large-scale model training.
Auto-Scaling: Configure the cluster to scale from 0 to N nodes so you only pay for compute during active jobs.Spot VMs: Leverage Azure Spot Virtual Machines on the cluster to reduce training costs by up to 80%.
Incorrect:
[Not E] Compute Instance (Development): Use this only as your workstation for writing code, Jupyter notebooks, and light debugging.
Reference:
https://medium.com/henkel-data-and-analytics/how-to-use-azure-ml-studio-an-eye-opening-model-training-tutorial-for-beginners-from-henkels-data-5035ee10a6d2

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135) Drag and Drop Question
You manage an Microsoft Foundry project.
You deploy a large language model from the model catalog.
You need to manually evaluate the model, collect the statistics, and be able to review the results later.
Which four actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Actions	Manually evaluate prompts in Microsoft Foundry playground
Automatically generate test data.	
Import data in CSV format.	
Provide thumbs up or down ratings to model responses.	
Evaluate the solution on 50 input rows of data.	
Save the evaluation results.	

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Answer:

Actions	Manually evaluate prompts in Microsoft Foundry playground
Automatically generate test data.	Import data in CSV format.
Import data in CSV format.	Evaluate the solution on 50 input rows of data.
Provide thumbs up or down ratings to model responses.	Provide thumbs up or down ratings to model responses.
Evaluate the solution on 50 input rows of data.	Save the evaluation results.
Save the evaluation results.	

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Explanation:
Step 1: Import Data in CSV Format
Supported formats: Azure AI Studio natively accepts .csv and .jsonl files for evaluation datasets.
Requirement: Your file must contain the input columns (e.g., user prompts) that you want to test against the model.
Step 2: Evaluate the Solution on 50 input Rows
Sample size: 50 rows is an excellent size for a manual, qualitative test baseline.
Execution: You will upload this dataset into the Evaluation blade of your project and map your data fields to the model's inputs.
Step 3: Provide thumbs up or down ratings to model responses
Manual UI: The platform features a manual review interface (often called human-in-the-loop evaluation).
Feedback: You can view the model's generated response for each of the 50 rows side-by-side with the input and log your binary (thumbs up/down) or detailed feedback.
Step 4: Save the evaluation results
Persistence: Once completed, the session is saved to your project's run history.
Review: You can return to the dashboard later to view aggregate statistics, check pass/fail rates, and export the annotated data for your records.
Reference:
https://oneuptime.com/blog/post/2026-02-16-how-to-perform-azure-migrate-at-scale-using-csv-import-for-large-datacenter-inventories/view

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136) Hotspot Question
You have an Azure Machine Learning workspace.
You plan to use Azure Machine Learning Python SDK v2 to define a pipeline component that trains an image classification model. The execution logic of the component is contained in the train() function in the file named model_train.py.
You write code to import all required libraries and store it as train_component.py in the same folder that contains model_train.py.
You need to complete the remaining code in train_component.py.
How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Azure Machine Learning Python SDK v2 code

```
@ [tool / pipeline / command_component] (
    name="train_image_classification",
    version="1",
    display_name="Train Image Classification",
    description="train image classification",
    environment=dict(
        conda_file=Path(__file__).parent / "conda.yaml",
        image="mcr.microsoft.com/azureml/openmpi4.1.0-ubuntu20.04",
    ),
)
def train_component(
    input_data: Input(type="uri_folder"),
    output_model: Output(type="uri_folder"),
    epochs=10,
):
    from [train / model_train / train_component] import train

    train(input_data, output_model, epochs)
```

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Azure Machine Learning Python SDK v2 code

```
@ [tool / pipeline / command_component] (
    name="train_image_classification",
    version="1",
    display_name="Train Image Classification",
    description="train image classification",
    environment=dict(
        conda_file=Path(__file__).parent / "conda.yaml",
        image="mcr.microsoft.com/azureml/openmpi4.1.0-ubuntu20.04",
    ),
)
def train_component(
    input_data: Input(type="uri_folder"),
    output_model: Output(type="uri_folder"),
    epochs=10,
):
    from [train / model_train / train_component] import train

    train(input_data, output_model, epochs)
```

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Azure Machine Learning Python SDK v2 code

```
@ [tool / pipeline / command_component] (
    name="train_image_classification",
    version="1",
    display_name="Train Image Classification",
    description="train image classification",
    environment=dict(
        conda_file=Path(__file__).parent / "conda.yaml",
        image="mcr.microsoft.com/azureml/openmpi4.1.0-ubuntu20.04",
    ),
)
def train_component(
    input_data: Input(type="uri_folder"),
    output_model: Output(type="uri_folder"),
    epochs=10,
):
    from [train / model_train / train_component] import train

    train(input_data, output_model, epochs)
```

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Explanation:
Box 1: command_component
The @command_component decorator transforms a standard Python function into a reusable pipeline component within Azure ML SDK v2.from model_ Box 2: model_train model_train import train: Because model_train.py resides in the same directory as train_component.py, you import the file directly by its module name (model_train) to access its execution logic inside the component function.
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-create-component-pipeline-python

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137) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear on the review screen.
You manage a Retrieval-Augmented Generation (RAG) application built on Microsoft Foundry.
The application retrieves documents from an indexed knowledge base and generates answers for internal users.
Recent feedback indicates that answers are fluent but sometimes include information that is not supported by the documents that were retrieved.
You need to evaluate whether a proposed change improves RAG answer quality by using supported and measurable techniques.
Solution: Measure token throughput and average response latency before and after applying the proposed change.
Does the solution meet the goal?

A. Yes

B. No

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137) Note: This question is part of a series of questions that present the same scenario. Each question in the series contains a unique solution that might meet the stated goals. Some question sets might have more than one correct solution, while others might not have a correct solution.
After you answer a question in this section, you will NOT be able to return to it. As a result, these questions will not appear on the review screen.
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You need to evaluate whether a proposed change improves RAG answer quality by using supported and measurable techniques.
Solution: Measure token throughput and average response latency before and after applying the proposed change.
Does the solution meet the goal?

A. Yes

B. No

Answer: B

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Correct:
* Review user feedback comments collected after deployment to determine whether answers appear more accurate.
Relying solely on user feedback comments is not the proper immediate action for evaluating this specific issue but it is still the best choice.
Evaluate whether your proposed change improves the Retrieval-Augmented Generation (RAG) system by reducing hallucinations (unsupported information), you need to measure faithfulness and context relevance using automated, quantifiable metrics.
The proper course of action is to implement an LLM-as-a-Judge framework using an open-source evaluation library like Ragas or TruLens.
Recommended Evaluation Plan
Establish a baseline: Run your current RAG pipeline through a test dataset of 50-100 representative user queries.
Capture the outputs: Save the user query, the exact retrieved document snippets, and the generated response for every test.
Apply the change: Deploy your proposed modification (e.g., altered prompt, different temperature, or re-ranking algorithm).Run the evaluation: Pass the test dataset through the updated pipeline to generate a new set of responses.
Compare the metrics: Use the framework to score both sets of data and mathematically verify if the change reduced unsupported claims.
Incorrect:
* Compare embedding vector dimensions used by the retrieval pipeline before and after the change.
Comparing embedding vector dimensions is not a valid or effective method for measuring RAG answer quality. Vector dimensions (e.g., 1536 or 3072) are static architectural properties of your embedding model. They do not reflect factual accuracy, semantic grounding, or the rate of hallucinations in your text generation.
* Measure token throughput and average response latency before and after applying the proposed change.
Measuring token throughput and latency is not the correct action to solve this specific problem.
Reference:
https://www.getmaxim.ai/articles/how-to-evaluate-your-rag-system/

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138) Hotspot Question
You create an Azure Machine Learning workspace and install the MLflow library.
You need to log different types of data by using the MLflow library.
Which method should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Log data	MLflow library method
matlotlib plot	log_metric log_text log_image log_figure
boolean value	log_metric log_text log_image log_figure

www.[redacted].com

138) Hotspot Question
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You need to log different types of data by using the MLflow library.
Which method should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer:



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Explanation:
Box 1: log_figure
matlotlib plot
To log a Matplotlib plot data using the MLflow library, you should use the mlflow.log_figure() method.
Alternatively, if you save the Matplotlib plot as an image object (such as a PIL image) or a local image file first, you can use mlflow.log_image() or mlflow.log_artifact(). However, mlflow.log_figure() is specifically built to accept a Matplotlib figure object directly.
Box 2: log_metric
Boolean value
To log a boolean value data using the MLflow library in an Azure Machine Learning workspace, you should use the mlflow.log_param() method.
Alternatively, if you want to track it as a numerical metric over time, you can map the boolean to an integer (1 for True, 0 for False) and log it using mlflow.log_metric().
Depending on your precise objective for tracking the boolean data, choose from the two main use cases:
mlflow.log_param(key, value): Best for static configuration data, flags, or hyperparameter switches (e.g., is_training=True). Parameters accept string representations of boolean values.
mlflow.log_metric(key, value): Best if the boolean represents a status or binary occurrence that shifts or updates throughout execution. MLflow metrics exclusively accept numeric values, meaning you must cast the boolean into a 1 or 0.
Reference:
https://docs.azure.cn/en-us/machine-learning/how-to-log-view-metrics

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139) Hotspot Question
You have an Azure Machine Learning workspace.
You plan to use Azure Machine Learning Python SDK v2 to register a component in the workspace. The component definition is stored in the local file ./components/train/train.yml.
You write code to connect to the workspace by using the ml_client object and import all required libraries.
You need to complete the remaining code.
How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Azure Machine Learning Python SDK v2 code

```
train_src_dir = "./components/train"
train_component = [from_config / load_component / create_or_update] (source=os.path.join(train_src_dir, "train.yml"))

train_component = ml_client.[from_config / load_component / create_or_update] (train_component)
```

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139) Hotspot Question
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You plan to use Azure Machine Learning Python SDK v2 to register a component in the workspace. The component definition is stored in the local file ./components/train/train.yml.
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How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer:

Answer Area

Azure Machine Learning Python SDK v2 code

```
train_src_dir = "./components/train"
train_component = [from_config / load_component / create_or_update] (source=os.path.join(train_src_dir, "train.yml"))

train_component = ml_client.[from_config / load_component / create_or_update] (train_component)
```

Explanation:
Box 1: load_component
train_component
Example:
Create component: training (using yaml definition)
Create and register the component. Registering it allows you to re-use it in other pipelines. Also, anyone else with access to your workspace can use the registered component.

```
# importing the Component Package
from azure.ai.ml import load_component
# Loading the component from the yml file
train_component = load_component(source=os.path.join(train_src_dir, "train.yml"))
# Now we register the component to the workspace
train_component = ml_client.create_or_update(train_component)
# Create (register) the component in your workspace
print(
f"Component {train_component.name} with Version {train_component.version} is registered" ) Box 2:
create_or_update ml_client Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/tutorial-pipeline-python-sdk
```

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140) You create an Azure Machine Learning workspace.
You must use the Python SDK v2 to implement an experiment from a Jupyter notebook in the workspace. The experiment must log a list of numerical metrics.
You need to implement a method to log a list of numerical metrics.
Which method should you use?

A. mlflow.log_batch()

B. mlflow.log_metric()

C. mlflow.log_artifact()

D. mlflow.log_image()

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140) You create an Azure Machine Learning workspace.
You must use the Python SDK v2 to implement an experiment from a Jupyter notebook in the workspace. The experiment must log a list of numerical metrics.
You need to implement a method to log a list of numerical metrics.
Which method should you use?

A. mlflow.log_batch()

B. mlflow.log_metric()

C. mlflow.log_artifact()

D. mlflow.log_image()

Answer: B

To log a list of numerical metrics using the Azure Machine Learning Python SDK v2, you should use the mlflow.log_metric() method within a loop, or mlflow.log_metrics() to log them simultaneously as a dictionary.
Reference:
https://learn.microsoft.com/en-us/answers/questions/1456554/downloading-azureml-experiment-metrics-logged-with

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141) You deploy a new model version to a managed online endpoint. You must test it with 10% traffic and automatically roll back if latency or error rate increases beyond threshold. What should you configure?

A. Separate endpoint for testing

B. Traffic splitting with monitoring alerts

C. Batch endpoint validation

D. Manual testing workflow

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141) You deploy a new model version to a managed online endpoint. You must test it with 10% traffic and automatically roll back if latency or error rate increases beyond threshold. What should you configure?

A. Separate endpoint for testing

B. Traffic splitting with monitoring alerts

C. Batch endpoint validation

D. Manual testing workflow

Answer: B

Traffic splitting enables controlled rollout of a new model version by directing a percentage of requests to it. Combined with monitoring alerts, it supports automated rollback when performance degrades. Separate endpoints lack built-in traffic management and do not provide seamless or automated rollback capabilities.

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142) A team uses a hosted Git repository to store training code and pipeline definitions of a machine learning experiment.
The team must ensure that access to the repository is granted without requiring each developer to store personal access tokens on their machines.
Repository access must be secure and centrally managed to reduce credential spread.
You need to enable secure access between an Azure Machine Learning workspace and the repository.
What should you do?

A. Share a repository deploy key across all developers on the team.

B. Generate a personal access token and store it in a pipeline variable.

C. Require each developer to authenticate locally before every pipeline run.

D. Configure the workspace to use a managed identity for repository access.

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C. Require each developer to authenticate locally before every pipeline run.

D. Configure the workspace to use a managed identity for repository access.

Answer: D

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143) An organization runs a customer-facing generative AI application built by using Microsoft Foundry. The application uses multiple prompts linked to multiple workflows to generate responses in production.
The application occasionally returns incomplete responses. The model call succeeds, but the final message sometimes stops early.
The issue cannot be reproduced reliably in development.
You need to identify where and why response generation is terminating early in production.
Which approach should you use?

A. Run a pre-release evaluation workflow to score groundedness and relevance on a test dataset.

B. Enable tracing and logging so that each workflow can be inspected.

C. Increase max_tokens and temperature to reduce the chance of early termination.

D. Replace the deployed model with a smaller model to reduce variability across responses.

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143) An organization runs a customer-facing generative AI application built by using Microsoft Foundry. The application uses multiple prompts linked to multiple workflows to generate responses in production.
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D. Replace the deployed model with a smaller model to reduce variability across responses.

Answer: B

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Enabling end-to-end tracing and logging is the most effective action to isolate this issue. Because the model call succeeds, the premature termination is likely caused by token limits, downstream application logic, or network timeouts rather than a failure in the LLM itself.

Here is how to isolate and fix the issue using tracing, along with the most likely culprits.

What to Log and Trace

To find the exact point of failure, your tracing system must capture specific metadata for every step of your production workflows:

Finish Reason: Check the finish_reason string returned in the API metadata. If it says length, the model hit a token limit. If it says stop, the model thinks it finished naturally.

Token Counts: Log completion_tokens, prompt_tokens, and total_tokens for every single prompt node.

Raw Prompt vs. Output: Capture the exact, fully rendered prompt string sent to the model, not just the template.

Workflow Node Timestamps: Log the exact entry and exit times for every workflow node to detect quiet timeouts.

Reference:

https://dev.to/utibe_okodi_339fb47a13ef5/your-ai-agent-just-failed-in-production-where-do-you-even-start-debugging-268

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144) Hotspot Question

You monitor an Azure Machine Learning classification training experiment named train_classification on Azure Notebooks.

You must store a table named table as an artifact in Azure Machine Learning Studio during model training.

You need to collect and list the metrics by using MLflow.

How should you complete the code segment? To answer, select the appropriate option in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

```
from mlflow.tracking import MlflowClient
import json
mlflow.set_experiment("train_classification")
mlflow_run = mlflow.start_run()
row1 = {"table.col1": 5, "table.col2": 10}
mlflow.                    (row1)
```

- log_artifact
- log_image
- log_metrics

```
with open("table.json", 'w') as f:
json.dump(table, f)
mlflow.                    ("table.json")
```

- log_artifact
- log_metric
- log_metrics

```
client = MlflowClient()
finished_mlflow_run =
MlflowClient().get_run(                                    )
```

- mlflow_run.info.run_id
- mlflow_run.info.run_id, path="accuracy_table.json"
- mlflow_run.info.experiment_id

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```

- log_artifact
- log_image
- log_metrics

```
with open("table.json", 'w') as f:
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mlflow.                    ("table.json")
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- log_artifact
- log_metric
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- mlflow_run.info.run_id, path="accuracy_table.json"
- mlflow_run.info.experiment_id

Answer:

Answer Area

```
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import json
mlflow.set_experiment("train_classification")
mlflow_run = mlflow.start_run()
row1 = {"table.col1": 5, "table.col2": 10}
mlflow.                    (row1)
```

- log_artifact
- log_image
- log_metrics ←

```
with open("table.json", 'w') as f:
json.dump(table, f)
mlflow.                    ("table.json")
```

- log_artifact ←
- log_metric
- log_metrics

```
client = MlflowClient()
finished_mlflow_run =
MlflowClient().get_run(                    )
```

- mlflow_run.info.run_id ←
- mlflow_run.info.run_id, path="accuracy_table.json"
- mlflow_run.info.experiment_id

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Explanation:
Box 1: log_metrics
Log Metrics dictionary
The dictionary row1 contains key-value pairs representing metrics. The mlflow.log_metrics() function is used to log multiple metrics simultaneously.
Box 2: log_artifact
Save Table Artifact
The code writes a JSON file locally and needs to upload it to the experiment run. The mlflow.log_artifact() function logs a local file or directory as an artifact in Azure Machine Learning Studio.
Box 3: mlflow_run.info.run_id
Retrieve Run ID
To fetch the finalized run data using the MlflowClient, you need to pass the unique run ID string.
This identifier is accessed via the active run object using mlflow_run.info.run_id. Note that you also need to use the instantiated client object variable instead of the class name MlflowClient to call the method correctly.
Reference:
https://levelup.gitconnected.com/mlops-mastering-mlflow-unlocking-efficient-model-management-and-experiment-tracking-d9d0e71cc697

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145) You use Azure Machine Learning to train a model.
You must use Bayesian sampling to tune hyperparameters.
You need to select a learning_rate parameter distribution.
Which two distributions can you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A. Choice

B. Uniform

C. QNormal

D. LogUniform

E. Normal

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145) You use Azure Machine Learning to train a model.
You must use Bayesian sampling to tune hyperparameters.
You need to select a learning_rate parameter distribution.
Which two distributions can you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A. Choice

B. Uniform

C. QNormal

D. LogUniform

E. Normal

Answer: A,B

Bayesian sampling in Azure Machine Learning supports only choice, uniform, and quniform parameter distributions over the search space. Any two of these (for example, Uniform and Choice) can be used for a learning_rate parameter.Supported Distributions for Bayesian SamplingUniform: Picks a value uniformly between a minimum and maximum value.Choice:
Picks a discrete value from a set of given options or a list.Quniform: Picks a discrete uniform value (uniform value multiplied by a rounding factor).

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147) Hotspot Question
You manage a Microsoft Foundry project. You build a solution that uses a set of PDF documents.
You require two large language models (LLMs):
- An embedding model must help categorize the documents.
- A general-purpose model must generate semantically and contextually
accurate output based on the documents.
You need to select benchmarks to observe the quality of the models.
Which metrics should you use for the benchmarks? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

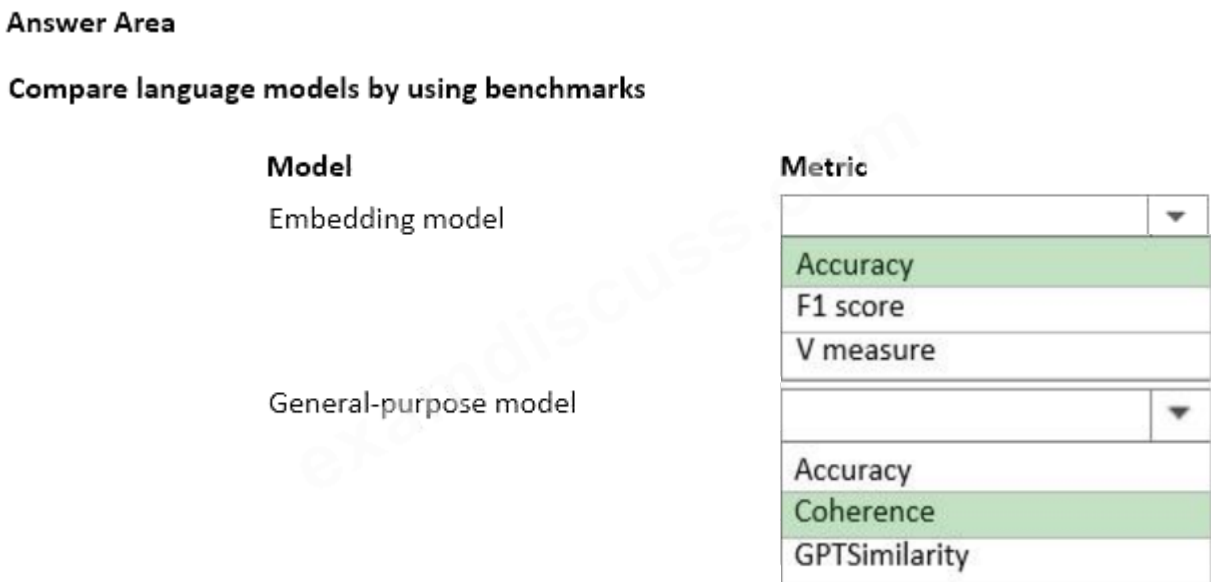
Compare language models by using benchmarks

Model	Metric
Embedding model	Accuracy F1 score V measure
General-purpose model	Accuracy Coherence GPTSimilarity

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You require two large language models (LLMs):
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You need to select benchmarks to observe the quality of the models.
Which metrics should you use for the benchmarks? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer:



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Explanation:
Box 1: Accuracy
Embedding model
An embedding model must help categorize the documents.
For an embedding model designed to categorize documents within Azure AI Foundry, cosine similarity is a crucial benchmark metric. It measures the semantic similarity between document embeddings, allowing you to assess how well the model groups related documents together.
Additionally, classification accuracy is important to evaluate how effectively the model assigns documents to predefined categories.
Classification Accuracy:
This metric measures the percentage of documents that are correctly categorized by the model. It directly reflects the model's ability to assign documents to the appropriate category based on their embeddings Box 2: Coherence General-purpose model A general-purpose model must generate semantically and contextually accurate output based on the documents.
Coherence evaluates how well the language model can produce output that flows smoothly, reads naturally, and resembles human-like language.
Incorrect:
* GPTsimilarity
GPT similarity refers to the ability of GPT models to assess and quantify the semantic similarity between pieces of text. This can be achieved by using the model to generate embeddings (numerical representations) of the text and then calculating the similarity (e.g., using cosine similarity) between these embeddings. Essentially, it allows you to determine how closely related two texts are in terms of their meaning.
Reference:
https://learn.microsoft.com/en-us/azure/ai-foundry/concepts/model-benchmarks
https://learn.microsoft.com/en-us/azure/machine-learning/prompt-flow/concept-model-monitoring-generative-ai-evaluation-metrics

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148) Hotspot Question
You manage an Azure Machine Learning workspace. You create an experiment named experiment1 by using the Azure Machine Learning Python SDK v2 and MLflow.
You are reviewing the results of experiment1 by using the following code segment:

```
runs = mlflow.search_runs(
    experiment_names=["experiment1"],
    max_results=5,
    order_by=["start_time ASC"])

runs[runs.status == "FAILED"]
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
Aborted runs are returned.	○	○
The runs returned have been completed with errors.	○	○
The jobs that are returned have been canceled or killed by the user or system.	○	○
All metrics and their values are returned for the returned experiment runs.	○	○

148) Hotspot Question
You manage an Azure Machine Learning workspace. You create an experiment named experiment1 by using the Azure Machine Learning Python SDK v2 and MLflow.
You are reviewing the results of experiment1 by using the following code segment:

```
runs = mlflow.search_runs(
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    order_by=["start_time ASC"])

runs[runs.status == "FAILED"]
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer:

Answer Area

Statements	Yes	No
Aborted runs are returned.	○	○
The runs returned have been completed with errors.	○	○
The jobs that are returned have been canceled or killed by the user or system.	○	○
All metrics and their values are returned for the returned experiment runs.	○	○

Explanation:
Box 1: No
Filter runs by status
You can also filter experiment by status. It becomes useful to find runs that are running, completed, canceled or failed.
Failed:The job/run has completed with errors.
KILLED: The job/run has been canceled or killed by the user/system.
The following example shows all the completed runs:
runs = mlflow.search_runs(experiment_ids=["1234-5678-90AB-CDEFG"])
runs[runs.status == "FINISHED"]
Box 2: No
Ordering runs
By default, experiments are ordered descending by start_time, which is the time the experiment was queue in Azure Machine Learning.
However, here we have order_by=["start_time ASC"]. The five oldest experiments are returned.
Use the argument max_results from search_runs to limit the number of runs returned.
Box 3: No
Failed: The job/run has completed with errors.
KILLED: The job/run has been canceled or killed by the user/system.
Box 4: No
Getting metrics, parameters, artifacts and models
The method search_runs returns a Pandas Dataframe containing a limited amount of information by default. You can get Python objects if needed, which may be useful to get details about them.
Use the output_format parameter to control how output is returned:

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```
runs = mlflow.search_runs(
experiment_ids=[ "1234-5678-90AB-CDEFG" ],
filter_string="params.num_boost_round='100'",
output_format="list",
)
```

Details can then be accessed from the info member. The following sample shows how to get the run_id:

```
last_run = runs[-1]
print("Last run ID:", last_run.info.run_id)
```

Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/how-to-track-experiments-mlflow

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149) DRAG DROP

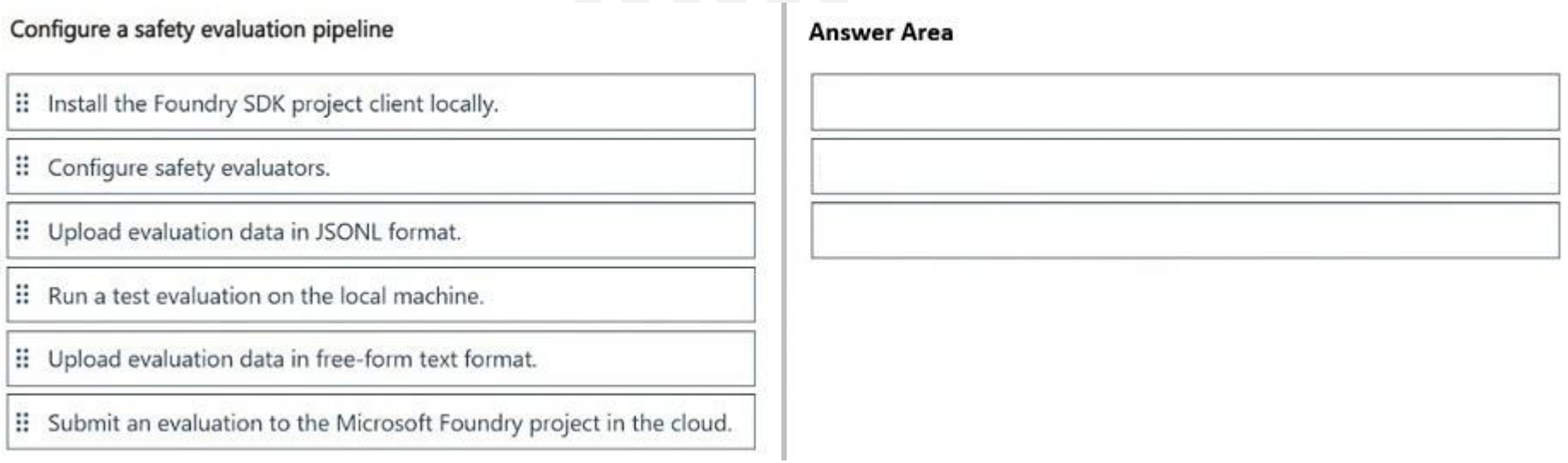
-

A team validates a generative AI application that produces free-form text responses by using Microsoft Foundry SDK.

The evaluation dataset is registered in the Microsoft Foundry environment.

You need to configure a safety evaluation pipeline that reliably evaluates model outputs for harmful content.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



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149) DRAG DROP

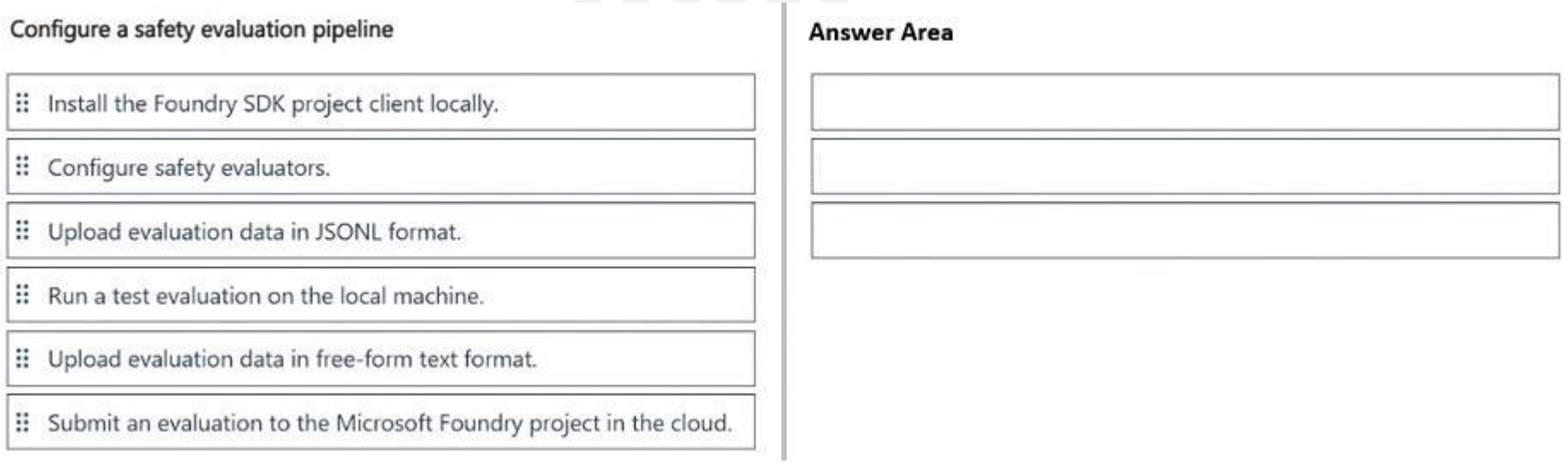
-

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Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



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Correct sequence

1. **Install the Foundry SDK project client locally.**
2. **Configure safety evaluators.**
3. **Submit an evaluation to the Microsoft Foundry project in the cloud.**

✅ **Answer:**

Install SDK → Configure safety evaluators → Submit evaluation to Microsoft Foundry

Why: Since the evaluation dataset is **already registered** in Microsoft Foundry, you don't need to upload it again. Safety evaluators such as **Violence, Sexual, Self-harm, and Hate/Unfairness** can be configured and then the evaluation can be submitted as a cloud evaluation run.

The **local test evaluation** is unnecessary because the requirement is to configure a reliable Foundry safety evaluation pipeline using the registered dataset.

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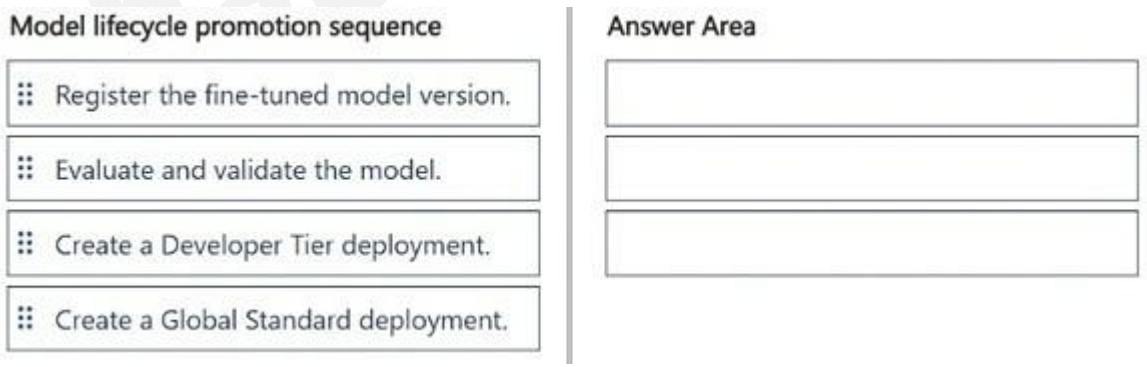
150) DRAG DROP
-

You complete the fine-tuning of a generative model in Microsoft Foundry. The fine-tuned model now appears as a new model variant in your development environment.

The deployment process must ensure that proper validation and control is maintained.

You need to promote the fine-tuned model from development to production.

Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



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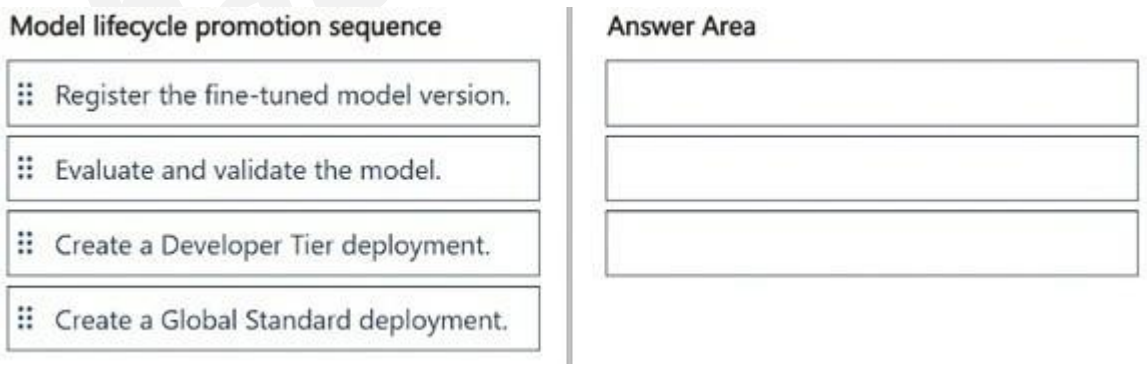
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Correct sequence

1. **Register the fine-tuned model version**
2. **Evaluate and validate the model**
3. **Create a Global Standard deployment**

✅ **Answer:**

Register model version → Evaluate & validate → Global Standard deployment

Why: The fine-tuned model first needs to be registered as a versioned model asset. After registration, you validate its quality/safety before promoting it to the production deployment. The **Global Standard deployment** is the production-oriented deployment choice.

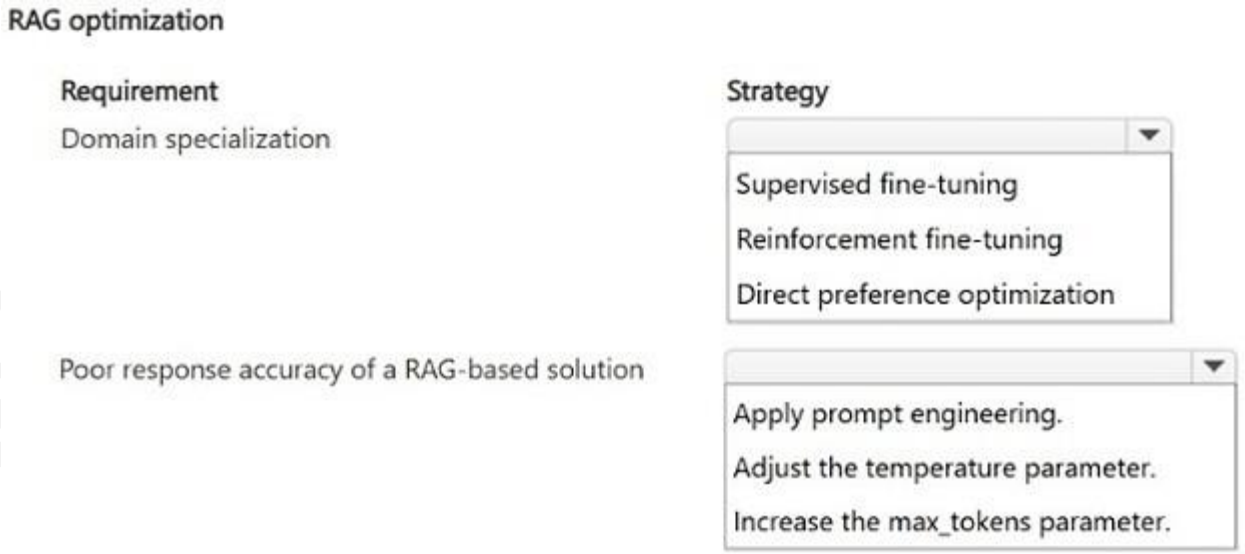
❌ **Developer Tier deployment** is not the final production promotion step in this scenario.

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151) You need to configure an optimization method to meet Fabrikam Inc.’s technical requirements.

Which strategy should you apply first? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



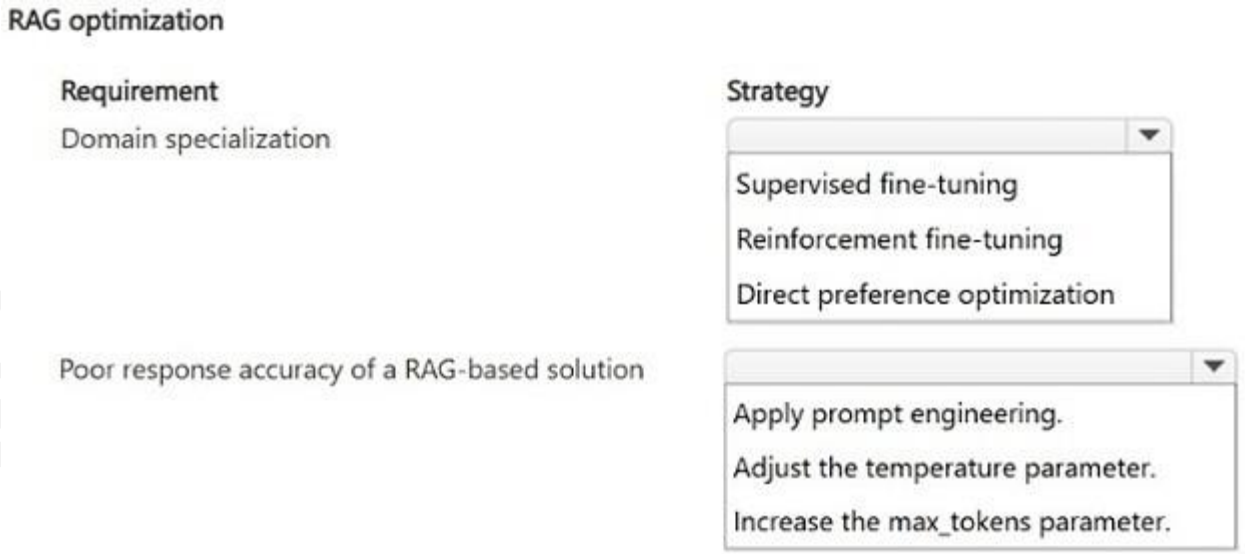
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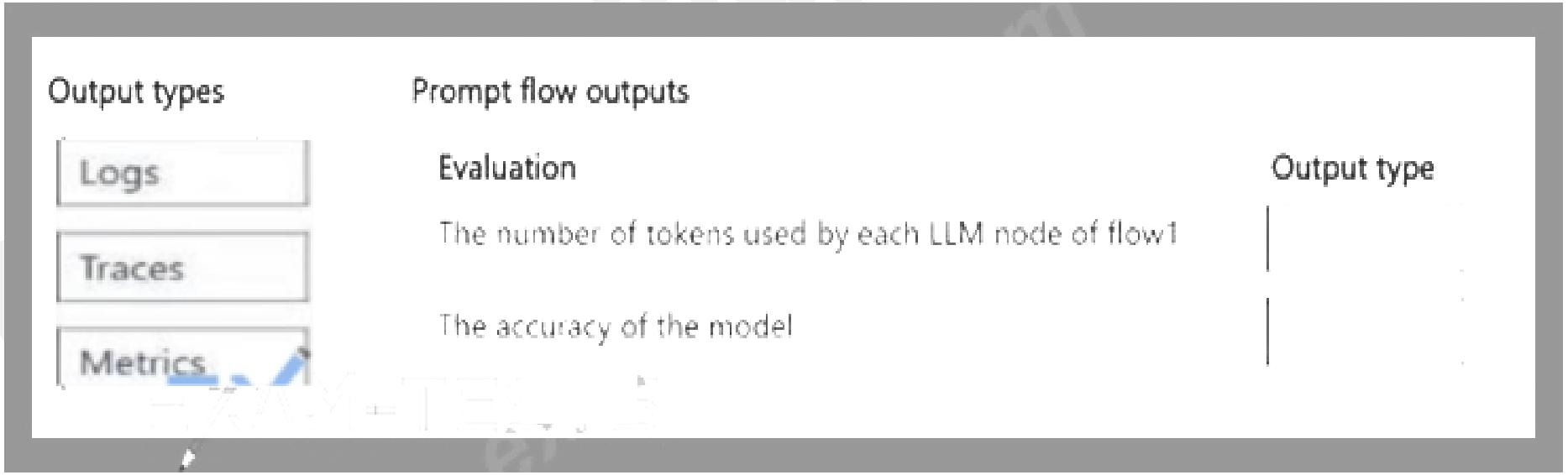
answers



Requirement	Strategy
Domain specialization	✅ Supervised fine-tuning
Poor response accuracy of a RAG-based solution	✅ Apply prompt engineering

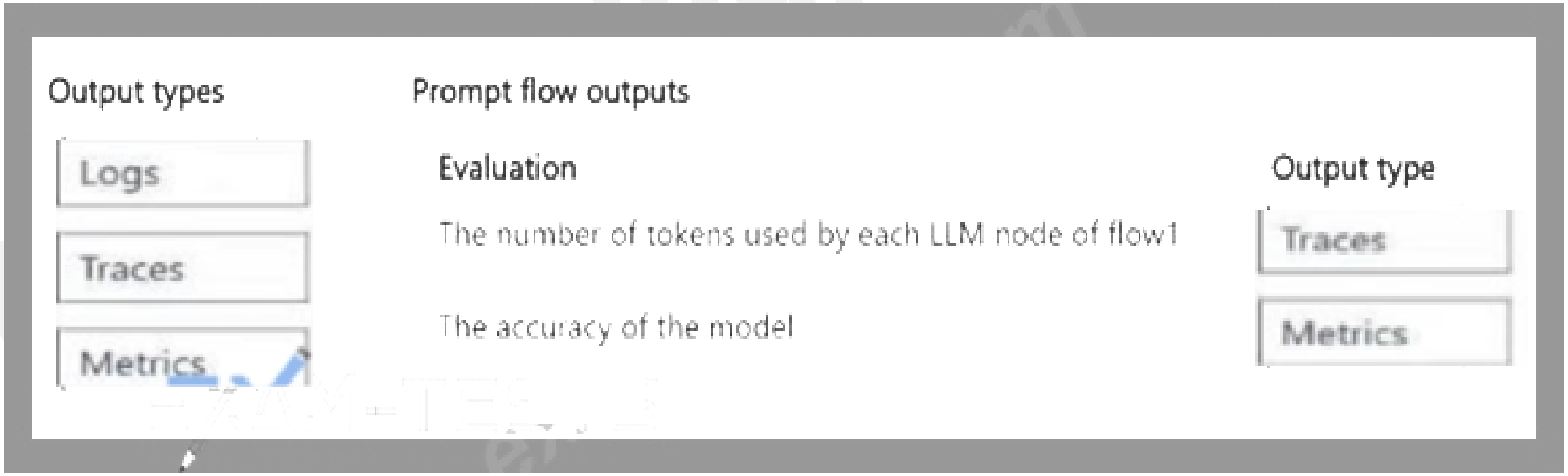
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152) Drag and Drop Question
You develop a flow for a Microsoft Foundry project.
You plan to use outputs generated by running the flow to determine the following information:
- the number of tokens used by each large language model (LLM) node of
the flow
- the accuracy of the model used by the flow
You need to examine the output that provides the required information.
Which output type should you examine? To answer, move the appropriate output types to the correct evaluations. You may use each output type once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



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152) Drag and Drop Question
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Explanation:
Box 1: Traces
The appropriate output flow type to determine the number of tokens used by each Large Language Model (LLM) node is Traces.
While Metrics provide a macro-level, aggregated overview of total token usage across an entire application or resource, Traces capture detailed, node-by-node execution details.
When a prompt flow runs in Microsoft Foundry, Traces track:
The precise execution path of individual items
Inputs and outputs for each specific LLM node
Detailed telemetry records-such as input_tokens, output_tokens, and total_tokens-bound to that exact step Box 2: Metrics The most appropriate output flow type to determine the accuracy of the model is Metrics.
Metrics are quantitative measurements (such as accuracy, F1-score, precision, recall, or mean squared error) specifically calculated by evaluating model predictions against ground truth data.
Reference:
https://huggingface.co/docs/evaluate/a_quick_tour

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153) Drag and Drop Question
You have an Azure Machine Learning workspace named WS1 and a GitHub account named account1 that hosts a private repository named repo1.
You need to clone repo1 to make it available directly from WS1. The configuration must maximize the performance of the repo1 clone.
Which four actions should you perform in sequence?

Clone configuration steps

- Add a private key to account1.
- Generate a Secure Shell (SSH) key pair.
- Create a compute instance.
- Add a public key to account1.
- Open a terminal window.

Answer area

1.
2.
3.
4.

153) Drag and Drop Question
You have an Azure Machine Learning workspace named WS1 and a GitHub account named account1 that hosts a private repository named repo1.
You need to clone repo1 to make it available directly from WS1. The configuration must maximize the performance of the repo1 clone.
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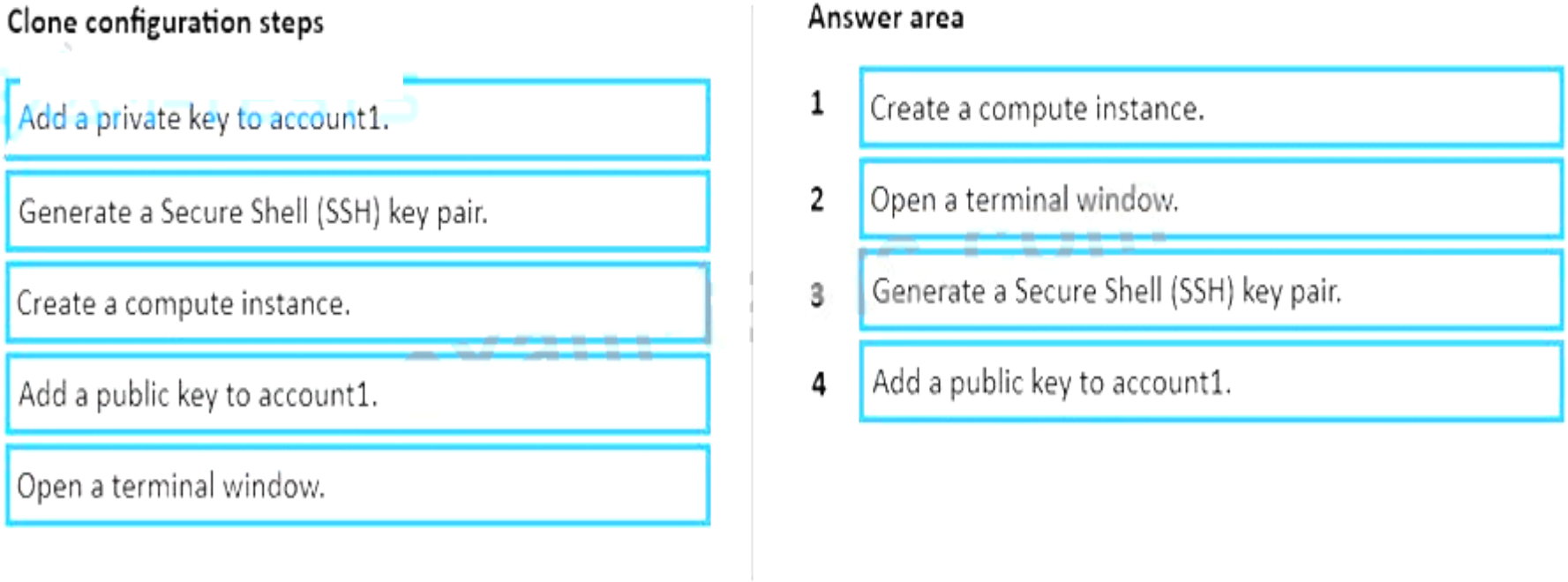
Clone configuration steps

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- Create a compute instance.
- Add a public key to account1.
- Open a terminal window.

Answer area

1.
2.
3.
4.

Answer:



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Explanation:
Step 1: Create a compute instance
Git integration for Azure Machine Learning
Clone Git repositories into your workspace file system
Azure Machine Learning provides a shared file system for all users in the workspace. To clone a Git repository into this file share, we recommend that you create a compute instance (Step 1) & open a terminal (Step 2). Once the terminal is opened, you have access to a full Git client and can clone and work with Git via the Git CLI experience.
We recommend that you clone the repository into your user directory so that others will not make collisions directly on your working branch.
Step 2: Open a terminal window
Step 3: Generate a Secure Shell (SSH) key pair
Authenticate your Git Account with SSH
Generate a new SSH key
1. Open the terminal window in the Azure Machine Learning Notebook Tab.
2. Paste the text below, substituting in your email address.
ssh-keygen -t rsa -b 4096 -C your_email@example.com
This creates a new ssh key, using the provided email as a label.
3. When you're prompted to "Enter a file in which to save the key" press Enter. This accepts the default file location.
4. Verify that the default location is '/home/azureuser/.ssh' and press enter. Otherwise specify the location '/home/azureuser/.ssh'.
5. At the prompt, type a secure passphrase. We recommend you add a passphrase to your SSH key for added security Step 4: Add a public key to account1.
Add the public key to Git Account

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1. In your terminal window, copy the contents of your public key file. If you renamed the key, replace id_rsa.pub with the public key file name.
cat ~/.ssh/id_rsa.pub
2. Select and copy the SSH key output to your clipboard.
3. Next, follow the steps to add the SSH key to your preferred account type:
Reference:
https://learn.microsoft.com/en-us/azure/machine-learning/concept-train-model-git-integration

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154) Hotspot Question

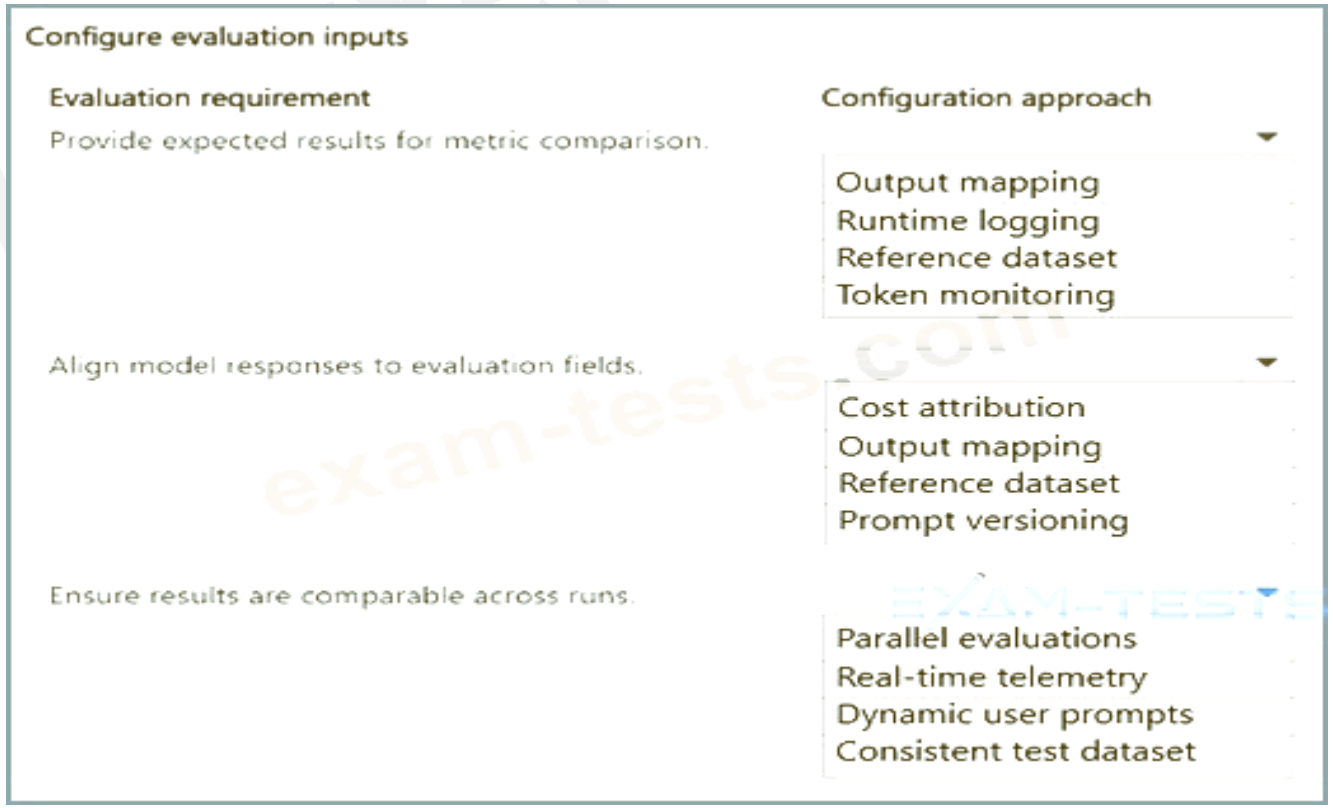
A team is preparing a generative AI application for production deployment. The application generates structured responses that must be evaluated for quality before each release.

The organization requires repeatable evaluation results that can be compared across builds and environments.

You need to configure evaluation inputs so quality metrics can be reliably calculated across test runs.

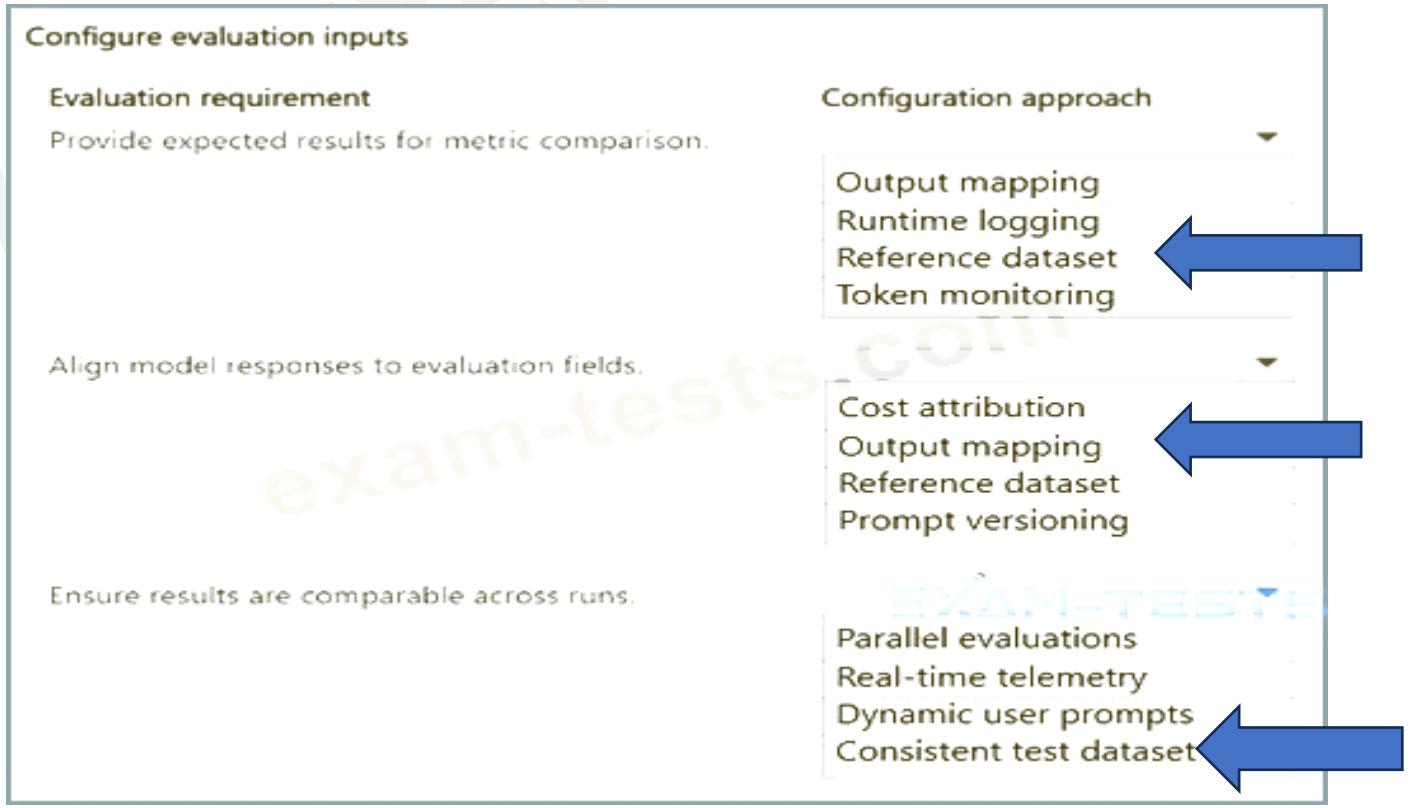
How should you prepare the evaluation inputs? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



154) Hotspot Question
A team is preparing a generative AI application for production deployment. The application generates structured responses that must be evaluated for quality before each release.
The organization requires repeatable evaluation results that can be compared across builds and environments.
You need to configure evaluation inputs so quality metrics can be reliably calculated across test runs.
How should you prepare the evaluation inputs? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer:



Explanation:
Box 1: Reference dataset
Provide expected results for metric comparison.
A reference dataset contains predefined input prompts matched with ground-truth "expected results" (or golden outputs). To calculate quality metrics reliably and compare them across builds and environments, the evaluation framework needs this baseline data to score the generated outputs against a consistent benchmark.
Box 2: Output mapping
Align model responses to evaluation fields.
Output mapping is the process of aligning generative model responses to specific evaluation fields. For structured outputs, your evaluation dataset needs to know exactly which part of the generated response corresponds to the ground truth or target metrics. Configuring output mapping ensures that data fields are aligned consistently across different test runs, making the evaluation repeatable and comparable across builds and environments.
Box 3: Consistent test dataset
Ensure results are comparable across runs.
A consistent test dataset is used to reliably compare evaluation metrics across different builds and environments, you must test the application using the exact same inputs each time. A fixed, high-quality test dataset ensures that changes in metrics reflect changes in the model or application logic rather than variations in the test data itself.
Reference:
https://testquality.com/llm-evaluation-metrics-testing-strategies/
https://www.freecodecamp.org/news/how-to-evaluate-and-select-the-right-llm-for-your-genai-application/

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155) You are implementing hyperparameter tuning by using Bayesian sampling for an Azure ML Python SDK v2-based model training from a notebook. The notebook is in an Azure Machine Learning workspace. The notebook uses a training script that runs on a compute cluster with 20 nodes.
The code implements Bandit termination policy with slackjactor set to 0.2 and a sweep job with max_concurrent_trials set to 10.
You must increase effectiveness of the tuning process by improving sampling convergence.
You need to select which sampling convergence to use.
What should you select?

A. Set the value of max_concurrent_trials to 20.
B. Set the value of max_concurrent_trials to 4.
C. Set the value of slack_factor of early_termination policy to 0.1.
D. Set the value of slack_factor of early_termination policy to 0.9.

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D. Set the value of slack_factor of early_termination policy to 0.9.

Answer: B

Explanation: (Only visible for Exam-tests members)

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156) A team manages prompts used by a generative AI application built on Microsoft Foundry. Prompt updates are frequent, and prior changes have caused unexpected behavior in production.
The team requires that new prompt versions can be tested and reviewed without affecting production behavior.
You need to implement a source control practice that allows safe prompt experimentation.
What should you do?

A. Create a GitHub Action workflow.
B. Create a feature git branch for each prompt change.
C. Revert the prompt in the production environment to a previous commit.

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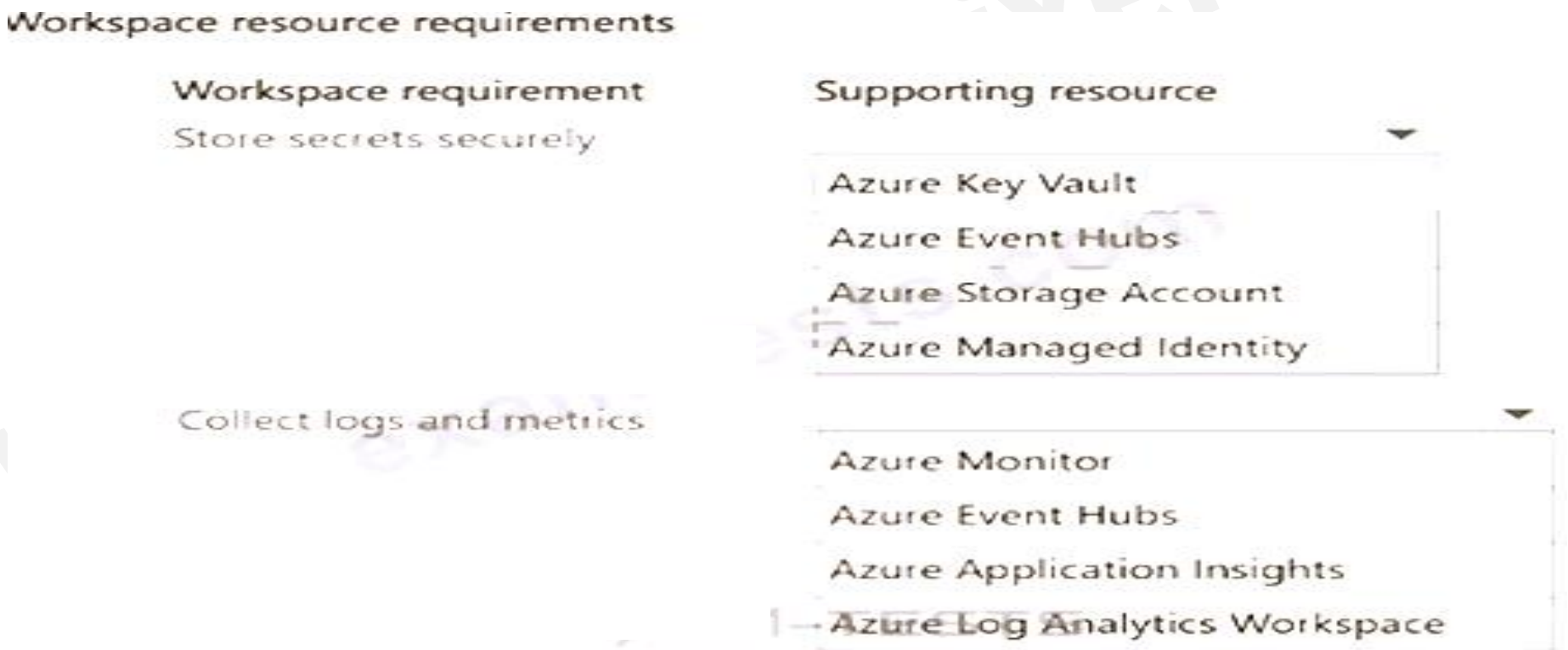
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157) Hotspot Question
A team is provisioning a new Azure Machine Learning workspace for a production project.
The workspace must support secure secret storage and operational monitoring. The team requires the workspace to be created with the correct dependent resources to meet security and monitoring requirements.
You need to configure the required dependencies when the team creates the workspace.
Which resources should you associate with the workspace? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



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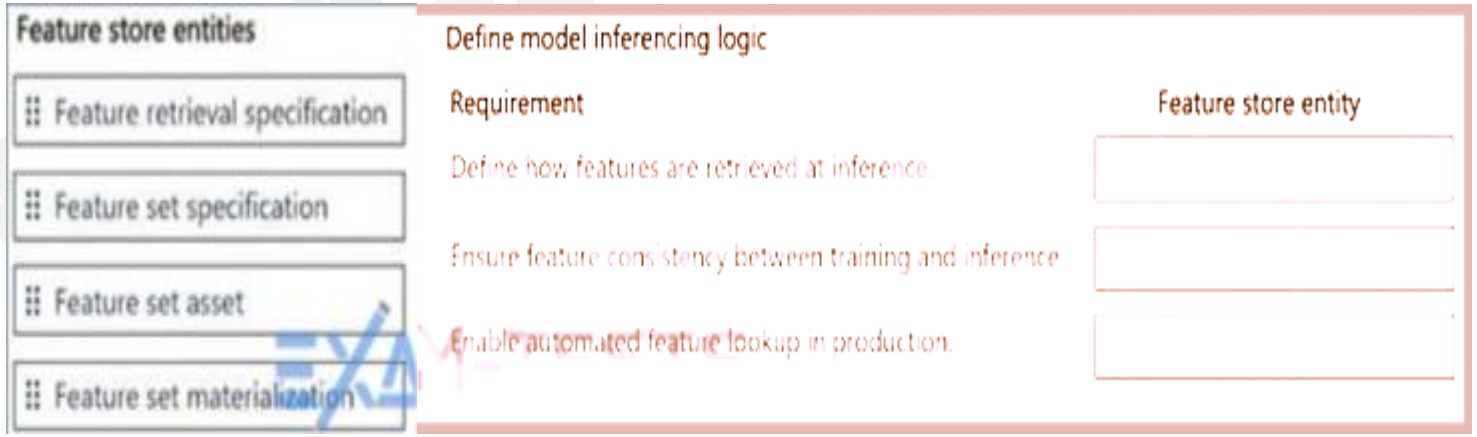
NOTE: Each correct selection is worth one point.

Answer:



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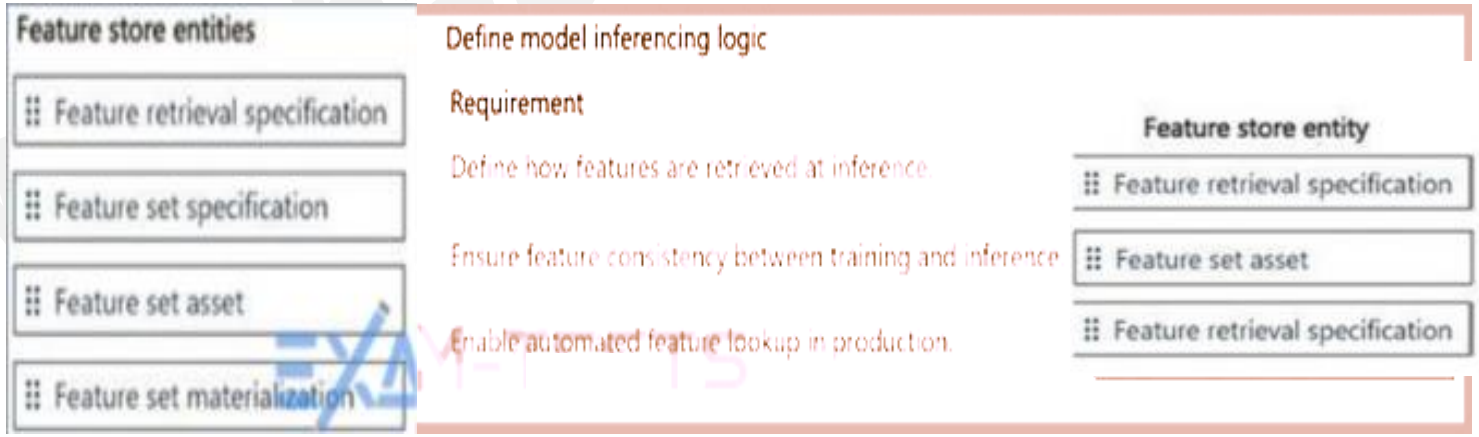
158) Drag and Drop Question
A data science team trains a model that depends on features that are stored in a managed feature store.
The model is registered in Azure Machine Learning and will be deployed to a real-time endpoint.
After deployment, the model must:
- Retrieve feature values dynamically at inference time.
- Use the same feature definitions that were used during training.
- Run without manual configuration changes across environments.
You need to define feature store entities so that feature retrieval behaves as expected when the model is deployed.
Which feature store entity should you select for each requirement? To answer, move the appropriate feature store entities to the correct requirements. You may use each feature store entity once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
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Answer:



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159) You need to improve a GPT-5 model performance based on Fabrikam Inc.'s technical requirements. Which action should you perform first?

A. Generate synthetic interaction data.
B. Evaluate the model output.
C. Fine-tune the model to improve accuracy.
D. Deploy the model to production to gather real-world feedback.

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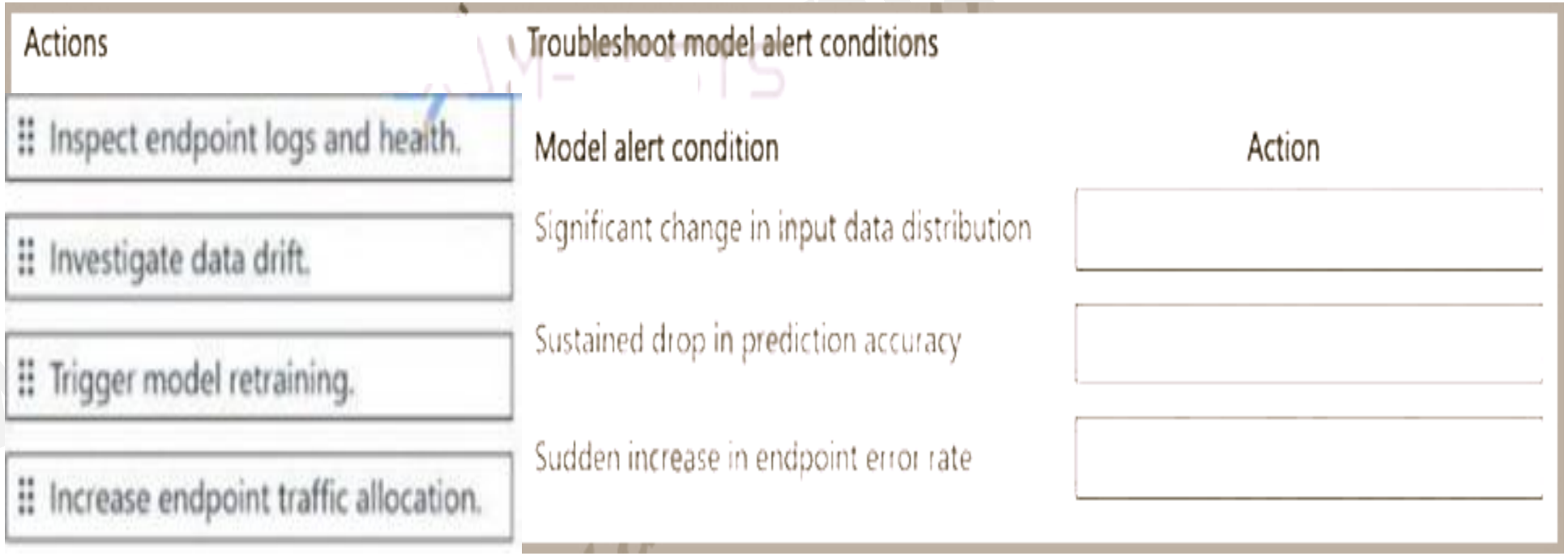
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Answer: B

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160) Drag and Drop Question
A team deploys a machine learning model to production and monitors it continuously. Alerts are configured on performance and data quality metrics.
Multiple alerts are triggered during normal operation.
You need to perform the appropriate action for each model alert condition.
Which action should you perform for each alert condition? To answer, move the appropriate actions to the correct model alert conditions. You may use each action once, more than once, or not at all. You may need to move the split bar between panes or scroll to view content.
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